# 12 Network Functions and Their Time-Domain and Frequency-Domain Response

## Introduction

*In any network, the functions relating the voltages and currents at different parts of the network are known as* network functions. *These functions act as operators which relate the response of the network to the source of the network.*

*In this chapter, the concept of impedance and admittance functions is discussed. Also, the time-domain and frequency domain behaviours of the systems are studied.*

## 12.1 TERMINAL AND TERMINAL PAIRS

A general network is represented by a box. If a conductor is connected to a node of the network and is brought out of the network for access then the end of this conductor is called a terminal.

A pair of terminals is commonly known as a port, across which a device can be connected.

Figure 12.1 shows a one-port and a two-port networks.

Figure 12.1 (c) shows a general *n*-port network. Here, the driving force (energy source) may be connected to one or more ports.

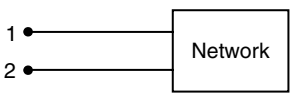



**Fig. 12.1 (a)** *One-port network*

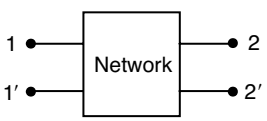



**Fig. 12.1 (b)** *Two-port network*

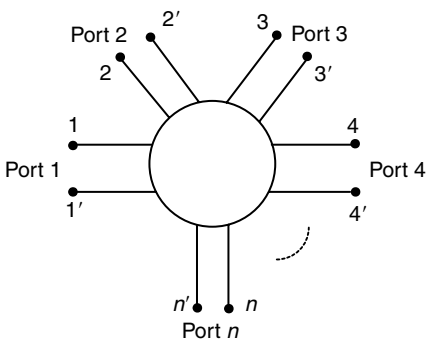



**Fig. 12.1 (c)** *General* n*-port network*

## 12.2 NETWORK FUNCTIONS FOR A ONE-PORT NETWORK

For a one-port network with zero initial condition and with no internal voltage or current source except the dependent sources, the *transform impedance* $Z(s)$ is defined as the ratio of the transform voltage $V(s)$ to the transform current $I(s)$ at the port, i. e., $Z(s)=\dfrac{V(s)}{I(s)}$.

The *transform admittance* is the reciprocal of the transform impedance, i. e., $Y(s)=\dfrac{1}{Z(s)}=\dfrac{I(s)}{V(s)}$.

## 12.3 NETWORK FUNCTIONS FOR TWO-PORT NETWORKS

Two-port network functions are broadly divided into two groups:

(I) Transfer function, and
(II) Driving point function.

### 12.3.1 Transfer Function

It is defined as the ratio of an output transform to an input transform, with zero initial condition and with no internal energy sources except the controlled sources.

For a two-port network, having the variables $I_1(s)$, $I_2(s)$, $V_1(s)$, and $V_2(s)$, the transfer function can take the following four forms:

*Voltage transfer function* $\quad G_{12}(s)=\dfrac{V_1(s)}{V_2(s)};\ \ G_{21}(s)=\dfrac{V_2(s)}{V_1(s)}$

*Current transfer function* $\quad \alpha_{12}(s)=\dfrac{I_1(s)}{I_2(s)};\ \ \alpha_{21}(s)=\dfrac{I_2(s)}{I_1(s)}$

*Transfer impedance function* $\quad Z_{12}(s)=\dfrac{V_1(s)}{I_2(s)};\ \ Z_{21}(s)=\dfrac{V_2(s)}{I_1(s)}$

*Transfer admittance function* $\quad Y_{12}(s)=\dfrac{I_1(s)}{V_2(s)};\ \ Y_{21}(s)=\dfrac{I_2(s)}{V_1(s)}$

Note
*i) For a one-port network, Z(s) = 1/Y(s); but for a two-port network, in general $Z_{12} \neq 1/Y_{12}$; $G_{12} \neq 1/\alpha_{12}$.*
*ii) Z and Y functions will become z and y parameters under the conditions of open-circuits or short-circuits.*

### 12.3.2 Driving Point Function

It takes two forms

***Driving Point Impedance [Z(s)]*** For a two-port network in zero state with no internal energy sources, the driving point impedance is the ratio of transform voltage at any port to the transform current at the same port.

$$Z_{11}(s)=\frac{V_1(s)}{I_1(s)};\ \ Z_{22}(s)=\frac{V_2(s)}{I_2(s)}$$

***Driving Point Admittance [Y(s)]*** For a two-port network in zero state with no internal energy sources, the driving point admittance is the ratio of transform current at any port to the transform voltage at the same port.

$$Y_{11}(s)=\frac{I_1(s)}{V_1(s)};\quad Y_{22}(s)=\frac{I_2(s)}{V_2(s)}$$

Note
*(i) Driving point impedance and admittance functions together are known as immittance function.*
*(ii) Z and Y functions will become z and y parameters under the conditions of open-circuits or short-circuits.*

## 12.4 POLES AND ZEROS OF NETWORK FUNCTIONS

We consider a network function given by the ratio of two polynomials as

$$F(s)=\frac{a_n s^n+a_{n-1}s^{n-1}+\cdots+a_2 s^2+a_1 s+a_0}{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_2 s^2+b_1 s+b_0} \tag{12.1}$$

It is often convenient to factor the polynomials in the numerator and denominator, and to write the transfer function in terms of those factors:

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)} \tag{12.2}$$

where, the numerator and denominator polynomials, *N(s)*and *D(s)*, have real coefficients defined by the system's differential equation and $K=\frac{a_n}{b_m}$ is a positive constant, known as scale factor.

From Eq. (12.2), we observe the following:

At $s = z_i, i = 1,2,3,\ldots, n$ the numerator polynomial $N(s) = 0$; these complex frequencies are known as the *zeros of the network function F(s)*. At zeros, the value of the network function is zero, i.e., $\lim_{s\to z_i} F(s)=0$.

At $s = p_i, i = 1,2,3,\ldots, m,$ the denominator polynomial $D(s) = 0$; these complex frequencies are known as the *poles of the network function F(s)*. At poles, the value of the network function is infinity, i.e., $\lim_{s\to p_i} F(s)=\infty$.

At any frequencies other than poles and zeros, the network function has a finite zero value. The values of poles and zeros of $F(s)$ and their locations in the $s$-plane completely specify a network function. All the coefficients of polynomials $N(s)$ and $D(s)$ are real, therefore the poles and zeros must be either purely real, or appear in complex conjugate pairs. In general for the poles, either $p_i = \sigma_i$, or else $p_i, p_{i+1} = \sigma_i \pm j\omega_i$. The existence of a single complex pole without a corresponding conjugate pole would generate complex coefficients in the polynomial $D(s)$. Similarly, the system zeros are either real or appear in complex conjugate pairs.

The poles and zeros are properties of the transfer function, and therefore of the differential equation describing the input–output system dynamics. Together with the gain constant $K$ they completely characterize the differential equation, and provide a complete description of the system.

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Example 12.1 ***A linear system is described by the differential equation,*** $\frac{d^2y}{dt^2}+5\frac{dy}{dt}+6y=2\frac{du}{dt}+1$

***Find the system poles and zeros.***

*Solution* From the differential equation the transfer function is $F(s)=\dfrac{2s+1}{s^2+5s+6}$

which may be written in factored form as $F(s)=\dfrac{2s+1}{s^2+5s+6}=\dfrac{1}{2}\left[\dfrac{s+1/2}{(s+3)(s+2)}\right]$

The system therefore has a single real zero at $s=-1/2$ and a pair of real poles at $s=-3$ and $s=-2$.

Example 12.2 ***A system has a pair of complex conjugate poles $p_1, p_2 = -1 \pm j2$ , a single real zero $z_1 = -4$, and a gain factor $k = 3$. Find the differential equation representing the system.***

*Solution* The network function is

$$F(s)=K\frac{(s-z)}{(s-p_1)(s-p_2)}=3\frac{s-(-4)}{\{s-(-1-j2)\}\{s-(-1+j2)\}}=3\frac{s+4}{(s+1-j2)(s+1-j2)}=3\frac{s+4}{s^2+2s+5}$$

and the differential equation is $\dfrac{d^2y}{dt^2}+2\dfrac{dy}{dt}+5y=3\dfrac{du}{dt}+12$

## 12.5 POLE ZERO DIAGRAM

The plot of poles and zeros of a network function in the *s*-plane is generally termed as pole-zero configuration or *p–z* pattern or pole–zero diagram or pole-zero plot of the network function.

A system is characterized by its poles and zeros in the sense that they allow reconstruction of the input/output differential equation. In general, the poles and zeros of a network function may be complex, and the system dynamics may be represented graphically by plotting their locations on the complex *s*-plane, whose axes represent the real and imaginary parts of the complex variable *s*. Such plots are known as pole–zero plots.

It is usual to mark a zero location by a circle (O) and a pole location by a cross (×). The location of the poles and zeros provide qualitative insights into the response characteristics of a system. Many computer programs are available to determine the poles and zeros of a system from either the network function or the system state equations.

Example 12.3 ***Draw the pole–zero plot of the network function given as $F(s)=\dfrac{s-a}{(s-b_1)(s+b_2)}$, where, $a$, $b_1$, and $b_2$ are positive constants.***

*Solution* The zero is at $s = a$ and the poles are at $s = b_1$ and $s = -b_2$.
The pole-zero plot is shown in Fig. 12.2.

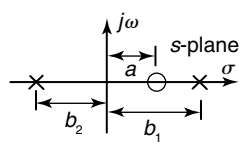



**Fig. 12.2** *Pole –zero plot of Example 12.3*

## 12.6 SIGNIFICANCE OF POLES AND ZEROS

The significance of poles and zeros for different types of network functions are explained below.

***For Driving Point Impedance Functions*** Here, the network function is $F(s)=Z(s)=\dfrac{V(s)}{I(s)}$

At poles, $Z(s) = \infty$ and at zeros, $Z(s) = 0$.

Therefore, poles of *Z(s)* imply–zero current for a finite voltage, i.e., open-*circuit*, and
zeros of *Z(s)* imply–zero voltage for a finite current, i.e., *short-circuit*.

Thus, a one-port network has an open-circuit at pole-frequencies and a short-circuit at zero-frequencies.

***For Driving Point Admittance Functions*** Here, the network function is $F(s)=Y(s)=\dfrac{I(s)}{V(s)}$

At poles, $Y(s) = \infty$, which implies zero voltage for a finite current, i.e., *short-circuit*, and at zeros, $Y(s) = 0$, which implies zero current for a finite voltage, i.e., *open-circuit*.

***For Transfer Functions*** We consider a voltage ratio transfer function,

$$G_{21}(s)=\frac{V_2(s)}{V_1(s)} \qquad \therefore V_2(s)=G_{21}(s)V_1(s)$$

If the input $V_1(s)$ or $v_1(t)$ is given and $G_{21}(s)$ is known, the output $V_2(s)$ or $v_2(t)$ can be obtained as follows:

$$V_2(s)=G_{21}(s)V_1(s)=\sum_{i=1}^{n}\frac{K_i}{(s-p_i)}+\sum_{j=1}^{m}\frac{K_j}{(s-p_j)}$$

where, $n$ and $m$ are the number of poles of $G_{21}(s)$ and $V_1(s)$, respectively.

Taking inverse Laplace transform, the output voltage is

$$v_2(t)=L^{-1}\left[G_{21}(s)V_1(s)\right]=L^{-1}\left[\sum_{i=1}^{n}\frac{K_i}{(s-p_i)}+\sum_{j=1}^{m}\frac{K_j}{(s-p_j)}\right]=\sum_{i=1}^{n}K_i e^{p_i t}+\sum_{j=1}^{m}K_j e^{p_j t}$$

The frequencies $p_i$ are the *natural frequencies of free oscillations*. These frequencies depend entirely upon the network transfer function $G_{21}(s)$.

The frequencies $p_j$ are the *frequencies of forced oscillations*, which depend entirely upon the source $V_1(s)$.

Thus, the poles determine the time-variation of the response, while the zeros determine the magnitudes of the response (given by the magnitudes of coefficients $K_i$ and $K_j$).

***For Other Network Functions*** Significance of poles and zeros in other network functions, namely, $Y_{21}(s)$, $Z_{21}(s)$ and $\alpha_{21}(s)$ is the same as discussed for voltage transfer ratio, $G_{21}(s)$.

## 12.7 NATURAL RESPONSE AND NATURAL FREQUENCIES

The *natural response* or *source-free response* of a network is the response with no source excitation; it is the response of the network to its initial conditions. This response is also known as the *transient response*.

The natural response of a system is also described as the response to unit impulse excitation. It is characterized by the *natural frequencies of oscillation* of the system.

We consider a driving-point impedance function as

$$Z_{11}(s)=\frac{V_1(s)}{I_1(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)} \tag{12.3}$$

For two different cases, two different natural frequencies are defined.

### 12.7.1 Open-Circuit Natural Frequencies (OCNF)

If $I_1 = 0$, i.e., the input is open-circuited, then from Eq. (12.3), $V_1 = 0$ unless $Z_{11}(s)$ is infinite. Therefore, with input open-circuited, for a voltage to exist, it is necessary that $Z_{11}(s)$ is infinite.

In this case, the poles of $Z_{11}(s)$, i.e., $p_1, p_2, \ldots, p_m$ are known as the *open-circuit natural frequencies (OCNF)* of the network.

### 12.7.2 Short-Circuit Natural Frequencies (SCNF)

If $V_1 = 0$, i.e., the input is short-circuited, then from Eq. (12.3), $I_1$ will be zero unless $Z_{11}(s)$ is zero. Therefore, with input short-circuited, for a current to exist, it is necessary that $Z_{11}(s)$ is zero.

In this case, the zeros of $Z_{11}(s)$, i.e., $z_1, z_2, \ldots, z_n$ are known as the *short-circuit natural frequencies (SCNF)* of the network.

We conclude that

(i) the poles of an impedance function are OCNF of the network
(ii) the zeros of an impedance function are SCNF of the network
(iii) the poles of an admittance function are SCNF of the network
(iv) the zeros of an admittance function are OCNF of the network

## 12.8 RELATION BETWEEN POLE POSITION, NATURAL RESPONSE AND STABILITY

### 12.8.1 Pole Position and Natural Response

As the transfer function completely represents a system differential equation, its poles and zeros effectively define the system response. In particular, the system poles directly define the components in the homogeneous or natural response. The unforced or natural response of a linear BIBO system to a set of initial conditions is

$$y(t) = \sum_{i=1}^{n} K_i e^{\lambda_i t} \tag{12.4}$$

where, the constants $K_i$ are determined from the given set of initial conditions and the exponents $\lambda_i$ are the roots of the characteristic equation of the system. The characteristic equation is

$$D(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 \tag{12.5}$$

and its roots are the system poles, that is $\lambda_i = p_i$.

Therefore, the natural response may be written as

$$y(t) = \sum_{i=1}^{n} K_i e^{p_i t} \tag{12.6}$$

### 12.8.2 Pole Position and System Stability

The stability of a linear system may be determined directly from its transfer function. An $n^{\text{th}}$ order linear system is asymptotically stable only if all of the components in the homogeneous response from a finite set of initial conditions decay to zero as time increases, i.e.,

$$\lim_{t \to \infty} \sum_{i=1}^{n} K_i e^{p_i t} = 0 \tag{12.7}$$

where $p_i$ are the system poles. In a stable system, all components of the homogeneous response must decay to zero as time increases. If any pole has a positive real part, there is a component in the output that increases without bound, causing the system to be unstable.

*For a linear system to be stable, all of its poles must have negative real parts, i.e., they must lie within the left-half of the s-plane.*

A system having one or more poles lying on the imaginary axis of the *s*-plane has non-decaying oscillatory components in its natural response, and is defined as a *marginally stable system.*

The location of the poles in the *s*-plane therefore defines the *n* components in the homogeneous response and the stability of the system as described below:

1. A real pole $p_i = -\sigma$ in the left-half of the *s*-plane defines an exponentially decaying component, $Ke^{-\sigma t}$, in the homogeneous response. The rate of the decay is determined by the pole location; poles far from the origin in the left-half plane correspond to components that decay rapidly, while poles near the origin correspond to slowly decaying components. In this case, the system is a stable system.
2. A pole at the origin $p_i = 0$ defines a component that is constant in amplitude and defined by the initial conditions. In this case, the system is a marginally stable system.
3. A real pole in the right-half *s*-plane corresponds to an exponentially increasing component $Ke^{\sigma t}$ in the homogeneous response; thus defining the system to be unstable. In this case, the system is an unstable system.
4. A complex conjugate pole pair ($\sigma \pm j\omega$) in the left-half of the *s*-plane combine to generate a response component that is a decaying sinusoid of the form $Ae^{-\sigma t}\sin(\omega t + \varphi)$where $A$ and $\varphi$ are determined by the initial conditions. The rate of decay is specified by $\sigma$; the frequency of oscillation is determined by $\omega$. The system is a stable system.
5. An imaginary pole pair, i.e., a pole pair lying on the imaginary axis, $\pm j\omega$, generates an oscillatory component with a constant amplitude determined by the initial conditions. The system is a marginally stable system
6. A complex pole pair in the right half-plane generates an exponentially increasing component. The system is an unstable system.

These results are summarized in Fig. 12.3.

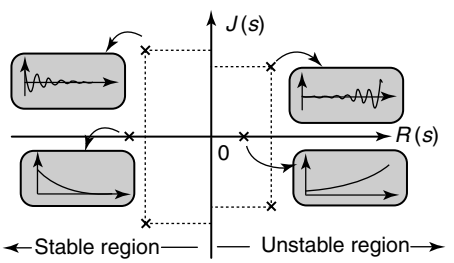



**Fig. 12.3** *Pole position and system stability*

## 12.9 RESTRICTION ON THE LOCATION OF THE POLES AND ZEROS IN THE *s*-PLANE

We consider a network function, $F(s) = \dfrac{N(s)}{D(s)}$ .

1. The coefficients in the polynomials $N(s)$ and $D(s)$ must be real and positive.
2. Complex and imaginary poles must be conjugate.
3. The real part of all poles and zeros must either be negative or zero. Thus, all poles and zeros must lie on the left half of the *s*-plane or on the imaginary axis.
4. If the real part of any pole or zero is zero then that pole or zero must be simple, not repeated. This includes the origin.
5. Polynomials $N(s)$ and $D(s)$ must not have missing terms between those of highest and lowest order unless all even terms or all odd terms are missing.
6. The degree of $N(s)$ and $D(s)$ may differ by zero or one only.
7. The lowest degree of $N(s)$ and $D(s)$ may differ by one at most.

## 12.10 NECESSARY CONDITIONS FOR DRIVING POINT FUNCTIONS (RESTRICTION ON POLE-ZERO LOCATIONS IN THE *s*-PLANE FOR DRIVING POINT FUNCTIONS)

1. The coefficients in the polynomials $N(s)$ and $D(s)$ must be real and positive.
2. Complex and imaginary poles must be conjugate.
3. The real part of all poles and zeros must either be negative or zero. Thus, all poles and zeros must lie on the left half of the $s$-plane or on the imaginary axis.
4. If the real part of any pole or zero is zero then that pole or zero must be simple, not repeated. This includes the origin.
5. Polynomials $N(s)$ and $D(s)$ must not have missing terms between those of highest and lowest degree unless all even terms or all odd terms are missing.
6. The degree of $N(s)$ and $D(s)$ may differ by zero or one only.
7. The lowest degree of $N(s)$ and $D(s)$ may differ by one at most.

### 12.10.1 Explanations of the Conditions

***The Coefficients in the Polynomials N(s) and D(s) Must be Real and Positive*** Any network function may be written as

$$F(s)=\frac{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0}{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_2 s^2 + b_1 s + b_0} \tag{12.1}$$

where the coefficients $a$'s and $b$'s are real and positive. Hence, as $s$ is real, $N(s)$ and $D(s)$ will also be real.

***Complex and Imaginary Poles Must be Conjugate*** Let $N(s)$ have a pair of conjugate zeros given as $(-a - jb)$ and $(-a + jb)$. $\therefore N(s)=(s+a+jb)(s+a-jb)=(s+a)^2+b^2=s^2+2as+(a^2+b^2)$

This $N(s)$ is real function as both $a$ and $b$ are real.

Now, let $N(s)$ have two complex zeros $(-a - jb)$ and $(-c + jb)$ which are not conjugate.

Then, $\therefore N(s)=(s+a+jb)(s+c+jd)=s^2+[a+c+j(b+d)]s+[ac-bd+j(bc+ad)]$

which is not a real function. Hence such zeros are inadmissible in network functions.

Therefore, we conclude that in a real network function, all complex and/or imaginary zeros and poles must occur in conjugate pairs.

***The Real Part of All Poles and Zeros Must Either be Negative or Zero*** With two conjugate zeros $(-a - jb)$ and $(-a + jb)$, the network function is $N(s) = s^2 + 2as + (a^2 + b^2)$.

In order that $N$(s) be real, the second term '$2as$' must be positive or '$a$' must be positive, i.e., $-a$ must be negative.

This restriction that the real part of a pole and zero be negative or zero, arises from the assumption that any network constituted by passive elements must be stable; i.e., bounded input must produce bounded output (BIBO stability).

We consider a driving point impedance $Z(s)$. Let the denominator contain a pole at $s = s_1 = (\sigma + j\omega)$. Then, the factor $(s - s_1)$ will cause a time-domain response given as

$$\text{response,} \quad y_1(t) = A_1 e^{s_1 t} = A_1 e^{\sigma t} e^{j\omega t}$$

As the network function is real, the denominator must also contain a factor $(s - s_2)$ where, $s_2 = (\sigma - j\omega)$. The corresponding time-domain response is

$$\text{response,} \quad y_2(t) = A_2 e^{s_2 t} = A_2 e^{\sigma t} e^{-j\omega t}$$

Combining these two, the total response is given as

$$\text{response,} \quad y(t) = y_1(t) + y_2(t) = e^{\sigma t} [A_1 e^{j\omega t} + A_2 e^{-j\omega t}] = A_0 e^{\sigma t} \sin(\omega t + \phi)$$

From this expression, it is observed that the response must be bounded if $\sigma \leq 0$.

Thus, we conclude that the real parts of all poles and zeros for a driving point function must be either negative or zero. This means that the poles and zeros must lie on the left half of the *s*-plane or on the imaginary axis; no pole or zero should lie on the right half of the *s*-plane.

***If the Real Part of Any Pole or Zero is Zero then that Pole or Zero Must be Simple, not Repeated*** For the poles or zeros located on the imaginary axis, the real parts are zero. The restriction for such poles or zeros is that they must be simple, not multiple or repeated.

This is because if the poles or zeros be of order '*n*' then the time-domain response contains terms $t^{n-1}$ and this term becomes infinite as $t \to \infty$; resulting in an unstable system.

Therefore, these poles or zeros must be simple.

***Polynomials N(s) and D(s) Must not have Missing Terms Between Those of Highest and Lowest Degree Unless All Even Terms or All Odd Terms are Missing*** The polynomials $N(s)$ and $D(s)$ have the factors:

$$K; (s + a); (s^2 + bs + c); (s^2 + d)$$

where $K, a, b, c$ and $d$ are all real and positive; '$a$' can be zero, too.

When the product of these factors is taken, there is no mechanism for a negative sign to be introduced and the coefficients of all the terms in $N(s)$ and $D(s)$ will be positive and real.

Also, since negative sign cannot be introduced, there exits no chance of cancellation of a positive term with a negative term. Thus, polynomials $N(s)$ and $D(s)$ have all coefficients non-zero. However, two conditions may happen:

(i) If the polynomial is composed of factors like $(s^2 + d)$ only; the polynomial contains only even-order terms, all odd-order terms being missing.

(ii) If the polynomial is composed of a simple (not multiple) pole or zero at the origin and other factors like $(s^2 + d)$ then the polynomial contains only odd terms, all even-order terms being missing.

***The Degree of N(s) and D(s) may Differ by Zero or One Only***

We consider the behaviour of a one-port network at very high and very low frequencies.

*At Very High Frequencies* Here, impedance of an inductor, $Z_L(s) = sL$ is very large compared to the impedance of any other element. The impedance of a capacitor, $Z_C(s) = \frac{1}{sC}$ is very small and may be neglected.

Hence, at very high frequencies, if a network contains an inductor, the impedance function of the network is represented by an equivalent inductance $L_{eq}$ as shown in Fig. 12.4 (a). If the network does not contain $L$ then either the resistor will dominate the impedance function of the network or the equivalent network would be short circuit $\left(\because \frac{1}{sC} \to 0 \quad \text{as} \quad s \to \infty\right)$, as shown in Fig. 12.4 (b).

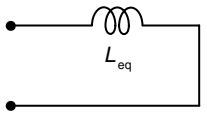



**Fig. 12.4 (a)** Z(s) *at high frequencies with* L *present*

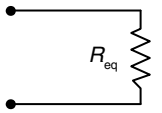



**Fig. 12.4 (b)** Z(s) *at high frequencies with* L *absent*

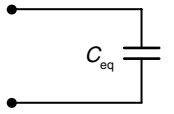



**Fig. 12.4 (c)** Y(s) *at high frequencies with* C *present*

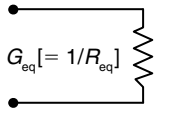



**Fig. 12.4 (d)** Y(s) *at high frequencies with* C *absent*

Similarly, at high frequencies, the admittances of an inductor and a capacitor will be,

$$Y_L(s)=\frac{1}{sL} \quad \text{and} \quad Y_C(s)=sC$$

and thus, $Y_L$ is very small and $Y_C(s)$ is very large. Thus, the admittance function may be represented by an equivalent capacitance $C_{eq}$. If the network does not contain a capacitor $C$ then the admittance function may be represented by a conductance $G_{eq}\left(=\frac{1}{R_{eq}}\right)$ or an open circuit.

Therefore, we conclude that the equivalent representation of a one-port network at high frequencies involves at most one kind of element.

*At Very Low Frequencies* At low frequencies, also, similar considerations may be drawn for impedance and admittance function of a one-port network.

• *Requisite Conditions for a Driving Point Voltage to Assume Equivalent Representation of Fig. 12.4 (a) to (d)* We consider a network function,

$$F(s)=\frac{a_n s^n+a_{n-1}s^{n-1}+\cdots+a_2 s^2+a_1 s+a_0}{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_2 s^2+b_1 s+b_0} \tag{12.1}$$

The value of the network function $F(s)$ when $s \to \infty$ is,

$$\lim_{s\to\infty}\left[F(s)\right]=\lim_{s\to\infty}\left[\frac{a_n s^n+a_{n-1}s^{n-1}+\cdots+a_2 s^2+a_1 s+a_0}{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_2 s^2+b_1 s+b_0}\right]=\lim_{s\to\infty}\left(\frac{a_n s^n}{b_m s^m}\right)=\lim_{s\to\infty}\left(\frac{a_n}{b_m}\right)s^{n-m}$$

But, we have already established that a one-port network assumes one of the three forms of Fig. 12.4 (a) to (d) at high frequencies (i.e., $s \to \infty$). Therefore, it is necessary that $\lim_{s\to\infty}\left[F(s)\right]$ assumes one of the three forms, namely $s$, 1 or $1/s$.

$\therefore$ $$|n-m| \le 1 \quad \text{or,} \quad (n-m)=-1, 0, \text{ or } 1$$

where $n$ and $m$ are the degrees of polynomials $N(s)$ and $D(s)$, respectively.

Hence, we conclude that the degrees of the polynomials $N(s)$ and $D(s)$ for a driving point function may differ by one at most.

***Lowest Degree of N(s) and D(s) may Differ by One at Most*** At low frequencies, as $s \to 0$, the higher order terms in $N(s)$ and $D(s)$ may be neglected, so that, the network function is

$\therefore$ $$F(s)=\frac{a_0+a_1 s+\cdots}{b_0+b_1 s+\cdots}$$

For this function to assume one of the four forms given by Fig. 12.4 (a) to (d), it is necessary that, $\lim_{s\to 0}\left[F(s)\right]$ is either $s$, a or $1/s$.

Three possibilities are there:

*Case (1)* $a_0 = 0$ and $b_0 \neq 0$

Here, $$\lim_{s\to 0}\left[F(s)\right] = \lim_{s\to 0}\left(\frac{a_1 s}{b_0 + b_1 s}\right) = \lim_{s\to 0}\left(\frac{a_1}{b_0}\right)s$$

Thus, the network function $F(s)$ represents

- an inductor, if $F(s)$ is a driving point impedance
- a capacitor, if $F(s)$ is a driving point admittance

*Case (2)* $b_0 \neq 0$ and $a_0 \neq 0$

Here, $$\lim_{s\to 0}\left[F(s)\right] = \lim_{s\to 0}\left(\frac{a_1 s + a_0}{b_1 s}\right) = \lim_{s\to 0}\left(\frac{a_0}{b_1}\right)\frac{1}{s}$$

Thus, the network function $F(s)$ represents

- a capacitor, if $F(s)$ is a driving point impedance
- an inductor, if $F(s)$ is a driving point admittance

*Case (3)* $a_0 \neq 0$ and $b_0 \neq 0$

Here, $$\lim_{s\to 0}\left[F(s)\right] = \lim_{s\to 0}\left(\frac{a_0 + a_1 s}{b_0 + b_1 s}\right) = \frac{a_0}{b_0}$$

Thus, the network function $F(s)$ becomes a constant which represents a resistor.

In all three cases, the difference of the degree of the terms in $N(s)$ and $D(s)$ is either one [Case (1) and Case (2)] or zero [Case (3)]. Therefore, we conclude that, in any driving point function F($s$), the terms of lowest degree in the numerator and denominator may differ by one at most.

---

Example 12.4 ***Verify whether the following network functions represent the driving point immittance:***

*(a)* $F(s) = \dfrac{s^4 + 2s^3 - 2s + 1}{s^3 + s^2 - 2s + 12}$ *(b)* $F(s) = \dfrac{s^2 + s + 2}{s^4 + 5s^3 + 6s}$

*(c)* $F(s) = \dfrac{s^2(s^2 - 2s + 6)}{s + 3}$ *(d)* $F(s) = \dfrac{s^2 + 5s + 4}{s^2 + 6s + 10}$

*Solution* (a) $F(s) = \dfrac{s^4 + 2s^3 - 2s + 1}{s^3 + s^2 - 2s + 12}$

This function does not represent a driving point function because of the following reasons:

(i) In $N(s)$, one coefficient is negative.
(ii) In $D(s)$, one coefficient is negative.
(iii) In $N(s)$, one coefficient is missing.
(iv) In $D(s)$, by factorization, $D(s) = (s + 3)(s^2 - 2s + 4)$, which has two poles on the right half of the $s$-plane.

(b) $F(s)=\dfrac{s^2+s+2}{s^4+5s^3+6s}$

This function does not represent a driving point function because of the following reasons:

(i) The difference of degrees of $N(s)$ and $D(s)$ is 2, which is more than the permitted maximum value of 1.

(ii) In $D(s)$, by factorization, $D(s) = s^2(s^2 + 5s + 6)$, which has two poles at origin, which is not allowed.

(c) $F(s)=\dfrac{s^2(s^2-2s+6)}{s+3}$

This function does not represent a driving point function because of the following reasons:

(i) The difference of lowest degree terms of $N(s)$ and $D(s)$ is 2, which is more than the permitted maximum value of 1.

(ii) In $N(s)$, a repeated zero is at origin, which is not allowed.

(iii) In $N(s)$, $(s^2 - 2s + 6)$ has two poles in the right-half of the $s$-plane.

(iv) The difference of degrees of $N(s)$ and $D(s)$ is 3, which is more than the permitted maximum value of 1.

(d) $F(s)=\dfrac{s^2+5s+4}{s^2+6s+10}$

This function represents driving point function because all conditions are satisfied.

## 12.11 NECESSARY CONDITIONS FOR TRANSFER FUNCTIONS (RESTRICTION ON POLE-ZERO LOCATIONS IN THE *s*-PLANE FOR TRANSFER FUNCTIONS)

1. The coefficients in the polynomials $N(s)$ and $D(s)$ must be real and positive.
2. Complex and imaginary poles must be conjugate.
3. The real part of all poles and zeros must either be negative or zero. Thus, all poles and zeros must lie on the left half of the $s$-plane or on the imaginary axis.
4. If the real part of any pole or zero is zero then that pole or zero must be simple, not repeated. This includes the origin.
5. The polynomial $D(s)$ may not have any missing terms between those of highest and lowest degree unless all even terms or all odd terms are missing.
6. The polynomial $N(s)$ may have missing terms between those of highest and lowest degree; some coefficients of $N(s)$ may be negative, too.
7. The degree of $N(s)$ may be as small as zero, independent of the degree of $D(s)$.
8. (a) For the voltage ratio transfer functions, $G$ and current ratio transfer functions, $\alpha$, the maximum degree of $N(s)$ is the degree of $D(s)$.
   (b) For transfer impedance $Z$ and transfer admittance $Y$, the maximum degree of $N(s)$ is the degree of $D(s)$ plus one.

## 12.12 TIME DOMAIN BEHAVIOUR FROM POLE–ZERO PLOT

The time-domain response of a given network can be determined from the pole-zero plot of the network function and the knowledge of the network sources.

We consider a network function as given in Eq. (12.2), i.e.,

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)} \tag{12.2}$$

where, $z_1, z_2, \ldots, z_n$ are the zeros, $p_1, p_2, \ldots, p_m$ are the poles and $K$ is a scale factor.

Assuming all the zeros and poles to be distinct, the network function can be written by partial fraction expansion as

$$F(s)=\frac{K_1}{s-p_1}+\frac{K_2}{s-p_2}+\cdots+\frac{K_m}{s-p_m} \tag{12.8}$$

Any of the coefficients can be found by the Heaviside method as

$$K_i=\left[(s-p_i)F(s)\right]_{s=p_i}=K\frac{(p_i-z_1)(p_i-z_2)(p_i-z_3)\cdots(p_i-z_n)}{(p_i-p_1)(p_i-p_2)(p_i-p_3)\cdots(p_i-p_{i-1})(p_i-p_{i+1})\cdots(p_i-p_m)} \tag{12.9}$$

In this equation, $p_i$, $z_n$ and $p_m$ all are complex numbers. Therefore, the factors like $(p_i - z_n)$ and $(p_i - p_m)$ are the difference between two complex numbers. This difference can be written as

$$\begin{aligned}(p_i-z_1)&=M_{1i}\angle\theta_{1i} & (p_i-p_1)&=N_{1i}\angle\alpha_{1i}\\ (p_i-z_2)&=M_{2i}\angle\theta_{2i} \quad \text{and} & (p_i-p_2)&=N_{2i}\angle\alpha_{2i}\\ &\vdots & &\vdots\\ (p_i-z_n)&=M_{ni}\angle\theta_{ni} & (p_i-p_m)&=N_{mi}\angle\alpha_{mi}\end{aligned}$$

where, $M$'s and $N$'s are the magnitudes of the phasors $(p_i - z_n)$ and $(p_i - p_m)$, respectively and $\theta$'s and $\alpha$'s are the corresponding phase angles.

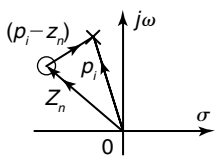



**Fig. 12.5** *Determination of coefficients of K's*

Figure 12.5 shows the difference of two complex quantities in the $s$-plane.

Now, expressing each term in terms of its magnitude and phase angle, the coefficients can be written as

$$K_i=K\frac{M_{1i}M_{2i}M_{3i}\cdots M_{ni}}{N_{1i}N_{2i}N_{3i}\cdots N_{mi}}\angle\left(\theta_{1i}+\theta_{2i}+\theta_{3i}+\cdots+\theta_{ni}\right)-\left(\alpha_{1i}+\alpha_{2i}+\alpha_{3i}+\cdots+\alpha_{mi}\right) \tag{12.10}$$

or, $$K_i=K\frac{\text{Product of all the directed lines from all zeros to } p_i}{\text{Product of all the directed lines from other remaining poles to } p_i}$$

Thus, all coefficients can be found out using Eq. (12.10). The time-domain response can be found out by taking the inverse Laplace transform,

$$f(t)=L^{-1}\left[F(s)\right]=L^{-1}\left[\frac{K_1}{s-p_1}+\frac{K_2}{s-p_2}+\cdots+\frac{K_m}{s-p_m}\right]$$

---

Example12.5 ***Draw poles and zeros for*** $V(s)=\dfrac{5s}{s^2+7s+12}$ ***and evaluate v(t) by making use of a pole–zero diagram.***

*Solution* $$V(s)=\frac{5s}{s^2+7s+12}=\frac{5s}{(s+3)(s+4)}$$

The function has a zero at $s = 0$ and two poles at $s = -3, -4$. The pole-zero diagram is shown in Fig. 12.6.

To find the inverse Laplace transform of $V(s)$, we use pole–zero diagram.

Since the factors (s + 3) and (s + 4) appear in the denominator polynomial, the time-domain response will be $v(t) = K_1 e^{-4t} + K_2 e^{-3t}$

The coefficients $K_1$ and $K_2$ are evaluated using the pole–zero diagram.

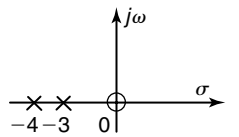



**Fig. 12.6** *Pole–zero diagram for* $\frac{5s}{(s+3)(s+4)}$

$$M_1 = 4 \qquad \theta_1 = 180° \qquad N_1 = 1 \qquad \alpha_1 = 180°$$

Hence, $$K_1 = K\frac{4\angle 180°}{1\angle 180°} = 5\times\frac{4\angle 180°}{1\angle 180°} = 20$$

$$M_2 = 3 \qquad \theta_1 = 180° \qquad N_1 = 1 \qquad \alpha_{01} = 0°$$

Hence, $$K_2 = K\frac{3\angle 180°}{1\angle 0°} = 5\times\frac{3\angle 180°}{1\angle 0°} = -15$$

Substituting the values of $K_1$ and $K_2$ we get $v(t) = 20e^{-4t} - 15e^{-3t}$

## 12.13 FREQUENCY DOMAIN BEHAVIOUR FROM POLE–ZERO PLOT

The steady-state response of a system for a sinusoidal input is known as *frequency response*. According to Fourier theorem, any function can be represented as a summation of sinusoids of different amplitudes, phases and frequencies. Therefore, the response of a network to any arbitrary excitation can be calculated by calculating the response to each of its sinusoidal components and then applying the superposition theorem.

The plots of the amplitude and phase of the response of a network against radian frequency ($\omega$) are known as *frequency response plots*.

### 12.13.1 Sinusoidal Network Function and Its Amplitude and Phase Plots

Let $F$(s) be a network function given by Eq. (12.1) as

$$F(s) = \frac{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0}{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_2 s^2 + b_1 s + b_0} \tag{12.1}$$

For sinusoidal steady-state response, we put $s = j\omega$ so that the function becomes

$$F(j\omega) = \frac{a_n (j\omega)^n + a_{n-1}(j\omega)^{n-1} + \cdots + a_2 (j\omega)^2 + a_1 (j\omega) + a_0}{b_m (j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_2 (j\omega)^2 + b_1 (j\omega) + b_0} \tag{12.11}$$

or, $$F(j\omega) = \left(\frac{a_n}{b_m}\right)\frac{(j\omega - z_1)(j\omega - z_2)(j\omega - z_3)\cdots(j\omega - z_n)}{(j\omega - p_1)(j\omega - p_2)(j\omega - p_3)\cdots(j\omega - p_m)} \tag{12.12}$$

This $F(j\omega)$, in general is a complex function and may be written as

$$F(j\omega) = \text{Re}[F((j\omega)] + \text{Im}[F((j\omega)] \tag{12.13}$$

or, in polar form, $F(j\omega) = |F(j\omega)|e^{\phi(\omega)}$ (12.14)

where, $|F(j\omega)|$ is the magnitude of the network function $F(j\omega)$ and $\phi(\omega)$ is the phase angle.

$$\therefore |F(j\omega)| = \sqrt{\left\{\text{Re}\left[F(j\omega)\right]\right\}^2 + \left\{\text{Im}\left[F(j\omega)\right]\right\}^2} \tag{12.15a}$$

$$\phi(\omega) = \tan^{-1}\left[\frac{\mathrm{Im}\left[F(j\omega)\right]}{\mathrm{Re}\left[F(j\omega)\right]}\right] \tag{12.15b}$$

A frequency response plot shows the variation of one or more coordinates of the network function $F(j\omega)$ with $\omega$. The four different coordinates are

1. Re$[F(j\omega)]$ versus $\omega$
2. Im$[F(j\omega)]$ versus $\omega$
3. $|F(j\omega)|$ versus $\omega$
4. $\phi(\omega)$ versus $\omega$

Note

*(i) The plot of amplitude and phase of the network function is referred as* complex loci plot.

*(ii) The plot of magnitude and phase in logarithmic scale is referred as* Bode plot.

## 12.13.2 Complex Locus Plot

For a network function $F(j\omega)$, having the general form given by Eq. (12.12) and with real and imaginary parts given by Eq. (12.13), (12.14) and (12.15), the complex locus plot is a plot which describes two informations, simultaneously, namely,

(i) the rectangular coordinates Re$[F(j\omega)]$ and Im$[F(j\omega)]$ with $\omega$ as the parameter, and

(ii) the polar coordinates $|F(j\omega)|$ and $\phi(\omega)$ with $\omega$ as the parameter.

From Fig. 12.7, it is observed that, for a given frequency $\omega$, a point $P$ represents the network function $F(j\omega)$ = Re$[F(j\omega)]$ + Im$[F(j\omega)]$ on a complex $F(j\omega)$-plane. As $\omega$ is varied, the tip of the point $P$ representing the network function $F(j\omega)$ moves tracing a locus on the complex frequency plane. This is known as *complex loci*.

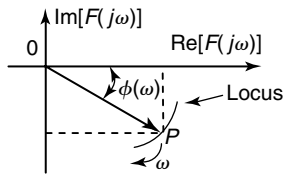



**Fig. 12.7** *Complex locus of* $|F(j\omega)|$ *and* $\phi(\omega)$ *with varying* $\omega$

Note

*The knowledge of complex locus is used to test* 'Nyquist criterion' *and for this reason complex locus is referred as the* 'Nyquist plot'.

## 12.13.3 Frequency Response from Pole–Zero Plot or *s*-plane Phasors

We consider a network function of the general form given by Eq. (12.16) as

$$F(j\omega) = K\frac{(j\omega - z_1)(j\omega - z_2)(j\omega - z_3)\cdots(j\omega - z_n)}{(j\omega - p_1)(j\omega - p_2)(j\omega - p_3)\cdots(j\omega - p_m)} \tag{12.16}$$

where, $z_1, z_2, \ldots, z_n$ are the finite zeros and $p_1, p_2, \ldots, p_m$ are the finite poles.

Each of these factors in the numerator and denominator is a complex quantity, and may be interpreted as a vector in the *s*-plane, originating from the point $z_i$ or $p_i$ and directed to the point $j\omega$ at which the frequency response is to be evaluated. Each of these vectors from the *n*-zeros and *m*-poles to a test point $s = j\omega$ has a magnitude and an angle given as

$$|j\omega - z_i| = \sqrt{\sigma_{zi}^{\,2} + (j\omega - \omega_{zi})^2} = r_i; \quad \angle(j\omega - z_i) = \tan^{-1}\left(\frac{\omega - \omega_{zi}}{-\sigma_{zi}}\right) = \phi_i$$

and, $\left|j\omega - p_i\right| = \sqrt{\sigma_{pi}{}^2 + \left(j\omega - \omega_{pi}\right)^2} = q_i$; $\angle\left(j\omega - p_i\right) = \tan^{-1}\left(\dfrac{\omega - \omega_{pi}}{-\sigma_{pi}}\right) = \theta_i$

where, for zero, $z_i = \left(\sigma_{zi} + j\omega_{zi}\right)$ and for a pole, $p_i = \left(\sigma_{pi} + j\omega_{pi}\right)$.

Therefore, the network function may be written as $F\left(j\omega\right) = K\dfrac{\text{Product of all zeros lines to } j\omega}{\text{Product of all poles lines to } j\omega}$

Hence, the magnitude and phase angle of the complete frequency response may be written as

$$\left|F\left(j\omega\right)\right| = K\frac{\text{Product of all zeros lines to } j\omega}{\text{Product of all poles lines to } j\omega} = K\frac{\prod_{i=1}^{n}\left|\left(j\omega - z_i\right)\right|}{\prod_{i=1}^{m}\left|\left(j\omega - p_i\right)\right|} = K\frac{r_1 r_2 \cdots r_n}{q_1 q_2 \cdots q_m} \tag{12.17}$$

and angle, $\angle F\left(j\omega\right) = \left(\text{Summation of angles of the vectors from zeros to } j\omega\text{-point}\right)$

$$-\left(\text{Summation of angles of the vectors from poles to } j\omega\text{-point}\right)$$

$$= \sum_{i=1}^{n}\angle\left(j\omega - z_i\right) - \sum_{i=1}^{m}\angle\left(j\omega - p_i\right) = \left(\phi_1 + \phi_2 + \phi_3 + \cdots + \phi_n\right) - \left(\theta_1 + \theta_2 + \theta_3 + \cdots + \theta_n\right) \tag{12.18}$$

It is explained in Fig. 12.8.

Thus, the network function at any value of $j\omega$ may be determined geometrically from the pole–zero plot, except for the overall gain factor, $K$. The magnitude of the function is proportional to the product of the geometric distances on the $s$-plane from each zero to the point $j\omega$ divided by the product of the distances from each pole to the point. The angle of the function is the sum of the angles of the vectors associated with the zeros minus the sum of the angles of the vectors associated with the poles.

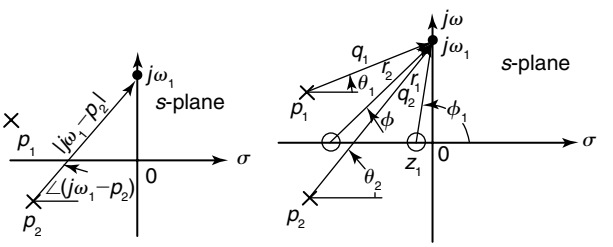



**Fig. 12.8** *Frequency response function from pole–zero plot:*
*(a) Vector from a pole (or zero)*
*(b) Vectors from all poles and zeros*

• ***Physical Interpretation of the Frequency Response from Pole–Zero Plot*** The graphical method can be very useful for deriving a qualitative picture of a system frequency response.

For example, we consider the sinusoidal response of a first-order system with a pole on the real axis at $s = -1/\tau$ as shown in Fig. 12.9 a, and its Bode plots in Fig. 12.9 b. Even though the gain constant $K$ cannot be determined from the pole–zero plot, the following observations may be made directly by noting the behavior of the magnitude and angle of the vector from the pole to the imaginary axis as the input frequency is varied:

1. At low frequencies, the gain approaches a finite value, and the phase angle has a small but finite lag.
2. As the input frequency is increased, the gain decreases (because the length of the vector increases), and the phase lag also increases (the angle of the vector becomes larger).
3. At very high input frequencies, the gain approaches zero, and the phase angle approaches $\pi/2$.

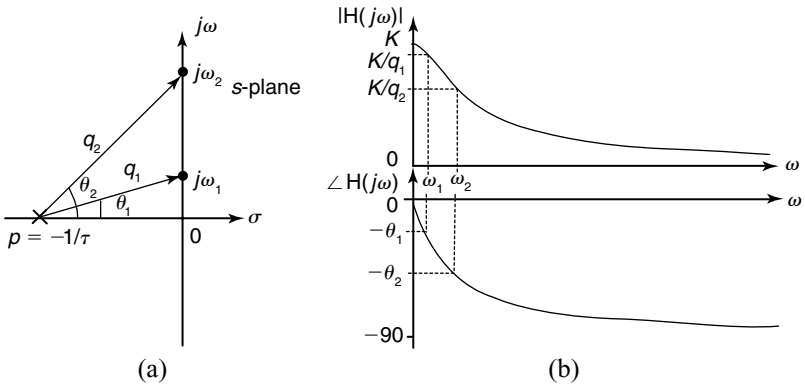



**Fig. 12.9** *The pole-zero plot of a first-order system and its frequency response functions*

As a second example, consider a second-order system, with the damping ratio chosen so that the pair of complex conjugate poles is located close to the imaginary axis as shown in Fig. 12.10 (a). In this case, there is a pair of vectors connecting the two poles to the imaginary axis, and the following conclusions may be drawn by noting how the lengths and angles of the vectors change as the frequency moves up the imaginary axis:

1. At low frequencies, there is a finite (but undetermined) gain and a small but finite phase lag associated with the system.
2. As the input frequency is increased and the test point on the imaginary axis approaches the pole, one of the vectors (associated with the pole in the second quadrant) decreases in length and at some point reaches a minimum. There is an increase in the value of the magnitude function over a range of frequencies close to the pole.
3. At very high frequencies, the lengths of both vectors tend to infinity, and the magnitude of the frequency response tends to zero, while the phase approaches an angle of $\pi$ radians because the angle of each vector approaches $\pi/2$.

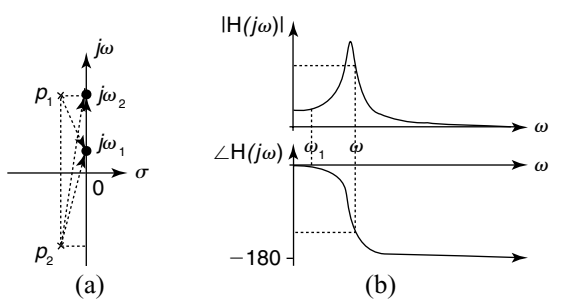



**Fig. 12.10** *Pole–zero plot of a first-order system and its frequency response functions*

The following generalizations may be made about the sinusoidal frequency response of a linear system, based upon the geometric interpretation of the pole–zero plot:

1. If a system has an excess of poles over the number of zeros, the magnitude of the frequency response tends to zero as the frequency becomes large. Similarly, if a system has an excess of zeros, the gain

increases without bound as the frequency of the input increases. This cannot happen in physical energetic systems because it implies an infinite power gain through the system.

2. If a system has a pair of complex conjugate poles close to the imaginary axis, the magnitude of the frequency response has a peak or resonance at frequencies in the proximity of the pole. If the pole pair lies directly upon the imaginary axis, the system exhibits an infinite gain at that frequency.
3. If a system has a pair of complex conjugate zeros close to the imaginary axis, the frequency response has a notch in its magnitude function at frequencies in the vicinity of the zero. If the pair of zeros lie directly upon the imaginary axis, the response is identically zero at the frequency of the zero, and the system does not respond at all to sinusoidal excitation at that frequency.
4. A pole at the origin of the *s*-plane (corresponding to a pure integration term in the network function) implies an infinite gain at zero frequency.
5. Similarly, a zero at the origin of the *s*-plane (corresponding to a pure differentiation) implies a zero gain for the system at zero frequency.

Example 12.6 ***A second-order system has a pair of complex conjugate poles at s = −2 ± j3 and a single zero at the origin of the s-plane. Find the network function and use the pole–zero plot to evaluate the function at s = 0 + j5.***

*Solution* From the given data, the network function is obtained as

$$F(s)=K\frac{s}{\big(s-\big(-2+j3\big)\big)\big(s-\big(-2-j3\big)\big)}=K\frac{s}{s^2+4s+13}$$

The pole-zero plot is shown in Fig. 12.11.

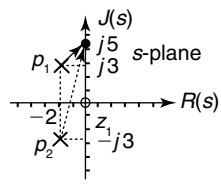



**Fig. 12.11** *Pole–zero plot*

From the figure, the magnitude and phase angle of the network function is obtained as

$$\big|F(s)\big|=K\frac{\sqrt{\big(0-5\big)^2}}{\sqrt{\big(0-\big(-2\big)\big)^2+\big(5-3\big)^2}\sqrt{\big(0-\big(-2\big)\big)^2+\big(5-\big(-3\big)\big)^2}}=K\frac{5}{4\sqrt{34}}$$

and phase angle, $\angle F(s)=\tan^{-1}\left(\frac{5}{0}\right)-\tan^{-1}\left(\frac{2}{2}\right)-\tan^{-1}\left(\frac{8}{2}\right)=-31^\circ$

## 12.13.4 Asymptotic Change of Magnitude with Frequency in Terms of Poles and Zeros—Bode Plot

Bode plot or Bode diagram is the curve that provides variation of magnitude and phase of a network function with frequency in a logarithmic scale.

We consider a network function given as

$$\begin{aligned}F(s)&=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)(s-z_3)\cdots(s-z_n)}{(s-p_1)(s-p_2)(s-p_3)\cdots(s-p_m)}\\&=\big|F(s)\big|\angle F(s)=M\angle\phi=Me^{j\phi}\end{aligned}\tag{12.19}$$

Taking the natural log,

$$\begin{aligned}\mathrm{Log}_e\big[F(s)\big]&=\mathrm{Log}_e\big[\big|F(s)\big|\angle F(s)\big]=\log_e\big[Me^{j\phi}\big]\\\ln\big[F(s)\big]&=\ln M+j\phi.\end{aligned}\tag{12.20}$$

where, In $M$ is the logarithmic gain or simply gain in neper, and $\phi$ is the phase angle in radian.

Alternately, the gain is expressed in decibel (dB), where, Gain in dB $= 20 \log_{10} M$

Note *Relationships (i) Gain in dB = 8.68 × gain in Neper*
*(ii) Angle in degree = 57.3 × angle in radian*

- ***Advantages of Bode Plot***

1. The product factors in $F(j\omega)$ become additive terms because of the use of logarithm.
2. The shape of the Bode plot is such that for most of the commonly used plots in servo systems, it can approximately represent the exact function by straight-line aysmptotes.

- ***Procedure for Bode Plot*** The way to make Bode plot is
  - to split the function into its constituent parts,
  - to plot the magnitude and phase of each part, and
  - to add them.

The different constituent parts of a network function are

1. A constant term,
2. A real pole,
3. A real zero,
4. A pole at the origin,
5. A zero at the origin,
6. A complex conjugate pair of poles, and
7. A complex conjugate pair of zeros.

For each part, procedure for plotting a Bode plot is explained below.

*A constant term* Consider a constant term, $H(s) = H(j\omega) = K$

*Magnitude* Clearly the magnitude is constant $|H(j\omega)| = |K|$

*Phase* The phase is also constant.
Phase is 0°, if $K$ is positive, or any even multiple of 180°.
Phase is ±180°, if $K$ is negative or any odd multiple of 180°.

Example 12.7 ***Plot the Bode diagram for the function H(s) = 15.***

*Solution* Here, $H(s) = 15$

$$|H(j\omega)| = 15 = 23.5 \text{ dB} \quad \angle \text{H}(j\omega) = 0°$$

The Bode plot is shown in Fig. 12.12.

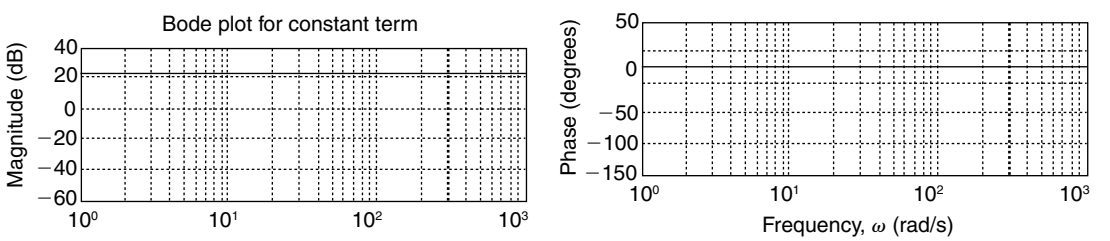



**Fig. 12.12** *Bode plot of a constant term*

*Key Concept: Bode Plot of Gain Term*

- For a constant term, the magnitude plot is a straight line.
- The phase plot is also a straight line, either at 0° (for a positive constant) or ±180° (for a negative constant).

*A Real Pole* Consider a simple real pole

$$H(s)=\frac{1}{1+\dfrac{s}{\omega_0}},\quad H(j\omega)=\frac{1}{1+j\dfrac{\omega}{\omega_0}}$$

The frequency $\omega_0$ is called the break frequency, the corner frequency or the 3 dB frequency.

*Magnitude* The magnitude is given by

$$\left|H(j\omega)\right|=\left|\frac{1}{1+j\dfrac{\omega}{\omega_0}}\right|=-20\log_{10}\left(\sqrt{1^2+\left(\frac{\omega}{\omega_0}\right)^2}\right),\quad \text{in dB}$$

We consider three cases for the value of the frequency:

**Case 1** $\omega << \omega_0$ This is the low-frequency case. We can write an approximation for the magnitude of the transfer function

$$\left|H(j\omega)\right|=-20\log_{10}\left(\sqrt{1^2+\left(\frac{\omega}{\omega_0}\right)^2}\right)=-20\log_{10}(1)=0,\quad \left[\because \omega<<\omega_0\right]$$

**Case 2** $\omega >> \omega_0$ This is the high-frequency case. We can write an approximation for the magnitude of the transfer function

$$\left|H(j\omega)\right|=-20\log_{10}\left(\sqrt{1^2+\left(\frac{\omega}{\omega_0}\right)^2}\right)=-20\log_{10}\left(\sqrt{\left(\frac{\omega}{\omega_0}\right)^2}\right)=-20\log_{10}\left(\frac{\omega}{\omega_0}\right)\quad \left[\because \omega>>\omega_0\right]$$

The high-frequency approximation is a straight line with a slope of −20 dB/decade going through the break frequency at 0 dB. That is, for every factor of 10 increase in frequency, the magnitude drops by 20 dB.

**Case 3** $\omega = \omega_0$ The break frequency. At this frequency

$$\left|H(j\omega)\right|=-20\log_{10}\left(\sqrt{1^2+\left(\frac{\omega}{\omega_0}\right)^2}\right),\quad \omega=\omega_0$$

$$=-20\log_{10}\left(\sqrt{2}\right)=-3.01\,\text{dB}\approx 3\,\text{dB}$$

This point is shown as a grey circle on the diagram in Fig. 12.13.

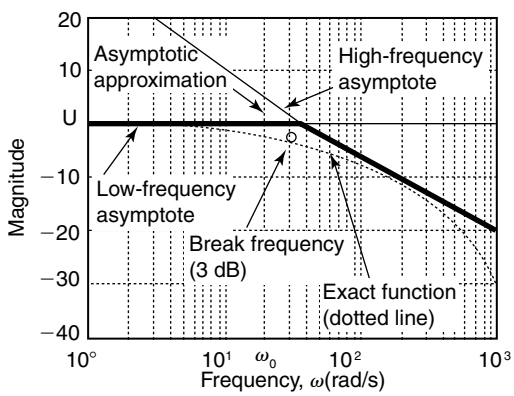



**Fig. 12.13** *Magnitude Bode plot of a real pole*

To draw a piecewise linear approximation, use the low-frequency asymptote up to the break frequency, and the high- frequency asymptote thereafter.

The resulting asymptotic approximation is shown by the thick line. The maximum error between the asymptotic approximation and the exact magnitude function occurs at the break frequency and is approximately 3 dB.

The rule for drawing the piecewise linear approximation for a real pole can be stated thus:

*For a simple real pole, the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then drops at 20 dB per decade (i.e., the slope is −20 dB/decade).*

*Phase* The phase of a single real pole is given by $\angle H(j\omega) = -\angle\left(1 + j\frac{\omega}{\omega_0}\right) = -\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$

We again consider three cases for the value of the frequency:

**Case 1** $\omega << \omega_0$ This is the low-frequency case. At these frequencies, we can write an approximation for the phase of the transfer function $\angle H(j\omega) \approx -\tan^{-1}(0) = 0° = 0$ radian

The low-frequency approximation is shown by the straight line in the middle of the diagram in Fig. 12.14.

**Case 2** $\omega >> \omega_0$ This is the high-frequency case. We can write an approximation for the phase of the transfer function $\angle H(j\omega) \approx -\tan^{-1}(\infty) = -90° = -\frac{\pi}{2}$ radian

The high frequency approximation is as shown by the line at the bottom of the diagram. It is a straight line with a slope at −90°.

**Case 3** $\omega = \omega_0$ At the break frequency, $\angle H(j\omega) \approx -\tan^{-1}(1) = -45° = -\frac{\pi}{4}$ radian

This point is shown as a circle on the diagram in Fig. 12.14.

A piecewise linear approximation is not as easy in this case because the high and low frequency asymptotes do not intersect. Instead, we use a rule that follows the exact function fairly closely, but is also arbitrary. Its main advantage is that it is easy to remember. The rule can be stated as follows:

*Follow the low-frequency asymptote until one-tenth the break frequency (0.1 $\omega_0$). Then decrease linearly to meet the high-frequency asymptote at ten times the break frequency (10$\omega_0$).*

This line is shown in Fig. 12.14. Note that there is no error at the break frequency and about 5.7° of error at one-tenth and ten times the break frequency.

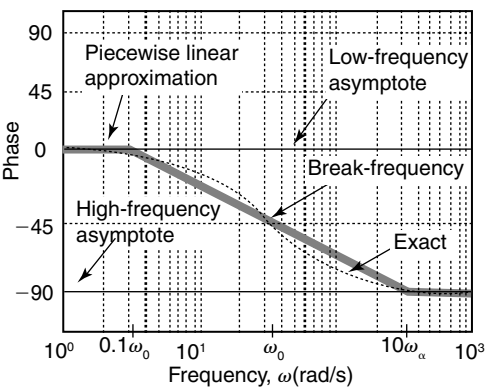



**Fig. 12.14** *Phase plot of a real pole*

Example 12.8 ***Real pole***

***A network function is given as*** $H(s) = \frac{1}{1 + s/10}$ ***. Sketch the Bode plot for this function.***

*Solution* The low-frequency asymptote is the dashed dark grey line, the exact function is the solid black line, and the dotted line represents 0.

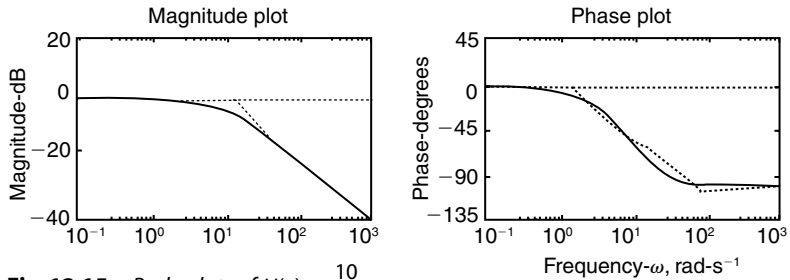



**Fig. 12.15** *Bode plots of* $H(s)=\dfrac{10}{s+10}$

## Example 12.9 *Repeated real pole*

***Find the Bode plot for the function,*** $H(s)=\dfrac{1}{\left(1+\frac{s}{30}\right)^2}$

*Solution* This function has a double pole at 30 radians per second. The slope of the asymptote is −40 dB/decade and the phase goes from 0 to −180°.

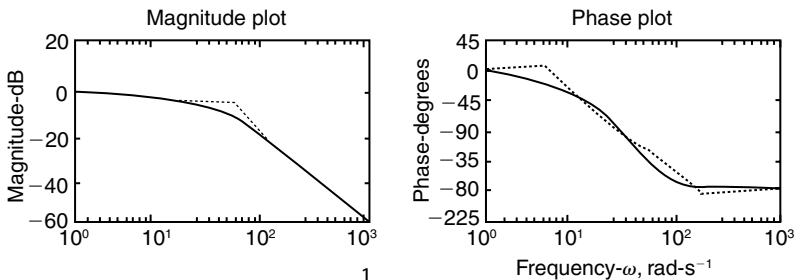



**Fig. 12.16** *Bode plots of* $H(s)=\dfrac{1}{\left(1+\frac{s}{30}\right)^2}$

---

*Key Concept: Bode Plot for Real Pole*

(i) For a simple real pole, the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then drops at 20 dB per decade (i.e., the slope is −20 dB/decade). An $n^{th}$ order pole has a slope of $-20n$ dB/decade.

(ii) The phase plot is at 0° until one-tenth the break frequency and then drops linearly to −90° at ten times the break frequency. An $n^{th}$ order pole drops to $-90°\cdot n$.

*A real zero* The piecewise linear approximation for a zero is much like that for a pole. Consider a simple zero:

$$H(s)=1+\frac{s}{\omega_0}, \qquad H(j\omega)=1+j\frac{\omega}{\omega_0}$$

*Magnitude* The development of the magnitude plot for a zero follows that for a pole. Refer the previous section for details. The magnitude of the zero is given by $\left|H(j\omega)\right|=\left|1+j\dfrac{\omega}{\omega_0}\right|$

Again there are three cases:

1. At low frequencies, $\omega << \omega_0$, the gain is approximately zero.
2. At high frequencies, $\omega >> \omega_0$, the gain increases at 20 dB/decade and goes through the break frequency at 0 dB.
3. At the break frequency, $\omega = \omega_0$, the gain is about 3 dB.

The rule for drawing the piecewise linear approximation for a real zero can be stated thus:

*For a simple real zero, the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then increases at 20 dB per decade (i.e., the slope is +20 dB/decade).*

*Phase* The phase of a simple zero is given by $\angle H(j\omega) = \angle\left(1 + j\frac{\omega}{\omega_0}\right) = \tan^{-1}\left(\frac{\omega}{\omega_0}\right)$

The phase of a single real zero also has three cases:

1. At low frequencies, $\omega << \omega_0$, the phase is approximately zero.
2. At high frequencies, $\omega >> \omega_0$, the phase is 90°.
3. At the break frequency, $\omega = \omega_0$, the phase is 45°.

The rule for drawing the phase plot can be stated thus:

*Follow the low-frequency asymptote until one-tenth the break frequency ($0.1\omega_0$). Then increase linearly to meet the high-frequency asymptote at ten times the break frequency ($10\ \omega_0$).*

---

Example 12.10 ***Determine Bode plot for the network function,*** $H(s) = 1 + \frac{s}{30}$.

*Solution* The low-frequency asymptote is the dashed dark grey line, the exact function is the solid black line, and the lightest line represents 0.

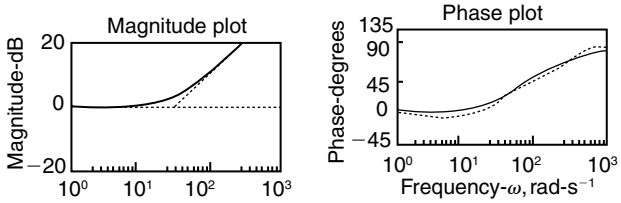



**Fig. 12.17** *Bode plot of* $H(s) = \left(1 + \frac{s}{30}\right)$

---

*Key Concept: Bode Plot of Real Zero*

- For a simple real *zero*, the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then *rises* at +20 dB per decade (i.e., the slope is +20 dB/decade). An $n^{th}$ order *zero* has a slope of $+20n$ dB/decade.
- The phase plot is at 0° until one-tenth the break frequency and then *rises* linearly to +90° at ten times the break frequency. An $n^{th}$ order *zero rises* to $+90°n$.

*A pole at the origin* A pole at the origin is easily drawn exactly. We consider

$$H(s) = \frac{1}{s}, \quad H(j\omega) = \frac{1}{j\frac{\omega}{\omega_0}} = -j\frac{\omega_0}{\omega}$$

*Magnitude* The magnitude is given by $\left|H(j\omega)\right| = \left|-j\frac{\omega_0}{\omega}\right| = \frac{\omega_0}{\omega}$

This function is represented by a straight line on a Bode plot with a slope of −20 dB/decade and going through 0 dB at 1rad/s. It also goes through 20 dB at 0.1rad/s, −20 dB at 10 rad/s.

The rule for drawing the magnitude for a pole at the origin can be thus:

*For a pole at the origin, draw a line with a slope of −20 dB/decade that goes through 0 dB at 1 rad/s.*

*Phase* The phase of a simple zero is given by $\angle H(j\omega) = \angle -j\frac{\omega_0}{\omega} = -90°$

The rule for drawing the phase plot for a pole at the origin can be stated thus:

*The phase for a pole at the origin is −90°.*

*Key Concept: Bode Plot for Pole at Origin*

- For a simple pole at the origin, draw a straight line with a slope of −20 dB per decade and going through 0 dB at 1 rad/s. An $n^{th}$ order pole has a slope of $-20n$ dB/decade.
- The phase plot is at −90°. An $n^{th}$ order pole is at $-90°n$.

*A zero at the origin* A zero at the origin is just like a pole at the origin but the magnitude increases, and the phase is positive.

*Key Concept: Bode Plot for Zero at Origin*

- For a simple *zero* at the origin, draw a straight line with a slope of +20 dB per decade and going through 0 dB at 1 rad/s. An $n^{th}$ order *zero* has a slope of $+20n$ dB/decade.
- The phase plot is at +90°. An $n^{th}$ order zero is at $+90°n$.

*A Complex Conjugate Pair of Poles* The magnitude and phase plots of a complex conjugate (under-damped) pair of poles are more complicated than those for a simple pole. We consider the transfer function:

$$H(s) = \frac{{\omega_0}^2}{s^2 + 2\xi\omega_0 s + {\omega_0}^2}, \quad \text{with} \quad 0 < \xi < 1$$

*Magnitude* The magnitude is given by

$$\left|H(j\omega)\right| = \frac{1}{\sqrt{\left(1-\left(\frac{\omega}{\omega_0}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_0}\right)^2}} = -20\log_{10}\left(\sqrt{\left(1-\left(\frac{\omega}{\omega_0}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_0}\right)^2}\right), \quad \text{in dB}$$

We consider three cases for the value of the frequency:

**Case 1** $\omega << \omega_0$ This is the low-frequency case. We can write an approximation for the magnitude of the transfer function

$$\left|H(j\omega)\right| = -20\log_{10}(1) = 0$$

The low-frequency approximation is shown in Fig. 12.18 (a).

**Case 2** $\omega >> \omega_0$ This is the high-frequency case. We can write an approximation for the magnitude of the transfer function

$$\left|H(j\omega)\right| = -20\log_{10}\left(\left(\frac{\omega}{\omega_0}\right)^2\right) = -40\log_{10}\left(\frac{\omega}{\omega_0}\right)$$

The high-frequency approximation is shown as straight line with a slope of −40 dB/decade going through the break frequency at 0 dB. That is, for every factor of 10 increase in frequency, the magnitude drops by 40 dB.

**Case 3** $\omega \approx \omega_0$ It can be shown that a peak occurs in the magnitude plot near the break frequency. The exact height and location can be determined by differentiating the expression for the magnitude of the transfer function with respect to frequency and setting it to zero.

$$\therefore \frac{d}{d\omega}\left[\frac{1}{\sqrt{\left(1-\left(\frac{\omega}{\omega_0}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_0}\right)^2}}\right] = 0 \Rightarrow \omega_r = \omega_0\sqrt{1-2\xi^2}$$

The peak has a magnitude of $\left|H(j\omega_r)\right| = -20\log_{10}\left(2\xi\sqrt{1-2\xi^2}\right)$, in dB

The actual peak frequency is not important when drawing Bode diagrams by hand because if the peak is large enough to draw, the peak frequency is very near the break frequency. This point is shown as a circle in Fig. 12.18 (a).

Note that the peak only exists for $0 < \xi < 0.707\left(= 1/\sqrt{2}\right)$, and the frequency of the peak is typically very near the break frequency. For $\xi = 0$, the peak is exactly at the resonant frequency but the peak frequency drops as $\xi$ increases. However, even for a fairly large $\xi = 0.3$ (a small peak of only 5 dB), the resonant frequency is $\omega_r = \omega_0\sqrt{1-2\xi^2} = \omega_0\sqrt{1-0.18} = 0.91\omega_0$ which is only a 9% deviation from the break frequency. It is generally accurate enough to put the peak at the resonant frequency.

*To Draw a Piecewise Linear Approximation*

- The low-frequency asymptote is used up to the break frequency,
- The high-frequency asymptote is used thereafter,
- A smooth curve is drawn between the low- and high-frequency asymptote that goes through the peak value.

The rule for drawing the piecewise linear approximation for a complex conjugate pair of poles can be stated thus:

*For the magnitude plot of complex conjugate poles, draw a line of 0 dB at low frequencies, go through a peak of height,* $\left|H(j\omega_r)\right| = -20\log_{10}\left(2\xi\sqrt{1-2\xi^2}\right)$, in dB *at the break frequency and then drop at 40 dB per decade (i.e., the slope is −40 dB/decade). The high frequency asymptote goes through the break frequency.*

*Phase* The phase of a complex conjugate pole is given by

$$\angle H(j\omega) = -\tan^{-1}\left(\frac{2\xi\dfrac{\omega}{\omega_0}}{1-\left(\dfrac{\omega}{\omega_0}\right)^2}\right)$$

We again consider three cases for the value of the frequency:

**Case 1** $\omega << \omega_0$ This is the low-frequency case. At these frequencies, we can write an approximation for the phase of the transfer function $\angle H(j\omega) = -\tan^{-1}(0) = 0° =$ radian

The low-frequency approximation is shown in Fig. 12.18 (b).

**Case 2** $\omega >> \omega_0$ This is the high-frequency case. We can write an approximation for the phase of the transfer function $\angle H(j\omega) = -180°$

The high frequency approximation is as shown in Fig. 12.18 (b). It is a straight line at −180°.

**Case 3** $\omega = \omega_0$ The break frequency. At this frequency, $\angle H(j\omega) = -90°$

This point is shown as a circle in Fig. 12.18 (b).

A piecewise linear approximation is not easy in this case, and there are no hard and fast rules for drawing it. The most common way is to look up a graph in a textbook with a chart that shows phase plots for many values of ζ. Another way is to connect the low-frequency asymptote to the high-frequency asymptote starting at

$$\omega = \omega_0 \frac{\log_{10}\left(\dfrac{2}{\xi}\right)}{2} \text{ and ending at } \omega = \omega_0 \frac{2}{\log_{10}\left(\dfrac{2}{\xi}\right)}$$

If $\zeta < 0.02$, the approximation can be simply a vertical line at the break frequency.

The rule for drawing phase of an underdamped pair of poles can be stated as follows:

*Follow the low-requency asymptote at 0° until* $\omega = \omega_0 \dfrac{\log_{10}\left(\dfrac{2}{\xi}\right)}{2}$. *Then decrease linearly to meet the high frequency asymptote at −180°* at $\omega = \omega_0 \dfrac{2}{\log_{10}\left(\dfrac{2}{\xi}\right)}$.

---

Example 12.11 ***Draw the Bode plot for the network function,*** $H(s) = \dfrac{100}{s^2+2s+100}$.

*Solution* Here, $H(s) = \dfrac{100}{s^2+2s+100}$

i.e., $H(s)=\dfrac{1}{\left(\dfrac{s}{\omega_0}\right)^2+2\xi\left(\dfrac{s}{\omega_0}\right)+1}$, $\omega_0=10$, $\xi=1$

The curve is shown in Fig. 12.18.

The peak will have an amplitude of 5.02 or 14 dB.

The resulting asymptotic approximation is shown as a black dotted line, and the exact response is a black solid line.

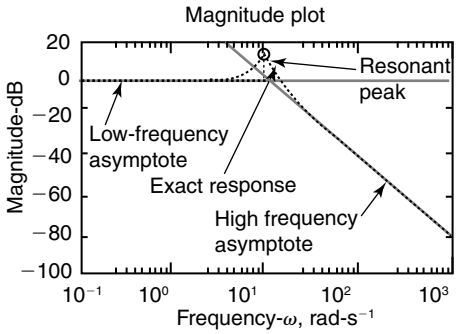



**Fig. 12.18 (a)** *Bode magnitude plot of* $H(s)=\dfrac{100}{s^2+2s+100}$

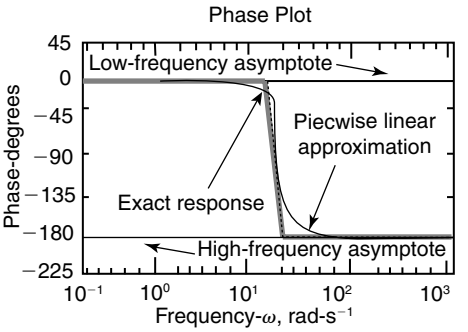



**Fig. 12.18 (b)** *Bode phase plot of* $H(s)=\dfrac{100}{s^2+2s+100}$

*Key Concept: Bode Plot for Complex Conjugate Poles*

- For the magnitude plot of complex conjugate poles, draw a 0 dB line at low frequencies, go through a peak of height, $\left|H\left(j\omega_r\right)\right|=-20\log_{10}\left(2\xi\sqrt{1-2\xi^2}\right)$, in dB at the break frequency and then drop at 40 dB per decade (i.e., the slope is $-40$ dB/decade). The high-frequency asymptote goes through the break frequency. Note that the peak only exists for $0<\xi<0.707\left(=1/\sqrt{2}\right)$.
- To draw the phase plot, simply follow the low-frequency asymptote at 0° until $\omega=\omega_0\dfrac{\log_{10}\left(\dfrac{2}{\xi}\right)}{2}$. Then decrease linearly to meet the high frequency asymptote at −180° at $\omega=\omega_0\dfrac{2}{\log_{10}\left(\dfrac{2}{\xi}\right)}$.

If $\zeta < 0.02$, the approximation can be simply a vertical line at the break frequency.

*A complex conjugate pair of zeros* A complex pair of zeros yields results similar to that for a complex pair of poles. The differences are that the magnitude has a dip instead of a peak, the magnitude increases above the break frequency and the phase increases rather than decreasing.

## Example 12.12 *Complex conjugate zero*

***Show the Bode plots for the function, H(s) = 0.01s² + 0.02s + 1.***

*Solution* The graph below corresponds to a complex conjugate zero with

$$H(s)=\left(\frac{s}{\omega_0}\right)^2+2\xi\left(\frac{s}{\omega_0}\right)+1, \quad \omega_0=10, \quad \xi=0.1$$

The dip in the magnitude plot will have a magnitude of 0.2 or −14 dB.

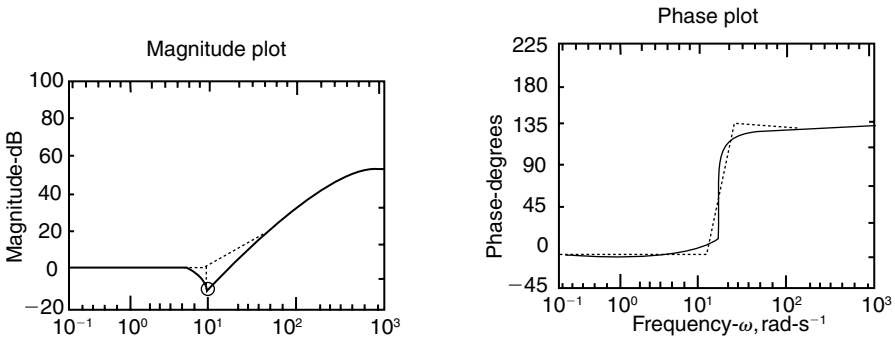



**Fig. 12.19** *Bode plots of* H(s) = *0.01s² + 0.02s + 1*

*Key Concept: Bode Plot of Complex Conjugate Zeros*

- For the magnitude plot of complex conjugate zeros, draw a 0-dB line at low frequencies, go through a dip of magnitude

$$\left|H\left(j\omega_r\right)\right|=20\log_{10}\left(2\xi\sqrt{1-2\xi^2}\right), \quad \text{in dB}$$

at the break frequency and then rise at +40 dB per decade (i.e., the slope is +40 dB/decade). The high-frequency asymptote goes through the break frequency. Note that the peak only exists for $0<\xi<0.707\left(=1/\sqrt{2}\right)$

- To draw the phase plot, simply follow the low-frequency asymptote at 0° until $\omega=\omega_0\dfrac{\log_{10}\left(\frac{2}{\xi}\right)}{2}$ then increase linearly to meet the high frequency asymptote at 180° at $\omega=\omega_0\dfrac{2}{\log_{10}\left(\frac{2}{\xi}\right)}$

### *Rules for Making Bode Plots*

The table below summarizes what to do for each type of term in a Bode plot.

**Table 12.1** *Rules for Making Bode Plots*

| Term | Magnitude | Phase |
|---|---|---|
| **Constant:** K | $20 \cdot \log_{10}(\lvert K \rvert)$ | $K > 0$: 0°<br>$K < 0$: ±180° |
| **Real pole:** $\dfrac{1}{\frac{s}{\omega_0}+1}$ | • Low-freq. asymptote at 0 dB<br>• High-freq. asymptote at −20 dB/dec<br>• Connect asymptotic lines at $\omega_0$. | • Low-freq. asymptote at 0°.<br>• High-freq. asymptote at −90°.<br>• Connect with straight line from $0.1\cdot\omega_0$ to $10\cdot\omega_0$. |
| **Real zero*:** $\dfrac{s}{\omega_0}+1$ | • Low-freq. asymptote at 0 dB<br>• High-freq. asymptote at +20 dB/dec.<br>• Connect asymptotic lines at $\omega_0$. | • Low-freq. asymptote at 0°.<br>• High-freq. asymptote at +90°.<br>• Connect with line from $0.1\cdot\omega_0$ to $10\cdot\omega_0$. |
| **Pole at origin:** $\dfrac{1}{s}$ | • −20 dB/dec; through 0 dB at $\omega = 1$. | • −90° for all $\omega$. |
| **Zero at origin*:** $s$ | • +20 dB/dec; through 0 dB at $\omega = 1$. | • +90° for all $\omega$. |
| **Underdamped poles:** $\dfrac{1}{\left(\frac{s}{\omega_0}\right)^2+2\zeta\left(\frac{s}{\omega_0}\right)+1}$ | • Low-freq. asymptote at 0 dB.<br>• High-freq. asymptote at −40 dB/dec.<br>• Connect asymptotic lines at $\omega_0$.<br>• Draw peak† at freq. $\omega_r = \omega_0\sqrt{1-2\zeta^2}$ with amplitude $H(j\omega_r) = -20\cdot\log_{10}\left(2\zeta\sqrt{1-\zeta^2}\right)$ | • Low-freq. asymptote at 0°.<br>• High-freq. asymptote at −180°.<br>• Connect with straight line from ‡ $\omega = \omega_0\dfrac{\log_{10}\left(\frac{2}{\zeta}\right)}{2}$ to $\omega = \omega_0\dfrac{2}{\log_{10}\left(\frac{2}{\zeta}\right)}$ |
| **Underdamped zeros*:** $\left(\frac{s}{\omega_0}\right)^2+2\zeta\left(\frac{s}{\omega_0}\right)+1$ | • Draw low-freq. asymptote at 0 dB.<br>• Draw high-freq. asymptote at +40 dB/dec.<br>• Connect asymptotic lines at $\omega_0$.<br>• Draw dip† at freq. $\omega_r = \dfrac{\omega_0}{\sqrt{1-2\zeta^2}}$ with amplitude $H(j\omega_r) = +20\cdot\log_{10}\left(2\zeta\sqrt{1-\zeta^2}\right)$. | • Low-freq. asymptote at 0°.<br>• Draw high-freq. asymptote at +180°.<br>• Connect with a straight line from ‡ $\omega = \omega_0\dfrac{\log_{10}\left(\frac{2}{\zeta}\right)}{2}$ to $\omega = \omega_0\dfrac{2}{\log_{10}\left(\frac{2}{\zeta}\right)}$ |

**Notes:**

* Rules for drawing zeros create the mirror image (around 0 dB, or 0°) of those for a pole with the same $\omega_0$.

† For underdamped poles and zeros, peak exists only for $0 < \zeta < 0.707 = \dfrac{1}{\sqrt{2}}$ and peak frequency is typically very near $\omega_0$.

‡ For underdamped poles and zeros, if $\xi < 0.02$ draw phase vertically from 0 to −180 degrees at $\omega_0$.

For $n^{th}$ order pole or zero make asymptotes, peaks and slopes $n$ times higher than shown (i.e., second-order asymptote is −40 dB/decade, and phase goes from 0 to −180°). Don't change frequencies, only plot values and slopes.

## Solved Problems

Problem 12.1 ***Calculate the driving-point impedance of the networks shown in Fig. 12.20, and obtain the pole–zero plot.***

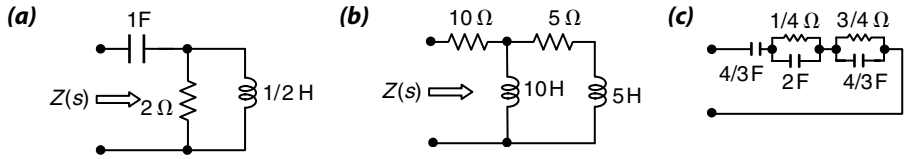



**Fig. 12.20**

*Solution*

(a) Here, driving point impedance, $Z(s)=\dfrac{1}{s}+\dfrac{\frac{s}{2}\times 2}{\frac{s}{2}+2}=\dfrac{1}{s}+\dfrac{2s}{s+4}=\dfrac{2s^2+s+4}{s(s+4)}$

(b) Driving point impedance, $Z(s)=10+\dfrac{10s(5+5s)}{10s+5+5s}=\dfrac{50+150s+50s+50s^2}{5+15s}=10\left(\dfrac{s^2+4s+1}{3s+1}\right)$

(c) Driving point impedance, $Z(s)=\dfrac{1}{\frac{4}{3}s}+\dfrac{\frac{1}{4}\times\frac{1}{3s}}{\frac{1}{4}+\frac{1}{3s}}+\dfrac{\frac{3}{4}\times\frac{3}{4s}}{\frac{3}{4}+\frac{3}{4s}}=\dfrac{3}{4s}+\dfrac{1}{2s+4}+\dfrac{9}{12s+12}=\dfrac{8s^2+17s+6}{4s(s+1)(s+2)}$

Note *The pole-zero plot is left as a task for the reader.*

Problem 12.2 ***Determine the driving point admittance function for the networks shown in Fig. 12.21 and plot the pole–zero pattern in the s-plane.***

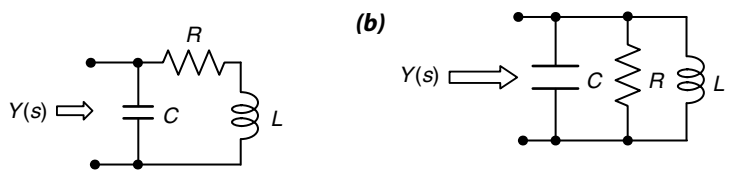



**Fig. 12.21**

*Solution*

(a) Here, driving point impedance, $Z(s)=\dfrac{\frac{1}{Cs}\times(R+sL)}{\frac{1}{Cs}+(R+sL)}=\dfrac{R+sL}{LCs^2+RCs+1}$

Therefore, driving point admittance, $Y(s)=\dfrac{1}{Z(s)}=\dfrac{LCs^2+RCs+1}{R+sL}$

The admittance function has two zeros at $s=-\dfrac{R}{2L}\pm\sqrt{\left(\dfrac{R}{2L}\right)^2-\dfrac{1}{LC}}$ and a pole at $s=-\dfrac{R}{L}$.

(b)Here, driving point impedance, $Z(s)=\dfrac{1}{\dfrac{1}{R}+\dfrac{1}{sL}+Cs}$

Therefore, driving point admittance, $Y(s)=\dfrac{1}{Z(s)}=\left(\dfrac{1}{R}+\dfrac{1}{sL}+Cs\right)=\dfrac{RLCs^2+sL+R}{sRL}$

The admittance function has two zeros at $s=-\dfrac{1}{2RC}\pm\sqrt{\left(\dfrac{1}{2RC}\right)^2-\dfrac{1}{LC}}$ and a pole at $s = 0$.

Problem 12.3 ***Determine the voltage transfer ratio*** $\dfrac{V_2(s)}{V_1(s)}$ ***of the networks shown.***

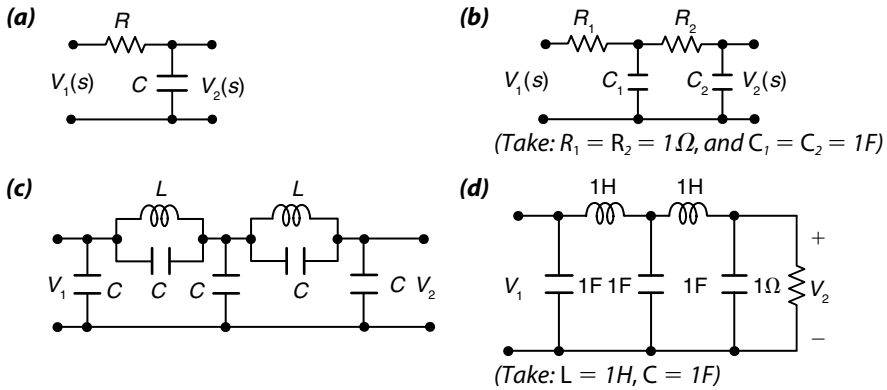



**Fig. 12.22**

*Solution*

(a) Here, the current in the circuit, $I(s)=\dfrac{V_1(s)}{R+\dfrac{1}{sC}}=\left(\dfrac{sC}{1+RCs}\right)V_1(s)$

$$\therefore V_2(s)=I(s)\times\frac{1}{sC}=\left(\frac{sC}{1+RCs}\right)V_1(s)\times\frac{1}{sC}$$

$$\therefore G_{21}(s)=\frac{V_2(s)}{V_1(s)}=\frac{1}{1+RCs}$$

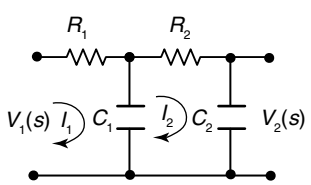



**Fig. 12.23**

(b) Consider Fig. 12.23. By KVL for two meshes, $\left(R_1+\dfrac{1}{sC_1}\right)I_1(s)-\dfrac{1}{sC_1}I_2(s)=V_1(s)$

and $$-\frac{1}{sC_1}I_1(s)+\left(R_2+\frac{1}{sC_1}+\frac{1}{sC_2}\right)I_2(s)=0$$

Solving for $I_2(s)$, $$I_2(s)=\frac{\begin{vmatrix}\left(R_1+\frac{1}{sC_1}\right) & V_1(s)\\ -\frac{1}{sC_1} & 0\end{vmatrix}}{\begin{vmatrix}\left(R_1+\frac{1}{sC_1}\right) & -\frac{1}{sC_1}\\ -\frac{1}{sC_1} & \left(R_2+\frac{1}{sC_1}+\frac{1}{sC_2}\right)\end{vmatrix}}=\frac{V_1(s)\left(\frac{1}{sC_1}\right)}{R_1R_2+\frac{1}{sC_1}(R_1+R_2)+\frac{R_1}{sC_2}+\frac{1}{s^2C_1C_2}}$$

$$\therefore V_2(s)=I_2(s)\times\frac{1}{sC_2}=\frac{V_1(s)\left(\frac{1}{s^2C_1C_2}\right)}{R_1R_2+\frac{1}{sC_1}(R_1+R_2)+\frac{R_1}{sC_2}+\frac{1}{s^2C_1C_2}}$$

Putting the values of $R_1 = R_2 = 1\ \Omega$ and $C_1 = C_2 = 1$ F, we get, $\therefore \frac{V_2(s)}{V_1(s)}=\frac{\frac{1}{s^2}}{1+\frac{3}{s}+\frac{1}{s^2}}=\frac{1}{s^2+3s+1}$

(c) The extreme left capacitance has no effect on the voltage ratio required. We can omit the capacitance. The transform impedance network is shown below.

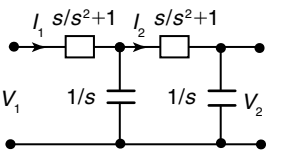


**Fig. 12.24**

Here, the driving point impedance at the Port 1 is

$$Z(s)=\frac{s}{s^2+1}+\frac{\frac{1}{s}\left(\frac{s}{s^2+1}+\frac{1}{s}\right)}{\frac{1}{s}+\frac{s}{s^2+1}+\frac{1}{s}}=\frac{s}{s^2+1}+\frac{\frac{2s^2+1}{s^2(s^2+1)}}{\frac{2}{s}+\frac{s}{s^2+1}}=\frac{s}{s^2+1}+\frac{2s^2+1}{s(3s^2+2)}$$

$$=\frac{3s^4+2s^2+2s^4+3s^2+1}{s(s^2+1)(3s^2+2)}=\frac{5s^4+5s^2+1}{s(s^2+1)(3s^2+2)}$$

$$\therefore \text{input current, } I_1(s)=\frac{V_1(s)}{Z(s)}=\frac{V_1(s)}{\frac{5s^4+5s^2+1}{s(s^2+1)(3s^2+2)}}=V_1(s)\left[\frac{s(s^2+1)(3s^2+2)}{(5s^4+5s^2+1)}\right]$$

$$\therefore I_2(s)=I_1(s)\times\frac{\frac{1}{s}}{\frac{1}{s}+\frac{s}{s^2+1}+\frac{1}{s}}=I_1(s)\times\frac{\frac{1}{s}}{\frac{2}{s}+\frac{s}{s^2+1}}=I_1(s)\times\frac{s^2+1}{3s^2+2}$$

$$=V_1(s)\left[\frac{s(s^2+1)(3s^2+2)}{(5s^4+5s^2+1)}\right]\times\frac{s^2+1}{3s^2+2}=V_1(s)\left[\frac{s(s^2+1)^2}{(5s^4+5s^2+1)}\right]$$

Now, $V_2(s) = I_2(s) \times \frac{1}{s} = V_1(s)\left[\frac{s(s^2+1)^2}{(5s^4+5s^2+1)}\right] \times \frac{1}{s} = V_1(s)\left[\frac{(s^2+1)^2}{(5s^4+5s^2+1)}\right]$

$$\therefore \frac{V_2(s)}{V_1(s)} = \frac{(s^2+1)^2}{(5s^4+5s^2+1)}$$

(d) The extreme left capacitor does not have any effect on the required voltage ratio. We may ignore it. Also, the 1-Ω resistor and 1-F capacitor in the output can be paralleled. So, the network becomes as shown in Fig. 12.25.

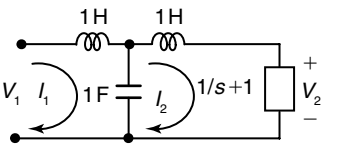



**Fig. 12.25**

By KVL,

$$\left(s+\frac{1}{s}\right)I_1(s) - \frac{1}{s}I_2(s)V_1(s) \;\Rightarrow\; (s^2+1)I_1(s) - I_2(s) = sV_1(s) \tag{i}$$

$$-\frac{1}{s}I_1(s) + \left(s+\frac{1}{s}+\frac{1}{s+1}\right)I_2(s) = 0 \;\Rightarrow\; -(s+1)I_1(s) + (s^3+s^2+2s+1)I_2(s) = 0. \tag{ii}$$

Solving for $I_2$, $I_2(s) = \dfrac{\begin{vmatrix}(s^2+1) & sV_1(s)\\ -(s+1) & 0\end{vmatrix}}{\begin{vmatrix}(s^2+1) & -1\\ -(s+1) & (s^3+s^2+2s+1)\end{vmatrix}} = \dfrac{s(s+1)V_1(s)}{(s^2+1)(s^3+s^2+2s+1)-(s+1)}$

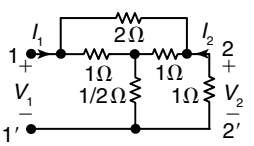
$$= \frac{s(s+1)V_1(s)}{s(s^4+s^3+3s^2+2s+1)} = \frac{(s+1)V_1(s)}{(s^4+s^3+3s^2+2s+1)}$$

$$\therefore V_2(s) = I_2(s) \times \frac{1}{s+1} = \frac{(s+1)V_1(s)}{(s^4+s^3+3s^2+2s+1)} \times \frac{1}{s+1} = \frac{V_1(s)}{(s^4+s^3+3s^2+2s+1)}$$

$$\therefore \frac{V_2(s)}{V_1(s)} = \frac{1}{(s^4+s^3+3s^2+2s+1)}$$

Problem 12.4 ***Find the current transfer ratio*** $\left(\frac{I_2}{I_1}\right)$ ***and voltage transfer ratio*** $\left(\frac{V_2}{V_1}\right)$ ***of the networks shown in Fig. 12.26.***

***(a)***




***(b)***

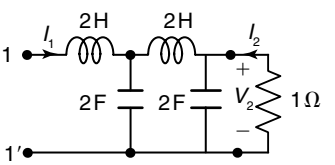



**Fig. 12.26**

*Solution*

(a) By KVL for the three meshes, we get, $\frac{3}{2}I_1(s)+\frac{1}{2}I_2(s)-I_3(s)=V_1(s)$

$$\frac{1}{2}I_1(s)+\frac{5}{2}I_2(s)+I_3(s)=0$$

$$-I_1(s)+I_2(s)+4I_3(s)=0$$

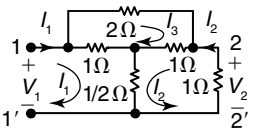



**Fig. 12.27**

Solving for $I_1$ and $I_2$,

$$I_1(s)=\frac{\begin{vmatrix} V_1(s) & \frac{1}{2} & -1 \\ 0 & \frac{5}{2} & 1 \\ 0 & 1 & 4 \end{vmatrix}}{\begin{vmatrix} \frac{3}{2} & \frac{1}{2} & -1 \\ \frac{1}{2} & \frac{5}{2} & 1 \\ -1 & 1 & 4 \end{vmatrix}}=\frac{9V_1(s)}{9}=V_1(s) \quad \text{and} \quad I_2(s)=\frac{\begin{vmatrix} \frac{3}{2} & V_1(s) & -1 \\ \frac{1}{2} & 0 & 1 \\ -1 & 0 & 4 \end{vmatrix}}{\begin{vmatrix} \frac{3}{2} & \frac{1}{2} & -1 \\ \frac{1}{2} & \frac{5}{2} & 1 \\ -1 & 1 & 4 \end{vmatrix}}=-\frac{3V_1(s)}{9}=-\frac{V_1(s)}{3}$$

$$\therefore \left(I_2/I_1\right)=-\frac{1}{3}$$

and $$\therefore \left(V_2/I_1\right)=-\frac{1\times I_2}{I_1}=\frac{1}{3}$$

(b) Here, by KVL for the three meshes, we get

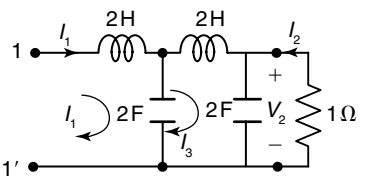



**Fig. 12.28**

$$\left(2s+\frac{1}{2s}\right)I_1(s)+0-\frac{1}{2s}I_3(s)=V_1(s) \;\Rightarrow\; \left(4s^2+1\right)I_1(s)+0-I_3(s)=V_1(s) \qquad \text{(i)}$$

$$-\frac{1}{2s}I_1(s)+\frac{1}{2s}I_2(s)+\left(2s+\frac{1}{2s}+\frac{1}{2s}\right)I_3(s)=0 \;\Rightarrow\; -I_1(s)+I_2(s)+\left(4s^2+2\right)I_3(s)=0 \qquad \text{(ii)}$$

$$0+\left(1+\frac{1}{2s}\right)I_2(s)+\frac{1}{2s}I_3(s)=0 \;\Rightarrow\; 0+(2s+1)I_2(s)+I_3(s)=0 \qquad \text{(iii)}$$

Solving for $I_1$ and $I_2$, we get,

$$I_1(s)=\frac{\begin{vmatrix} V_1(s) & 0 & -1 \\ 0 & 1 & (4s^2+2) \\ 0 & (2s+1) & 1 \end{vmatrix}}{\begin{vmatrix} (4s^2+1) & 0 & -1 \\ -1 & 1 & (4s^2+2) \\ 0 & (2s+1) & 1 \end{vmatrix}}=\frac{V_1(s)\left[1-(2s+1)(4s^2+2)\right]}{(4s^2+1)\left[1-(2s+1)(4s^2+2)\right]+1\times(2s+1)}$$

$$=-\frac{V_1(s)\left[8s^3+4s^2+4s+1\right]}{16s^4+8s^3+12s^2+4s+1}$$

$$I_2(s)=\frac{\begin{vmatrix} (4s^2+1) & V_1(s) & -1 \\ -1 & 0 & (4s^2+2) \\ 0 & 0 & 1 \end{vmatrix}}{\begin{vmatrix} (4s^2+1) & 0 & -1 \\ -1 & 1 & (4s^2+2) \\ 0 & (2s+1) & 1 \end{vmatrix}}=\frac{V_1(s)}{(4s^2+1)\left[1-(2s+1)(4s^2+2)\right]+1\times(2s+1)}$$

$$=\frac{V_1(s)}{16s^4+8s^3+12s^2+4s+1}$$

$$\therefore \left(I_2/I_1\right)=-\frac{1}{8s^3+4s^2+4s+1}$$

$$\therefore \left(V_2/V_1\right)=-\frac{1\times I_2}{V_1}=\frac{1}{16s^4+8s^3+12s^2+4s+1}$$

Problem 12.5 ***Find the transfer function*** $\frac{V(s)}{I(s)}$ ***of the network shown in Fig. 12.29.***

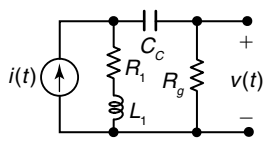


**Fig. 12.29**

*Solution* Here, current through the resistance $R_g$ is

$$I_{R_g}=I(s)\frac{R_1+sL_1}{R_1+sL_1+\frac{1}{sC_C}+R_g}$$

$$=I(s)\frac{sC_C(R_1+sL_1)}{s^2L_1C_C+sC_C(R_1+R_g)+1}$$

$$\therefore V(s) = I_{R_g} \times R_g = I(s)\frac{sR_gC_C(R_1 + sL_1)}{s^2L_1C_C + sC_C(R_1 + R_g) + 1}$$

$$\therefore \frac{V(s)}{I(s)} = \frac{sR_gC_C(R_1 + sL_1)}{s^2L_1C_C + sC_C(R_1 + R_g) + 1}$$

Problem 12.6 ***Find the driving point impedance at the terminals 1−1′ of the ladder networks shown in Fig. 12.30.***

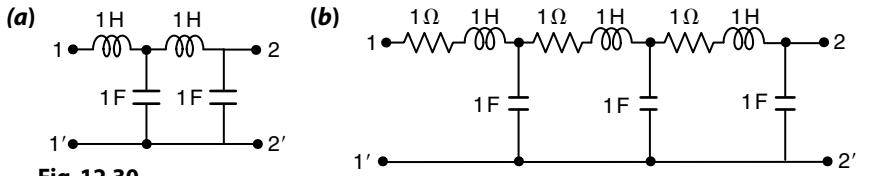



**Fig. 12.30**

*Solution*

(a) The driving point impedance at $1-1'$ is,

$$Z_{11}(s) = s + \cfrac{1}{s + \cfrac{1}{s + \cfrac{1}{s}}} = \frac{s^4 + 3s^2 + 1}{s^2 + 2s}$$

(b) The driving point impedance at $1-1'$ is,

$$Z_{11} = (s+1) + \cfrac{1}{s + \cfrac{1}{(s+1) + \cfrac{1}{s + \cfrac{1}{(s+1) + \cfrac{1}{s}}}}} = \frac{s^6 + 3s^5 + 8s^4 + 11s^3 + 11s^2 + 6s + 1}{s^5 + 2s^4 + 5s^3 + 4s^2 + 3s}$$

Problem 12.7 ***Determine the network functions $Y_{21}$ and $Z_{21}$ for the network shown.***

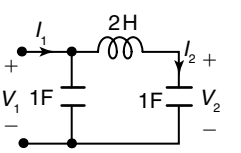



**Fig. 12.31**

*Solution* Writing two mesh equations,

$$(I_1 - I_2)\frac{1}{s} = V_1 \;\Rightarrow\; I_1 - I_2 = sV_1 \qquad \text{(i)}$$

$$-\frac{1}{s}I_1 + \left(2s + \frac{2}{s}\right)I_2 = 0 \;\Rightarrow\; -I_1 + (2s^2 + 2)I_2 = 0 \qquad \text{(ii)}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 \\ -1 & (2s^2+2) \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} sV_1 \\ 0 \end{bmatrix}$$

$$\therefore I_2 = \frac{\begin{vmatrix} 1 & sV_1 \\ -1 & 0 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ -1 & \left(2s^2+2\right) \end{vmatrix}} = \frac{sV_1}{2s^2+1}$$

$$\therefore Y_{21} = \frac{I_2}{V_1} = \frac{s}{2s^2+1}$$

$$\therefore V_2 = \frac{1}{s} I_2 = \frac{V_1}{2s^2+1} \quad \text{and} \quad I_1 = \frac{\begin{vmatrix} sV_1 & -1 \\ 0 & \left(2s^2+2\right) \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ -1 & \left(2s^2+2\right) \end{vmatrix}} = \frac{2\left(s^2+1\right)sV_1}{2s^2+1}$$

$$\therefore Z_{21} = \frac{V_2}{I_1} = \frac{V_1}{2s^2+1} \times \frac{2s^2+1}{2\left(s^2+1\right)sV_1} = \frac{1}{2s\left(s^2+1\right)}$$

Problem 12.8 ***Determine driving point impedance, $Z_{11}$, transfer impedance, $Z_{21}$ and voltage transfer function, $G_{21}$ for the network shown.***

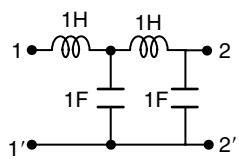


**Fig. 12.32**

*Solution* The driving point impedance at $1-1'$ is,

$$Z_{11} = s + \frac{1}{s + \dfrac{1}{s + \dfrac{1}{s}}} = \frac{s^4+3s^2+1}{s^2+2s}$$

To find the transfer impedance, $Z_{21}$, we start from the right end,

$$\therefore I_2 = V_2 \times s$$
$$\therefore V' = I_2 \times s + V_2 = \left(s^2+1\right)V_2$$
$$\therefore I_1 = I_2 + V's = V_2 s + \left(s^3+s\right)V_2 = \left(s^3+2s\right)V_2$$
$$\therefore Z_{21} = \frac{V_2}{I_1} = \frac{1}{s^3+2s}$$

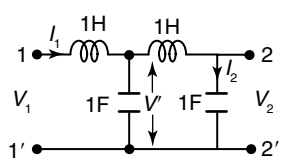


**Fig. 12.33**

Also, $V_1 = I_1 \times s + V' = \left(s^4+2s^2\right)V_2 + \left(s^2+1\right)V_2 = \left(s^4+3s^2+1\right)V_2$

$$\therefore G_{21} = \frac{V_2}{V_1} = \frac{1}{s^4+3s^2+1}$$

Problem 12.9 ***Determine the current transfer ratio, $\alpha_{21}$ and driving point impedance, $Z_{21}$ for the circuit shown.***

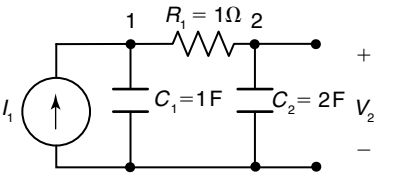



**Fig. 12.34**

*Solution* By KCL at the node 1,

$$\frac{V_1 - V_2}{1} + sV_1 = I_1 \quad \text{or,} \quad V_1(1+s) - V_2 = I_1 \tag{i}$$

By KCL at the node 2, $\frac{V_2 - V_1}{1} + 2sV_2 = 0$ or, $-V_1 + V_2(1+2s) = 0$ (ii)

Solving for $V_2$,

$$V_2 = \frac{\begin{vmatrix} (1+s) & I_1 \\ -1 & 0 \end{vmatrix}}{\begin{vmatrix} (1+s) & -1 \\ -1 & (1+2s) \end{vmatrix}} = \frac{I_1}{2s^2+3s}$$

$$\therefore Z_{21} = \frac{V_2}{I_1} = \frac{1}{2s^2+3s} = \frac{0.5}{s(s+1.5)}$$

$$\therefore \alpha_{21} = \frac{I_2}{I_1} = \frac{V_2 \times 2s}{I_1} = \frac{0.5 \times 2s}{s(s+1.5)} = \frac{1}{s+1.5}$$

Problem 12.10 ***Obtain $\frac{V(s)}{I(s)}$ and $\frac{V_{ab}(s)}{V(s)}$ for the network shown in Fig. 12.35.***

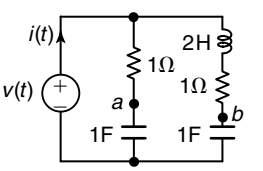



**Fig. 12.35**

*Solution* Here, the current,

$$I(s) = \frac{V(s)}{\frac{\left(1+\frac{1}{s}\right)\left(1+2s+\frac{1}{s}\right)}{\left(1+\frac{1}{s}\right)+\left(1+2s+\frac{1}{s}\right)}} = \frac{V(s)\left(2+2s+\frac{2}{s}\right)}{\left(\frac{s+1}{s}\right)\left(\frac{2s^2+s+1}{s}\right)}$$

$$\therefore \frac{V(s)}{I(s)} = \frac{(s+1)(2s^2+s+1)}{2s(s^2+s+1)}$$

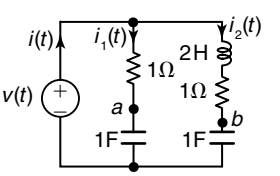



**Fig. 12.36**

Also, the voltage between the points *a* and *b* is,

$$V_{ab}(s) = I_2(s)(2s+1) - I_1(s) \times 1 = \left(\frac{V(s)}{2s+1+\frac{1}{s}}\right) \times (2s+1) - \frac{V(s)}{1+\frac{1}{s}} = V(s)\left[\frac{s(2s+1)}{2s^2+s+1} - \frac{s}{s+1}\right]$$

$$= \frac{V(s)s}{(s+1)(2s^2+s+1)}\left[(s+1)(2s+1) - (2s^2+s+1)\right] = \frac{V(s)2s^2}{(s+1)(2s^2+s+1)}$$

$$\therefore \frac{V_{ab}(s)}{V(s)} = \frac{2s^2}{(s+1)(2s^2+s+1)}$$

Problem 12.11 ***Find out*** $\frac{V_1(s)}{I_1(s)}$ ***and*** $\frac{I_2(s)}{V_1(s)}$ ***for the network shown in Fig. 12.37. Consider the mutual inductance to be negligible.***

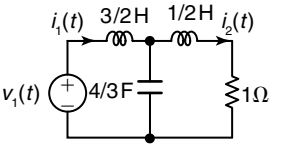



**Fig. 12.37**

*Solution* Here, $\frac{V_1(s)}{I_1(s)}$ is the driving point impedance, $Z(s)$.

$$Z(s) = \frac{3}{2}s + \frac{\frac{3}{4s}\times\left(\frac{1}{2}s+1\right)}{\frac{3}{4s}+\frac{1}{2}s+1} = \frac{3}{2}s + \frac{\frac{3}{2}s+1}{2s^2+4s+3} = \frac{3(s^3+2s^2+2s+1)}{2s^2+4s+3}$$

Also, by current division rule,

$$I_2(s) = I_1(s)\times\frac{\frac{3}{4s}}{\frac{3}{4s}+\frac{1}{2}s+1} = \frac{V_1(s)}{Z(s)}\times\frac{3}{2s^2+4s+3}$$

$$\therefore \frac{I_2(s)}{V_1(s)} = \frac{1}{\frac{3(s^3+2s^2+2s+1)}{2s^2+4s+3}}\times\frac{3}{2s^2+4s+3} = \frac{1}{s^3+2s^2+2s+1}$$

Problem 12.12 ***For the given bridged T network, find the driving point admittance*** $Y_{11}$ ***and the transfer admittance*** $Y_{21}$ ***with a 2-Ω load resistor connected across the Port 2.***

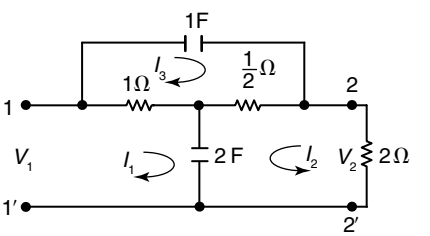



**Fig. 12.38**

*Solution* By KVL,

$$\left(1+\frac{2}{s}\right)I_1 + \frac{2}{s}I_2 - I_3 = V_1$$

$$\frac{2}{s}I_1 + \left(\frac{1}{2}+\frac{2}{s}\right)I_2 + \frac{1}{2}I_3 = V_2$$

$$-I_1 + \frac{1}{2}I_2 + \left(\frac{3}{2}+\frac{2}{s}\right)I_3 = 0$$

$$\therefore \Delta = \begin{vmatrix} \left(1+\frac{1}{s}\right) & \frac{1}{s} & -1 \\ \frac{1}{s} & \left(1+\frac{1}{s}\right) & 1 \\ -1 & 1 & \left(2+\frac{1}{s}\right) \end{vmatrix} = \frac{s+2}{s^2}$$

$$\Delta_{11}=(-1)^{1+1}\begin{vmatrix}\left(1+\frac{1}{s}\right) & 1\\ 1 & \left(2+\frac{1}{s}\right)\end{vmatrix}=\frac{s^2+3s+1}{s^2}$$

$$\Delta_{12}=(-1)^{1+2}\begin{vmatrix}\frac{1}{s} & 1\\ 1 & \left(2+\frac{1}{s}\right)\end{vmatrix}=-\frac{s^2+2s+1}{s^2}$$

$$\therefore Y_{11}=\frac{\Delta_{11}}{\Delta}=-\frac{s^2+3s+1}{s^2}\times\frac{s^2}{s+2}=\frac{s^2+3s+1}{s+2}$$

$$\therefore Y_{21}=\frac{\Delta_{21}}{\Delta}=-\frac{s^2+2s+1}{s^2}\times\frac{s^2}{s+2}=-\frac{s^2+2s+1}{s+2}$$

Problem 12.13 ***Determine the voltage transfer function of the symmetrical lattice network shown in Fig. 12.39.***

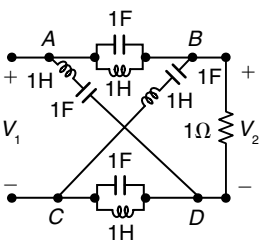



**Fig. 12.39**

**Sol:** Let, $Z_1$ = Series arm impedance $=\dfrac{s\times\frac{1}{s}}{s+\frac{1}{s}}=\dfrac{s}{s^2+1}$

$Z_2$ = Shunt arm impedance $=s+\dfrac{1}{s}=\dfrac{s^2+1}{s}$

Rearranging the figure, we have Fig. 12.40 as shown.

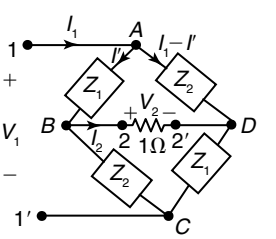



**Fig. 12.40** *Equivalent network*

Applying KVL to the mesh $1ABDC1'$, we get,

$$V_1(s)=I'(s)Z_1(s)+I_2(s)\times 1+Z_1(s)\left[I_1(s)-I'(s)+I_2(s)\right]$$

or, $$V_1(s)=I_2(s)\left[Z_1(s)+1\right]+Z_1(s)I_1(s) \quad \text{(i)}$$

Applying KVL to the mesh $1ADBC1'$, we get,

$$V_1(s)=Z_2(s)\left[I_1(s)-I'(s)\right]-I_2(s)\times 1+Z_2(s)\left[I'(s)-I_2(s)\right]$$

or, $$V_1(s)=Z_2(s)I_1(s)-\left[Z_2(s)+1\right]I_2(s) \quad \text{(ii)}$$

Multiplying (i) by $Z_2(s)$ and (ii) by $Z_1(s)$ and subtracting (ii) from (i),

$$V_1(s)\left[Z_2(s)-Z_1(s)\right]=I_2(s)\left[Z_1(s)+Z_2(s)+2Z_1(s)Z_2(s)\right]$$

or, $$\frac{I_2(s)}{V_1(s)}=\frac{Z_2(s)-Z_1(s)}{Z_1(s)+Z_2(s)+2Z_1(s)Z_2(s)}$$

Now the output voltage $V_2(s) = I_2(s) \times 1$

$$\frac{V_2(s)}{V_1(s)}=\frac{Z_2(s)-Z_1(s)}{Z_1(s)+Z_2(s)+2Z_1(s)Z_2(s)}=\frac{\dfrac{s^2+1}{s}-\dfrac{s}{s^2+1}}{\dfrac{s}{s^2+1}+\dfrac{s^2+1}{s}+2\times\dfrac{s^2+1}{s}\times\dfrac{s}{s^2+1}}$$

$$=\frac{\left(s^2+1\right)^2-s^2}{s^2+\left(s^2+1\right)^2+2s\left(s^2+1\right)}=\frac{\left(s^2+1+s\right)\left(s^2+1-s\right)}{\left(s^2+1+s\right)^2}\ \Rightarrow\ \frac{V_2(s)}{V_1(s)}=\frac{\left(s^2-s+1\right)}{\left(s^2+s+1\right)}$$

Note *This symmetrical lattice network is used as an all-pass network because it has the property for sinusoidal inputs that every thing which comes in goes out without any change in magnitude but distortion in phase.*

Problem 12.14 ***Determine the condition under which the input impedance of the network shown will be equal to* R.**

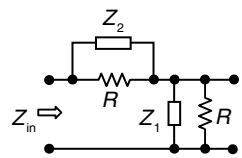


**Fig. 12.41**

*Solution* Here, input impedance for the network,

$$Z_{\text{in}}=\frac{Z_2R}{Z_2+R}+\frac{Z_1R}{Z_1+R}=\frac{\left(Z_1+Z_2\right)R^2+2Z_1Z_2R}{\left(Z_1+R\right)\left(Z_2+R\right)}$$

According to the given condition,

$$Z_{\text{in}}=R$$

$$\frac{\left(Z_1+Z_2\right)R^2+2Z_1Z_2R}{\left(Z_1+R\right)\left(Z_2+R\right)}=R\ \Rightarrow\ R=\sqrt{Z_1Z_2}$$

Problem 12.15 ***(a) If*** $Z_aZ_b = Z_0^2$ ***and*** $R_0 = Z_0$***, show that the voltage transfer function of the network is given by,*** $\dfrac{V_2}{V_1}=\dfrac{1}{1+Z_a/R_0}$.

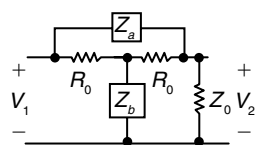


**Fig. 12.42**

***(b) Under the condition*** $Z_aZ_b = R_0^2$ ***and*** $R_0 = Z_0$***, show that the input impedance of the network of Fig. 12.42 is*** $R_0$.

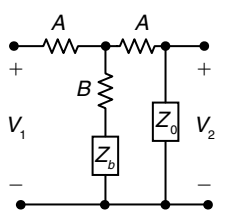


**Fig. 12.43**

*Solution* Replacing the $R_0$, $R_0$, and $Z_a$ delta by equivalent star, we have,

$$A=\frac{R_0Z_a}{R_0+Z_a}\qquad B=\frac{R^2}{2R_0+Z_a}$$

(a) Let, $I_1$ = Input current,

$I_2$ = current through $Z_0$.

$$\therefore V_1=I_1Z_{\text{in}}=I_1Z_0$$

and $$V_2=I_2Z_0=I_1\frac{B+Z_b}{A+B+Z_0+Z_b}Z_0$$

So, the voltage transfer function is $\frac{V_2}{V_1} = I_1 \frac{B+Z_b}{A+B+Z_0+Z_b} Z_0 \times \frac{1}{I_1 Z_0} = \frac{1}{1+\frac{A+Z_0}{B+Z_b}}$

Now,

$$\frac{A+Z_0}{B+Z_b} = \frac{\frac{R_0 Z_a}{R_0+Z_a}+Z_0}{\frac{R_0^2}{2R_0+Z_a}+Z_b} = \frac{R_0Z_a+2R_0Z_0+Z_aZ_0}{R_0^2+2R_0Z_b+Z_aZ_b} = \frac{R_0Z_a+2R_0Z_0+Z_aR_0}{Z_aZ_b+2Z_0Z_b+Z_aZ_b}$$

$$= \frac{2R_0(Z_0+Z_a)}{2Z_b(Z_0+Z_a)} \quad [\text{putting the conditions, } R_0 = Z_0, \quad Z_aZ_b = Z_0^2]$$

$$= \frac{Z_0}{Z_b} = \frac{Z_0}{Z_0^2 / Z_a} = \frac{Z_a}{Z_0}$$

$$\therefore \frac{V_2}{V_1} = \frac{1}{1+\frac{A+Z_0}{B+Z_b}} = \frac{1}{1+\frac{Z_a}{Z_0}} \quad \text{(Proved)}$$

(b) So, the input impedance, $Z_{in} = A + \frac{(A+Z_0)(B+Z_b)}{A+Z_0+B+Z_b} = \frac{A(A+B+Z_0+Z_b)+(A+Z_0)(B+Z_b)}{A+B+Z_0+Z_b}$

$$= \frac{Z_0(A+B+Z_0+Z_b)+2A(B+Z_b)+A^2-Z_0^2}{A+B+Z_0+Z_b} = Z_0 + \frac{2A(B+Z_b)+A^2-Z_0^2}{A+B+Z_0+Z_b}$$

$$= Z_0 + \frac{2R_0Z_a}{2R_0+Z_a}\left(\frac{R_0^2}{2R_0+Z_a}+Z_b\right) + \frac{R_0^2Z_a^2}{(2R_0+Z_a)^2} - Z_0^2$$

$$= Z_0 + \frac{2R_0^3Z_a}{(2R_0+Z_a)^2} + \frac{2R_0Z_aZ_b}{2R_0+Z_a} + \frac{R_0^2Z_a^2}{(2R_0+Z_a)^2} - Z_0^2$$

$$= Z_0 + \frac{2R_0^3Z_a}{(2R_0+Z_a)^2} + \frac{R_0^2Z_a^2}{(2R_0+Z_a)^2} + \frac{2R_0Z_aZ_b}{2R_0+Z_a} - Z_0^2$$

$$= Z_0 + \frac{R_0^2Z_a(2R_0+Z_a)}{(2R_0+Z_a)^2} + \frac{2R_0Z_aZ_b}{2R_0+Z_a} - Z_0^2 = Z_0 + \frac{R_0^2Z_a}{2R_0+Z_a} + \frac{2R_0Z_aZ_b}{2R_0+Z_a} - Z_0^2$$

$$Z_{in} = Z_0 + \frac{Z_0^2Z_a}{2Z_0+Z_a} + \frac{2Z_0Z_0^2}{2Z_0+Z_a} - Z_0^2 \quad [\text{putting the conditions, } R_0 = Z_0, \quad Z_aZ_b = Z_0^2]$$

$$= Z_0 + \frac{Z_0^2(2Z_0+Z_a)}{2Z_0+Z_a} - Z_0^2 = Z_0 + Z_0^2 - Z_0^2 \Rightarrow Z_{in} = Z_0$$

Problem 12.16 ***A network and a pole–zero pattern for driving point impedance Z(s) are shown in Fig. 12.44.***

***(a) Write an expression for Z(s).***

***(b) Specify the parameters R, L, G and C as a function of a, b and c if Z(0) = 1.***

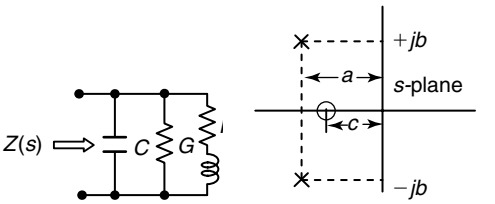



**Fig. 12.44**

*Solution*

(a) From the pole zero pattern it is seen that the driving point impedance $Z(s)$ has a zero at $s = -c$ and two complex conjugate poles at $s = -a \pm jb$.

$$Z(s) = K\frac{s+c}{(s+a+jb)(s+a-jb)} = K\frac{(s+c)}{(s+a)^2+b^2}$$

Given: $Z(0) = 1$. $\therefore 1 = K\dfrac{c}{a^2+b^2} \Rightarrow K = \left(\dfrac{a^2+b^2}{c}\right)$

$$\therefore Z(s) = \left(\frac{a^2+b^2}{c}\right)\frac{s+c}{(s+a)^2+b^2}$$

$$= \frac{(a^2+b^2)s+(a^2+b^2)c}{cs^2+2acs+(a^2+b^2)c} \quad \text{(i)}$$

(b) For the network shown, the admittance function is

$$Y(s) = \frac{1}{Z(s)} = G+Cs+\frac{1}{R+sL} = \frac{LCs^2+(GL+RC)s+1+GR}{R+sL}$$

$$\therefore Z(s) = \frac{R+sL}{LCs^2+(GL+RC)s+1+GR} \quad \text{(ii)}$$

From equations (i) and (ii), comparing the coefficients of numerator and denominator,

$$\boxed{\begin{aligned} R &= (a^2+b^2)c \\ L &= (a^2+b^2) \end{aligned}}$$

$LC = c \Rightarrow C = \dfrac{c}{L} = \dfrac{c}{(a^2+b^2)} \therefore \boxed{C = \dfrac{c}{L} = \dfrac{c}{(a^2+b^2)}}$

Also, $(1+GR) = (a^2+b^2)c \Rightarrow \therefore \boxed{G = \dfrac{(a^2+b^2)c-1}{c(a^2+b^2)}}$

Problem 12.17 ***Write down the driving-point impedance Z(s) of the network shown in Fig. 12.45 Locate the poles and zeros of Z(s) on the s-plane.***

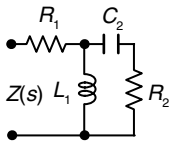


**Fig. 12.45**

*If* $v(t) = e^{-5t}$ *is applied to the network at* $t = 0$, *calculate* $i_2(t)$ *in* $R_2$, *while* $R_1 = R_2 = 1\ \Omega$, $L_1 = \frac{1}{2}$ *H and* $C_2 = \frac{1}{2}$ *F.*

*Solution* Here, driving point impedance,

$$Z(s)=1+\frac{\frac{s}{2}\left(\frac{2}{s}+1\right)}{\frac{s}{2}+\left(\frac{2}{s}+1\right)}=1+\frac{s(s+2)}{s^2+2s+4}=\frac{2(s^2+2s+2)}{s^2+2s+4}=\frac{2\left[(s+1)^2+1\right]}{(s+1)^2+3}$$

The driving-point impedance $Z(s)$ has two complex conjugate zeros at, $s = -1 \pm j$ and two complex conjugate poles at, $s=-1\pm j\sqrt{3}$.

For a source of $V_1(s)$, the current $I_1(s)$ is, $I_1(s)=\frac{V_1(s)}{Z(s)}$

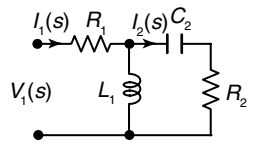


**Fig. 12.46 (a)**

The current $I_2(s)$ is

$$I_2(s)=I_1(s)\times\frac{\frac{s}{2}}{\frac{s}{2}+1+\frac{2}{s}}=I_1(s)\times\frac{s^2}{s^2+2s+4}$$

$$=\frac{V_1(s)}{Z(s)}\times\frac{s^2}{s^2+2s+4}=V_1(s)\times\frac{s^2}{2(s^2+2s+2)}$$

For the given voltage, $v(t)=e^{-5t} \Rightarrow V_1(s)=\frac{1}{s+5}$

$$\therefore I_2(s)=\frac{s^2}{2(s+5)(s^2+2s+2)}=\frac{k_1}{s+5}+\frac{k_2}{s+1+j1}+\frac{k_2^*}{s+1-j1}$$

where, $k_1=\left.\frac{s^2}{2(s^2+2s+2)}\right|_{s=-5}=\frac{25}{34}$

$$k_2=\left.\frac{s^2}{2(s+5)(s+1-j1)}\right|_{s=-1-j1}=-\frac{4+j1}{34}=-\frac{2}{17}-j\frac{1}{34}$$

$$k_2^*=\left.\frac{s^2}{2(s+5)(s+1+j1)}\right|_{s=-1+j1}=-\frac{2}{17}+j\frac{1}{34}$$

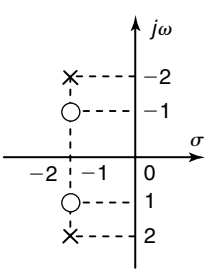


**Fig. 12.46 (b)**

$$\therefore I_2(s)=\frac{25/34}{s+5}+\frac{-\frac{2}{17}-j\frac{1}{34}}{s+1+j1}+\frac{-\frac{2}{17}+j\frac{1}{34}}{s+1-j1}$$

$$\therefore i_2(t)=\frac{25}{34}e^{-5t}+\left(-\frac{2}{17}-j\frac{1}{34}\right)e^{-(1+j1)t}+\left(-\frac{2}{17}+j\frac{1}{34}\right)e^{-(1-j1)t}$$

$$=\frac{25}{34}e^{-5t}-\frac{2}{17}e^{-t}\left(e^{jt}+e^{-jt}\right)+j\frac{1}{34}e^{-t}\left(e^{jt}-e^{-jt}\right)$$

$$=\frac{25}{34}e^{-5t}-\frac{4}{17}e^{-t}\cos t-\frac{1}{34}e^{-t}\sin t$$

Problem 12.18 ***A two-terminal network consists of a coil having an inductance*** **L** ***and a resistance*** **R** ***shunted by a capacitance*** **C.** ***The poles and zero of the driving point function*** **Z(s)** ***of this network are***

$$\text{Pole at } -\frac{1}{2}\pm \text{j}\frac{\sqrt{3}}{2};\quad \text{Zero at } -1-\text{j}0$$

If Z(j0) = 1, determine the values of R, L and C.

*Solution* Here, driving-point impedance, $Z(s)$ is

$$Z(s)=\frac{\left(R+sL\right)\times\frac{1}{sC}}{\left(R+sL\right)+\frac{1}{sC}}=\frac{R+sL}{LCs^2+RCs+1} \tag{i}$$

From the given values of poles and zero, we get the driving-point impedance function as

$$Z(s)=K\frac{(s+1)}{\left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)\left(s+\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)}=K\frac{(s+1)}{s^2+s+1}$$

Given that: $Z(j0)=1 \;\Rightarrow\; 1=K\frac{0+1}{0+0+1} \;\Rightarrow\; K=1$

$$\therefore Z(s)=\frac{s+1}{s^2+s+1} \tag{ii}$$

Comparing the coefficients of numerators and denominators of (i) and (ii), we get,

$$R=1\ \Omega \quad L=1\ \text{H} \quad C=1\ \text{F}$$

Problem 12.19 ***The transform voltage of a network is given as***

$$\text{V}(\text{s})=\frac{3\text{s}}{(\text{s}+2)(\text{s}^2+2\text{s}+2)}$$

*Draw its pole-zero diagram and hence obtain* v(t).

*Solution* Here, $V(s)=\frac{3s}{(s+2)(s^2+2s+2)}=\frac{3s}{(s+2)\left[(s+1)^2+1\right]}$

Zero is at origin (s = 0)

Poles are at $s = -2, \quad (-1 \pm j1)$

The pole–zero diagram is shown.

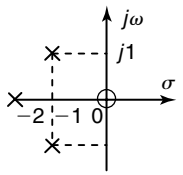



**Fig. 12.47**

$$V(s)=\frac{3s}{(s+2)(s^2+2s+2)}=\frac{k_1}{(s+2)}+\frac{k_2}{s+1+j1}+\frac{k_2^*}{s+1-j1}$$

From the pole-zero diagram, the constants $k_1$, $k_2$ and $k_2$* can be evaluated as follows.

$$k_1 = K\frac{\text{Product of all the directed lines from all zeros to } p_i}{\text{Product of all the directed lines from other remaining poles to } p_i}$$

$$=3\frac{2\angle 180^\circ}{\sqrt{2}\angle 45^0\times\sqrt{2}\angle -45^\circ}=-3$$

$$\therefore k_2=3\frac{\sqrt{2}\angle 45^\circ}{\sqrt{2}\angle 135^\circ\times 2\angle -90^\circ}=\frac{3}{2}$$

$$\therefore k_2^*=\frac{3}{2}$$

$$V(s)=\frac{-3}{(s+2)}+\frac{\frac{3}{2}}{s+1+j1}+\frac{\frac{3}{2}}{s+1-j1}$$

$$v(t)=\frac{3}{2}e^{-(1+j1)t}+\frac{3}{2}e^{-(1-j1)t}-3e^{-2t}=3e^{-t}\left(\frac{e^{jt}-e^{-jt}}{2}\right)-3e^{-2t}=3e^{-t}\cos t-3e^{-2t}$$

Problem 12.20 ***Draw poles and zeros for the transform voltage***

$$V(s)=\frac{s^2+3s+2}{s^2+7s+12}$$

and evaluate v(t) either analytically or by making use of pole–zero diagram.

*Solution* Here, $V(s)=\dfrac{s^2+3s+2}{s^2+7s+12}=\dfrac{(s+1)(s+2)}{(s+3)(s+4)}$

The function has poles at $s = -3$ and $-4$ and zeros at $s = -1$ and $-2$.

The pole-zero diagram is shown in Fig. 12.48.

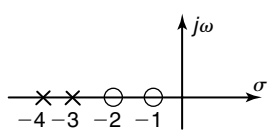



**Fig. 12.48** *Pole–zero diagram of* $\dfrac{s^2+3s+2}{s^2+7s+12}$

Since the numerator and denominator of $V(s)$ have the same power of $s$, the numerator has to be brought down on one degree less to evaluate $v(t)$.

$$\therefore V(s)=\frac{s^2+3s+2}{s^2+7s+12}=1-\frac{4s+10}{s^2+7s+12}=1-4\frac{s+5/2}{(s+3)(s+4)}=1-V_1(s) \qquad \text{(i)}$$

where, $V_1(s)=4\dfrac{s+5/2}{(s+3)(s+4)}$

This function has one zero at $s = -5/2$ and two poles at $s = -3$ and $s = -4$. The pole–zero diagram of $V_1(s)$ is shown in Fig. 12.49.

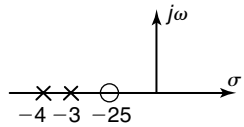


**Fig. 12.49**

Let, $V_1(s) = 4\dfrac{s+5/2}{(s+3)(s+4)} = \dfrac{k_1}{s+3} + \dfrac{k_2}{s+4}$

From pole-zero diagram, $k_1 = 4\dfrac{0.5\angle 180^\circ}{1\angle 0^\circ} = -2 \qquad k_2 = 4\dfrac{1.5\angle 180^\circ}{1\angle 180^\circ} = 6$

$$\therefore V_1(s) = \frac{6}{s+4} - \frac{2}{s+3} \quad \text{and} \quad \therefore v_1(t) = 6e^{-4t} - 2e^{-3t}$$

The inverse Laplace of 1 cannot be evaluated from the pole–zero diagram. However, we know that inverse Laplace of 1 is impulse function, $\delta(t)$.

Therefore, from (i), $v(t) = \delta(t) - v_1(t) = \delta(t) + 2e^{-3t} - 6e^{-4t}$

Problem 12.21 ***The locations of the poles and zeros of the driving point impedance Z(s) of the circuit shown in Fig. 12.50 (a) are shown in Fig. 12.50 (b). Calculate the values of the components. It is given that Z(j0) = 5.***

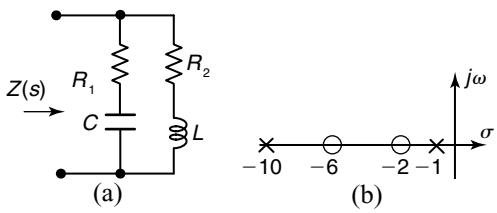


**Fig. 12.50**

*Solution* From Fig. 12.50 (a), the driving point impedance is given as

$$Z(s) = \frac{\left(R_2 + sL\right)\left(R_1 + \dfrac{1}{Cs}\right)}{R_1 + sL + R_2 + \dfrac{1}{Cs}} = \frac{R_1 LCs^2 + \left(R_1 R_2 C + L\right)s + R_2}{LCs^2 + \left(R_1 + R_2\right)Cs + 1}$$

$$= \frac{s^2 + \left(\dfrac{R_1 R_2 C + L}{R_1 LC}\right)s + \dfrac{R_2}{R_1 LC}}{\dfrac{1}{R_1}s^2 + \left(\dfrac{R_1 + R_2}{R_1 L}\right)s + \dfrac{1}{R_1 LC}} \qquad \text{(i)}$$

Again, from Fig. 12.50 (b), the driving point impedance function is written as

$$Z(s) = 5\frac{(s+2)(s+6)}{(s+1)(s+10)} = \frac{s^2 + 8s + 12}{\dfrac{1}{5}s^2 + \dfrac{11}{5}s + 2} \qquad \text{(ii)}$$

From (i) and (ii), comparing the coefficients of the numerators and denominators, $R_1 = 5$

$$\frac{R_2}{R_1 LC} = 12 \;\Rightarrow\; R_2 = 60\,CL \qquad \text{(iii)}$$

and

$$\frac{1}{R_1 LC} = 2 \;\Rightarrow\; CL = 0.1 \qquad \text{(iv)}$$

Form (iii) and (iv), $R_2 = 6$

Also, $\dfrac{R_1+R_2}{R_1 L}=\dfrac{11}{5} \Rightarrow \dfrac{5+6}{5L}=\dfrac{11}{5} \Rightarrow L=1$

$$\therefore C=\frac{0.1}{L}=0.1$$

Therefore, the component values are $R_1=5\,\Omega$, $R_2=6\,\Omega$, $L=1\,\text{H}$, and $C=0.1\,\text{F}$

Problem 12.22 ***The pole–zero diagram of the driving point impedance function of the network of Fig. 12.51 (a) is shown in Fig. 12.51 (b). At dc the input impedance is resistive and equals to 2 Ω. Determine the values of R, L and C.***

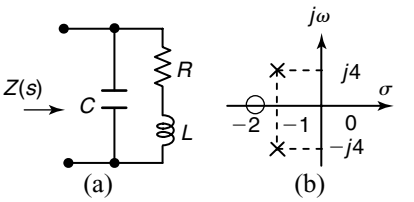



**Fig. 12.51**

*Solution* From Fig. 12.51 (a), the driving point impedance is given as

$$Z(s)=\frac{(R+sL)\times\dfrac{1}{Cs}}{R+sL+\dfrac{1}{Cs}}=\frac{s+\dfrac{R}{L}}{Cs^2+\dfrac{R}{L}s+\dfrac{1}{L}} \qquad \text{(i)}$$

From Fig. 12.51 (b), the driving-point impedance function is written as,

$$Z(s)=K\frac{(s+2)}{(s+1+j4)(s+1-j4)}=K\frac{s+2}{s^2+2s+17} \qquad \text{(ii)}$$

Since dc input impedance is 2 Ω, $\therefore Z(0)=2 \Rightarrow 0=K\times\dfrac{2}{17} \Rightarrow K=17$

$$\therefore Z(s)=17\frac{(s+2)}{(s+1+j4)(s+1-j4)}=\frac{s+2}{\dfrac{1}{17}s^2+\dfrac{2}{17}s+1} \qquad \text{(iii)}$$

Comparing the coefficients of the numerators and denominators of (i) and (iii), we get, $R=2\,\Omega$, $L=1\,\text{H}$, and $C=\dfrac{1}{17}\,\text{F}$

Problem 12.23 ***Draw the asymptotic Bode plot for the following functions:***

***(a)*** $G(s)=\dfrac{s+3}{s(s+1)(s+2)}$ ***(b)*** $G(s)=\dfrac{20(s+2)}{s(s^2+4s+16)}$ ***(c)*** $G(s)=\dfrac{200(s+1)}{(s+10)^2}$

*Solution*

(a) $G(s)=\dfrac{s+3}{s(s+1)(s+2)}=\dfrac{3\left(1+\dfrac{s}{3}\right)}{2s(1+s)\left(1+\dfrac{s}{2}\right)}=\dfrac{1.5\left(1+\dfrac{s}{3}\right)}{s(1+s)\left(1+\dfrac{s}{2}\right)}$

For this function,

(i) For the constant $K = 1.5$, magnitude $= 20 \log_{10}(1.5) = 3.52$ dB and phase $= 0°$

(ii) For a simple zero at $s = -3$

$$\text{Magnitude} = 20\log_{10}\left(\sqrt{1+\frac{\omega^2}{9}}\right); \quad \text{Phase} = \tan^{-1}\left(\frac{\omega}{3}\right)$$

The magnitude contributes to a straight line of slope 20 dB/decade for $\omega > 3$.

(iii) For a simple pole at origin ($s = 0$),

Magnitude $= -20 \log_{10} \omega$

Phase $= -90°$

This magnitude contributes to a straight line of slope $-20$ dB/decade passing through the intersection of 0 dB at $\omega = 1$.

| Range of $\omega$ | Factor | Resultant slope |
|---|---|---|
| starting | Pole at origin | $-20$ dB/decade |
| $0 < \omega < 1$ | – | $-20$ dB/decade |
| $1 < \omega < 2$ | Pole at $s = -1$ | $-40$ dB/decade |
| $2 < \omega < 3$ | Pole at $s = -2$ | $-60$ dB/decade |
| $3 < \omega < \infty$ | Pole at $s = -3$ | $-40$ dB/decade |

(iv) For the simple pole at $s = -1$,

$$\text{Magnitude} = -20\log_{10}\left(\sqrt{1+\omega^2}\right)$$

Phase $= -\tan^{-1}(\omega)$

The magnitude contributes to a straight line of slope $-20$ dB/decade for $\omega > 1$.

(v) For a simple pole at $s = -2$,

$$\text{Magnitude} = -20\log_{10}\left(\sqrt{1+\frac{\omega^2}{4}}\right); \quad \text{Phase} = -\tan^{-1}\left(\frac{\omega}{2}\right)$$

The magnitude contributes to a straight line of slope $-20$ dB/decade for $\omega > 2$.

Therefore, the slopes for different ranges of values of $\omega$ are as given below.

Also, the phase angle,

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{3}\right) - 90° - \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right)$$

The Bode magnitude and phase plots are shown in Fig. 12.52.

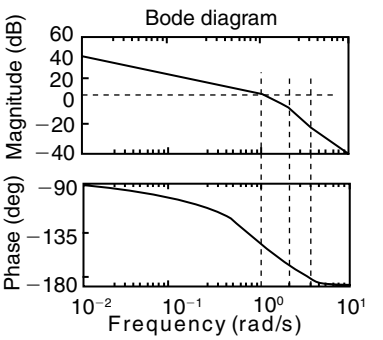



**Fig. 12.52**

(b) $$G(s) = \frac{20(s+2)}{s(s^2+4s+16)} = \frac{20\times 2\left(1+\frac{s}{2}\right)}{16s\left(\frac{s^2}{16}+\frac{s}{4}+1\right)} = \frac{2.5\left(1+\frac{s}{2}\right)}{s\left[\left(\frac{s}{4}\right)^2+\frac{s}{4}+1\right]}$$

(i) For the constant $K = 2.5$, magnitude $= 20 \log_{10}(2.5) = 7.96$ dB and phase $= 0°$.

(ii) For a simple zero at $s = -2$,

$$\text{Magnitude} = 20\log_{10}\left(\sqrt{1+\frac{\omega^2}{4}}\right); \quad \text{Phase} = \tan^{-1}\left(\frac{\omega}{2}\right).$$

The magnitude contributes to a straight line of slope 20 dB/decade for $\omega > 2$.

(iii) For a simple pole at origin ($s = 0$),
Magnitude $= -20 \log_{10} \omega$; Phase $= -90°$

This magnitude contributes to a straight line of slope $-20$ dB/decade passing through the intersection of 0 dB at $\omega = 1$.

(iv) For the complex conjugate poles, $\xi = 0.5$ and $\omega_0 = 4$ rad/s

$$\therefore \text{Magnitude} = -20\log_{10}\left[\sqrt{\left(1-\frac{\omega}{4}\right)^2+\left(\frac{\omega}{4}\right)^2}\right]$$

$$\text{Phase} = -\tan^{-1}\left[\frac{\left(\omega/4\right)}{1-\left(\omega/4\right)^2}\right] = -\tan^{-1}\left(\frac{4\omega}{16-\omega^2}\right)$$

This magnitude contributes to a straight line of slope $-40$ dB/decade for $\omega > 4$.

Also, phase angle is,

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{2}\right) - 90° - \tan^{-1}\left(\frac{4\omega}{16-\omega^2}\right)$$

The Bode magnitude and phase plots of the function are shown in Fig. 12.53.

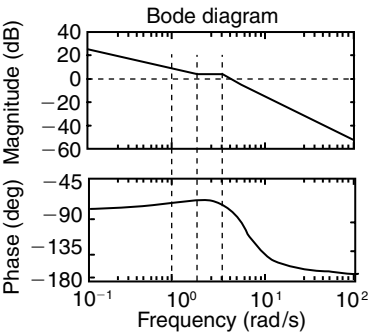



**Fig. 12.53**

(c) $$G(s) = \frac{200(s+1)}{(s+10)^2} = \frac{2(1+s)}{\left(1+\frac{s}{10}\right)^2}$$

(i) For the constant $K = 2$, magnitude $= -20 \log_{10}(2) = 6$ dB; and phase $= 0°$.

(ii) For the simple zero at $s = -1$,

$$\text{Magnitude} = 20\log_{10}\left(\sqrt{1+\omega^2}\right); \quad \text{Phase} = \tan^{-1}(\omega)$$

This contributes to a straight line of slope 20 dB/decade for $\omega > 1$.

(iii) For the double poles at $s = -10$,

$$\text{Magnitude} = -20\log_{10}\left(\sqrt{1+\left(\frac{\omega}{10}\right)^2}\right)^2 = -40\log_{10}\left(\sqrt{1+\left(\frac{\omega}{10}\right)^2}\right); \quad \text{Phase} = -2\tan^{-1}\left(\frac{\omega}{10}\right)$$

This magnitude contributes to a straight line of slope $-40$ dB/decade for $\omega > 10$.

Therefore, the slopes of the magnitude curve for different ranges of values of $\omega$ are given below.

| Range of $\omega$ | Factor | Resultant slope |
|---|---|---|
| starting | Constant $K = 2$ | 0 dB/decade |
| $0 < \omega < 1$ | – | 0 dB/decade |
| $1 < \omega < 10$ | Zero at $s = -1$ | 20 dB/decade |
| $10 < \omega < \infty$ | Repeated poles at $s = -10$ | $-40$ dB/decade |

The Bode magnitude and phase plots of the function are shown in Fig. 12.54 below.

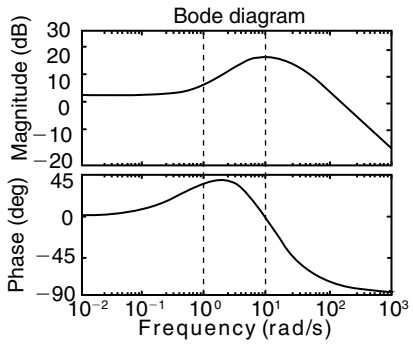



**Fig. 12.54**

Problem 12.24 ***The Bode diagram of a function is shown in Fig. 12.55. Find the function.***

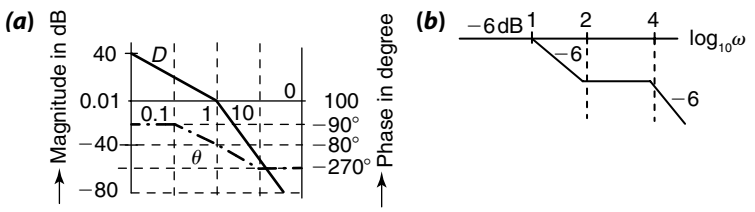



**Fig. 12.55**

*Solution* (a) From the Bode plot, we get the following information:

(i) At low frequency, phase $= -90°$ and magnitude decreases at a rate of $-20$ dB/decade.
This implies that there is a simple pole at origin ($s = 0$).

(ii) At $\omega = 1$ rad/s, the slope changes to $-40$ dB/decade. The phase curve is a straight line of slope $-45°$ per decade passing through $-180°$ at $\omega = 1$
This implies that there is a simple pole at $s = -1$.

Hence, the function can be written as $F(s) = \dfrac{K}{s(s+1)}$

To find the constant $K$, we consider the value of magnitude at $\omega = 1$. From Bode diagram, at $\omega = 1$,

$$20 = 20\log_{10} K - 20\log_{10}(1) \;\Rightarrow\; K = 1$$

$$F(s) = \frac{1}{s(s+1)}$$

(b) From the Bode plot, it is seen that the function has poles at $s = -1$ and $-4$ and zero at $s = -2$ [$\because -6$ dB/octave $\equiv -20$ dB/decade]. Hence, the function can be written as

$$F(s) = K\frac{\left(1+\frac{s}{2}\right)}{\left(1+\frac{s}{1}\right)\left(1+\frac{s}{4}\right)} = 2K\frac{(s+2)}{(s+1)(s+4)}$$

Also, to find the constant $K$, it is seen that at $\omega = 1$, $|F(j\omega)| = -6$ dB

$$\therefore 20\log_{10} K = -6 \;\Rightarrow\; K = 0.5$$

Therefore, the network function becomes, $F(s) = 2K\dfrac{(s+2)}{(s+1)(s+4)} = 2\times 0.5\dfrac{(s+2)}{(s+1)(s+4)} = \dfrac{(s+2)}{(s+1)(s+4)}$

Problem 12.25 ***Construct the Bode magnitude and phase plot of the impedance function of the network shown in Fig. 12.56.***

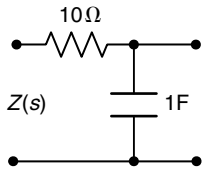


**Fig. 12.56**

*Solution* Here, $Z(s) = 10 + \dfrac{1}{s} = \dfrac{(1+10s)}{s}$

Following the steps, as mentioned in Prob. 12.24, Bode plots are drawn and shown in Fig. 12.57 below.

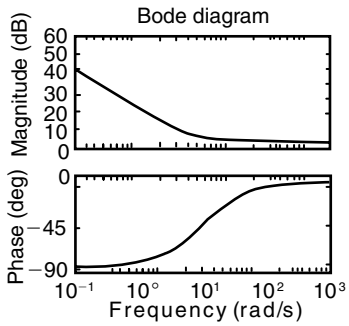
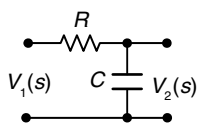


**Fig. 12.57**

Problem 12.26 ***Draw the complex locus plot of the voltage ratio transfer function $G_{21}$ of the following* RC *networks:***

***(a)***




***(b)***

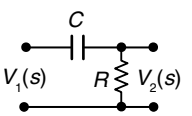



**Fig. 12.58**

*Solution*

(a) Here, the voltage ratio transfer function is $G_{21}(s) = \dfrac{1}{1+RCs}$

For steady-state, $s = j\omega$.

$$\therefore G_{21}(j\omega) = \frac{1}{1+j\omega RC} = \frac{1}{1+\omega^2R^2C^2} - j\frac{\omega RC}{1+\omega^2R^2C^2}$$

$$\therefore \mathrm{Re}\left[G_{21}(j\omega)\right] = X = \frac{1}{1+\omega^2R^2C^2} \tag{i}$$

and

$$\mathrm{Im}\left[G_{21}(j\omega)\right] = Y = -\frac{\omega RC}{1+\omega^2R^2C^2} \tag{ii}$$

For complex locus plot, we have to evaluate $\omega$ from (i) and (ii).

By (ii) ÷ (i), we get,

$$\frac{Y}{X}=\omega RC \;\Rightarrow\; \omega=\frac{Y}{XRC}$$

Substituting this value of $\omega$ in (i), we get,

$$X=\frac{1}{1+\left(\frac{Y}{XRC}\right)^2 R^2C^2}=\frac{X^2}{X^2+Y^2} \quad \text{or,} \quad \left(X-\frac{1}{2}\right)^2+Y^2=\left(\frac{1}{2}\right)^2$$

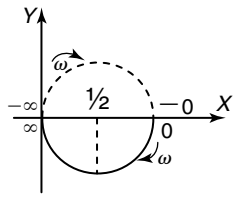


**Fig. 12.59**

This is a circle with centre at (½, 0) and radius ½.

The upper half of the circle corresponds to negative frequencies, ranging from $\omega = -\infty$ to $\omega = -0$.

The locus for negative frequencies is the image of that for positive frequencies. This is because the real part Re[$G_{21}(j\omega)$] has even symmetry while the imaginary Im[$G_{21}(j\omega)$] part has odd symmetry.

At $\omega=\frac{1}{RC}$, the magnitudes of real and imaginary parts are equal and $\text{Re}\left[G_{21}\left(j\omega\right)\right]=\text{Im}\left[G_{21}\left(j\omega\right)\right]=\frac{1}{2}$

so that, $\left|G_{21}\left(j\omega\right)\right|=\frac{1}{\sqrt{2}}$. This frequency ($\omega=\frac{1}{RC}$) is known as *half-power frequency*.

(b) Here, $G_{21}\left(s\right)=\frac{RCs}{1+RCs}$

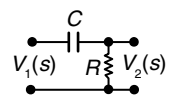


**Fig. 12.60**

For steady-state, $s = j\omega$.

$$\therefore G_{21}\left(j\omega\right)=\frac{j\omega RC}{1+j\omega RC}=\frac{\omega^2R^2C^2}{1+\omega^2R^2C^2}+j\frac{\omega RC}{1+\omega^2R^2C^2}$$

$$\therefore \text{Re}\left[G_{21}\left(j\omega\right)\right]=X=\frac{\omega^2R^2C^2}{1+\omega^2R^2C^2} \qquad \text{(i)}$$

and $$\text{Im}\left[G_{21}\left(j\omega\right)\right]=Y=\frac{\omega RC}{1+\omega^2R^2C^2} \qquad \text{(ii)}$$

For complex locus plot, we have to evaluate $\omega$ from (i) and (ii).

By (ii) ÷ (i), we get,

$$\frac{Y}{X}=\frac{1}{\omega RC} \;\Rightarrow\; \omega=\frac{X}{YRC}$$

Substituting this value of $\omega$ in (i), we get,

$$X=\frac{\left(\frac{X}{YRC}\right)^2 R^2C^2}{1+\left(\frac{X}{YRC}\right)^2 R^2C^2}=\frac{X^2}{X^2+Y^2} \quad \text{or,} \quad \left(X-\frac{1}{2}\right)^2+Y^2=\left(\frac{1}{2}\right)^2$$

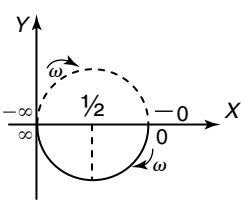


**Fig. 12.61**

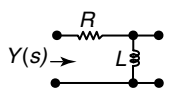
This is a circle with centre at (½, 0) and radius ½.

The upper half of the circle corresponds to negative frequencies, ranging from $\omega = -\infty$ to $\omega = -0$. The locus for negative frequencies is the image of that for positive frequencies. This is because the real part $\text{Re}[G_{21}(j\omega)]$ has even symmetry while the imaginary $\text{Im}[G_{21}(j\omega)]$ part has odd symmetry.

At $\omega = \frac{1}{RC}$, the magnitudes of real and imaginary parts are equal and $\text{Re}\left[G_{21}(j\omega)\right] = \text{Im}\left[G_{21}(j\omega)\right] = \frac{1}{2}$ so that, $\left|G_{21}(j\omega)\right| = \frac{1}{\sqrt{2}}$. This frequency ( $\omega = \frac{1}{RC}$ ) is known as *half-power frequency*.

Problem 12.27 ***Draw the complex locus plot of the admittance function of the networks shown below.***

***(a)***




***(b)***

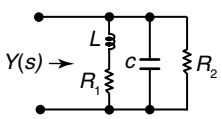



**Fig. 12.62**

*Solution* The admittance function is $Y(s) = \frac{1}{R + sL}$

$$\therefore Y(j\omega) = \frac{1}{R + j\omega L} = \frac{R}{R^2 + \omega^2 L^2} - j\frac{\omega L}{R^2 + \omega^2 L^2}$$

$$\therefore \text{Re}\left[Y(j\omega)\right] = X = \frac{R}{1 + \omega^2 L^2} \tag{i}$$

and

$$\text{Im}\left[Y(j\omega)\right] = Y = -\frac{\omega L}{1 + \omega^2 L^2} \tag{ii}$$

For complex locus plot, we have to evaluate $\omega$ from (i) and (ii).

By (ii) ÷ (i), we get $\frac{Y}{X} = \frac{\omega L}{R} \Rightarrow \omega = \frac{YR}{XL}$

Substituting this value of $\omega$ in (i), we get

$$X = \frac{R}{R^2 + \left(\frac{YR}{XL}\right)^2 L^2} = \frac{X^2}{R\left(X^2 + Y^2\right)} \quad \text{or,} \quad \left(X - \frac{1}{2R}\right)^2 + Y^2 = \left(\frac{1}{2R}\right)^2$$

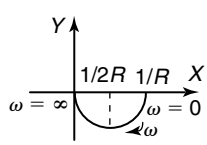



**Fig. 12.63**

This is a circle with centre at $\left(\frac{1}{2R}, 0\right)$ and radius $\frac{1}{2R}$.

It is observed that at $\omega = 0$ $\left|Y(j\omega)\right| = \frac{1}{R}$ and as $\omega \to \infty$, $\left|Y(j\omega)\right| = 0$

(b) This network may be considered to be parallel connection of three sub-networks:

(i) a resistor $R_2$

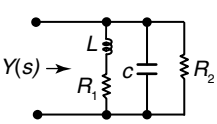



**Fig. 12.64**

(ii) a capacitor $C$

(iii) series combination of $R_1$ and $L$

Total admittance is given as

$$Y(j\omega)=\frac{1}{R_2}+j\omega C+\frac{1}{R_1+j\omega L}=Y_{R_2}(j\omega)+Y_C(j\omega)+Y_{R_1-L}(j\omega)$$

The complex loci for the three admittances can be drawn separately as shown in Fig. 12.65 below.

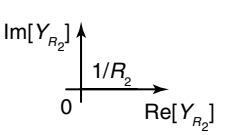



**Fig. 12.65 (a)** *Complex locus for* $R_2$

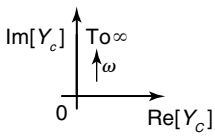



**Fig. 12.65 (b)** *Complex locus for* C

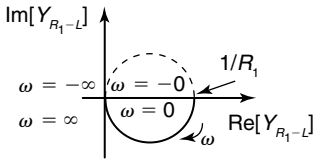



**Fig. 12.65 (c)** *Complex locus for* $R_1$ -L

The point-by-point phasor addition of these three loci gives the locus of the complete admittance of the network given. This is shown below in Fig. 12.65.

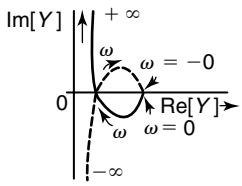



**Fig. 12.65 (d)** *Complex locus for the entire network*

Problem 12.28 ***Find the frequency response plot of the following network functions:***

***(a)*** $F(s)=\dfrac{25}{(s+3)(s+4)}$ ***(b)*** $F(s)=\dfrac{25}{s^2+s+9.25}$ ***(c)*** $F(s)=\dfrac{s^2+1.03}{s^2+1.23}$

*Solution*

(a) $F(s)=\dfrac{25}{(s+3)(s+4)}$

Here, $F(s)$ has two poles at $s=-3,-4$

For steady-state $s=j\omega$,

$$F(j\omega)=\frac{25}{(j\omega+3)(j\omega+4)}=|F(j\omega)|\angle F(j\omega)$$

$$\therefore |F(j\omega)|=\frac{25}{\sqrt{(9+\omega^2)(16+\omega^2)}} \quad \text{and} \quad \angle F(j\omega)=-\tan^{-1}\left(\frac{\omega}{3}\right)-\tan^{-1}\left(\frac{\omega}{4}\right)$$

The values of magnitude and phase angles are tabulated as follows:

| $\omega$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $\lvert F(j\omega)\rvert$ | 2.06 | 1.95 | 1.57 | 1.18 | 0.915 | 0.675 | 0.525 | 0.412 | 0.332 |
| $\angle F(j\omega)$ | 0 | $-33°$ | $-60°$ | $-83°$ | $-99°$ | $-110°$ | $-120°$ | $-127°$ | $-133°$ |

Also, as $\omega \to \infty$, $\lvert F(j\omega)\rvert \to 0$ and $\angle F(j\omega) \to -180°$.

The magnitude and phase plots are shown in Fig. 12.66.

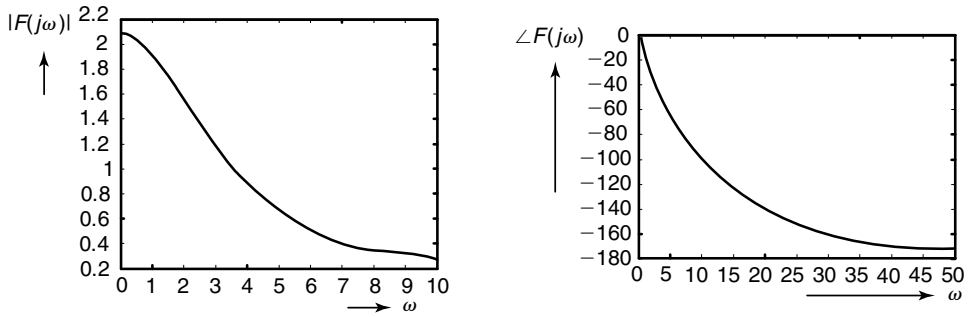



**Fig. 12.66**

(b) $$F(s) = \frac{25}{s^2 + s + 9.25} = \frac{25}{\left(s + \frac{1}{2}\right)^2 + 3^2}$$

Here, $F(s)$ has two poles at: $s = -\frac{1}{2} \pm j3$

For steady state $s = j\omega$,

$$F(j\omega) = \frac{25}{(j\omega)^2 + j\omega + 9.25} = \frac{25}{9.25 - \omega^2 + j\omega} = \lvert F(j\omega)\rvert \angle F(j\omega)$$

$$\therefore \lvert F(j\omega)\rvert = \frac{25}{\sqrt{\left(9.25 - \omega^2\right)^2 + \omega^2}} \quad \text{and} \quad \angle F(j\omega) = -\tan^{-1}\left(\frac{\omega}{9.25 - \omega^2}\right)$$

The values of magnitude and phase angles are tabulated as follows:

| $\omega$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\lvert F(j\omega)\rvert$ | 2.71 | 3.0 | 4.38 | 8.8 | 3.2 | 2.56 |
| $\angle F(j\omega)$ | 0° | $-7°$ | $-25°$ | $-85°$ | $-159.5°$ | $-173°$ |

Also, as $\omega \to \infty$, $\lvert F(j\omega)\rvert \to 0$ and $\angle F(j\omega) \to -180°$.

The magnitude and phase plots are shown in Fig. 12.67.

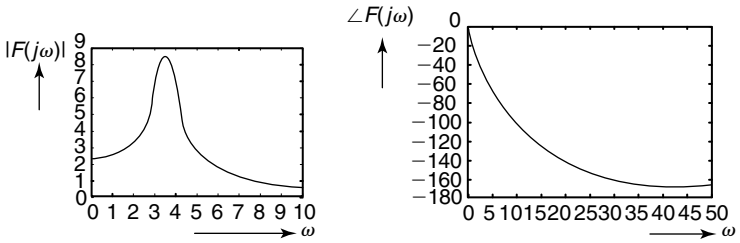



**Fig. 12.67**

Problem 12.29 ***Find the frequency response plot from the pole–zero plot for the following function:***

$$F(s)=\frac{5s}{s^2+6s+25}$$

*Solution* Here, $F(s)=\dfrac{5s}{s^2+6s+25}=\dfrac{5s}{(s+3)^2+4^2}$

The function has poles at $s=-3\pm j4$ and zero at $s=0$.

The pole–zero diagram of the function is shown in Fig. 12.68.

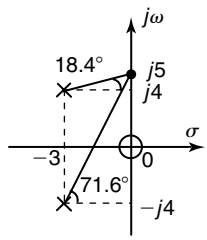



**Fig. 12.68**

To find the magnitude of the function at a point (say, $j5$, i.e., $\omega=5$ rad/s), we have,

$$F(j5)=\frac{5\times5}{\sqrt{10}\times\sqrt{90}}=\frac{5}{6}=0.83$$

$$\angle F(j5)=90°-18.4°-71.6°=0°$$

Also, at $\omega=0°$, $|F(j0)|$ and $\angle F(j0)=90°$ and at $\omega\to\infty$, since the vector length from all poles and zeros are all tending to infinity and there are two poles, $|F(j\infty)|=0$ and $\angle F(j\infty)=90°-90°-90°=-90°$

Similarly, at some other frequency, like $\omega=10$,

$$F(j10)=\frac{5\times10}{\sqrt{45}\times\sqrt{205}}=0.52$$

$$\angle F(j10)=90°-63.5°-77.9°=-51.4°$$

In this way, the magnitudes and phases for different values of $\omega$ can be calculated from the pole–zero plot. The plots are shown in Fig. 12.69 below.

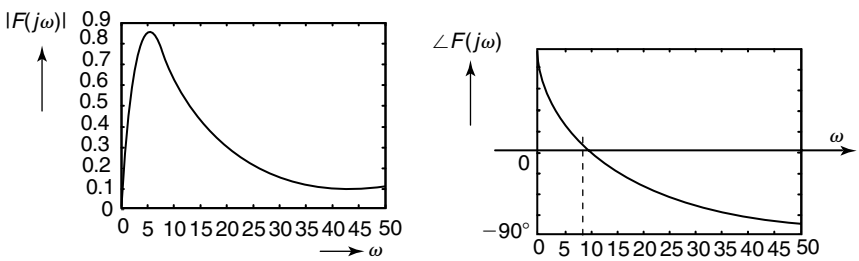



**Fig. 12.69**

## Summary

1. In any network, the functions relating the voltages and currents at different parts of the network are known as *network functions*.
2. A pair of terminals is commonly known as a port, across which a device can be connected.
3. For a two-port network, there are two types of network functions, transfer functions and driving point functions.
4. Transfer function is the ratio of an output transform to an input transform, with zero initial condition and with no internal energy sources except the controlled sources.
5. Four types of transfer function are

   voltage transfer function,

$$G_{12}(s)=\frac{V_1(s)}{V_2(s)};\quad G_{21}(s)=\frac{V_2(s)}{V_1(s)}$$

   current transfer function, ,

$$\alpha_{12}(s)=\frac{I_1(s)}{I_2(s)};\quad \alpha_{21}(s)=\frac{I_2(s)}{I_1(s)}$$

   transfer impedance function

$$Z_{12}(s)=\frac{V_1(s)}{I_2(s)};\quad Z_{21}(s)=\frac{V_2(s)}{I_1(s)}\text{, and}$$

   transfer admittance function,

$$Y_{12}(s)=\frac{I_1(s)}{V_2(s)};\quad Y_{21}(s)=\frac{I_2(s)}{V_1(s)}$$

6. Driving point immittance is the ratio of transform voltage or current at any port to the transform voltage or current at the same port, in zero state with no internal energy sources.
7. Four types of driving point immittance functions are

$$Z_{11}(s)=\frac{V_1(s)}{I_1(s)};Z_{22}(s)=\frac{V_2(s)}{I_2(s)}\quad\text{and}\quad Y_{11}(s)=\frac{I_1(s)}{V_1(s)};Y_{22}(s)=\frac{I_2(s)}{V_2(s)}$$

8. For a network function of the form $F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)}$, at the complex frequencies, $s = z_i$, $i = 1, 2, 3, \ldots, n$, the numerator polynomial $N(s) = 0$, and these frequencies are known as the *zeros of the network function* $F(s)$. Similarly, at the complex frequencies $s = p_i$, $i = 1, 2, 3, \ldots, m$, the denominator polynomial $D(s) = 0$ and these frequencies are known as the *poles of the network function* $F(s)$.
9. The plot of poles and zeros of a network function in the *s*-plane is known as pole–zero diagram of the network function. The locations of poles and zeros in the *s*-plane describes the behaviour of the *s* system.
10. The *natural response* or *transient response* or *source-free response* of a network is the response with no source excitation; it is the response of the network to its initial conditions or response due to unit impulse excitation.
11. For a stable system, all poles must lie in the left half of the *s*-plane. For a marginally stable system, the poles may lie at the origin.
12. For driving point and transfer function, the pole–zero locations in the *s*-plane must satisfy some necessary conditions as given in sections 12.10 and 12.11.
13. The time-domain response of a given network can be determined from the pole–zero plot of the network function and the knowledge of the network sources. For a network function,

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)}=\frac{K_1}{s-p_1}+\frac{K_2}{s-p_2}+\cdots+\frac{K_m}{s-p_m}$$

   Coefficients,

$$K_i=\left[(s-p_i)F(s)\right]_{s=p_i}=K\frac{(p_i-z_1)(p_i-z_2)(p_i-z_3)\cdots(p_i-z_n)}{(p_i-p_1)(p_i-p_2)(p_i-p_3)\cdots(p_i-p_{i-1})(p_i-p_{i+1})\cdots(p_i-p_m)}$$

   If $(p_i - z_n) = M_{ni}\angle\,\theta_{ni}$ and $(p_i - p_m) = N_{mi}\angle\,\alpha_{mi}$, where $M$'s and $N$'s are the magnitudes of the phasors $(p_i - z_n)$ and $(p_i - p_m)$, respectively and $\theta$'s and $\alpha$'s are the corresponding phase angles, the coefficients can be written as

$$K_i=K\frac{M_{1i}M_{2i}M_{3i}\cdots M_{ni}}{N_{1i}N_{2i}N_{3i}\cdots N_{mi}}\angle\left(\theta_{1i}+\theta_{2i}+\theta_{3i}+\cdots+\theta_{ni}\right)\cdot -\left(\alpha_{1i}+\alpha_{2i}+\alpha_{3i}+\cdots+\alpha_{mi}\right)$$

$$=K\frac{\text{Product of all the directed lines from all zeros to } p_i}{\text{Product of all the directed lines from other remaining poles to } p_i}$$

14. The frequency-domain response of a given network can be determined from the pole–zero plot of the network function and the knowledge of the network sources. For a network function,

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)}\Bigg|_{s=j\omega}$$
$$=K\frac{(j\omega-z_1)(j\omega-z_2)(j\omega-z_3)\cdots(j\omega-z_n)}{(j\omega-p_1)(j\omega-p_2)(j\omega-p_3)\cdots(j\omega-p_m)}$$

If, $|j\omega-z_i|=\sqrt{\sigma_{zi}{}^2+(j\omega-\omega_{zi})^2}=r_i$

$$\angle(j\omega-z_i)=\tan^{-1}\left(\frac{\omega-\omega_{zi}}{-\sigma_{zi}}\right)=\phi_i$$

and, $|j\omega-p_i|=\sqrt{\sigma_{pi}{}^2+(j\omega-\omega_{pi})^2}=q_i$

$$\angle(j\omega-p_i)=\tan^{-1}\left(\frac{\omega-\omega_{pi}}{-\sigma_{pi}}\right)=\theta_i$$

then the network function may be written as

$$F(j\omega)=K\frac{\text{Product of all zeros lines to } j\omega}{\text{Product of all poles lines to } j\omega}$$

Hence, the magnitude and phase angle of the complete frequency response may be written as

$$|F(j\omega)|=K\frac{\text{Product of all zeros lines to } j\omega}{\text{Product of all poles lines to } j\omega}$$
$$=K\frac{\prod_{i=1}^{n}|(j\omega-z_i)|}{\prod_{i=1}^{m}|(j\omega-p_i)|}K\frac{r_1r_2\cdots r_n}{q_1q_2\cdots q_m}$$

and angle

$\angle F(j\omega)$ = (summation of angles of the vectors from zeros to $j\omega$-point) − (summation of angles of the vectors from poles to $j\omega$-point)

$$=\sum_{i=1}^{n}\angle(j\omega-z_i)-\sum_{i=1}^{m}\angle(j\omega-p_i)$$
$$=(\phi_1+\phi_2+\phi_3+\cdots+\phi_n)-(\theta_1+\theta_2+\theta_3+\cdots+\theta_n)$$

15. The variation of magnitude and phase of a network function with frequency in logarithmic scale is known as Bode plot.

## Short-Answer Questions

**1. What are the poles and zeros? What information do they provide in respect of the network to which they relate?**

We consider a network function given by the ratio of two polynomials as

$$F(s)=\frac{a_ns^n+a_{n-1}s^{n-1}+\cdots+a_2s^2+a_1s+a_0}{b_ms^m+b_{m-1}s^{m-1}+\cdots+b_2s^2+b_1s+b_0} \quad (1)$$

It is often convenient to factor the polynomials in the numerator and denominator, and to write the transfer function in terms of those factors:

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)} \quad (2)$$

where, the numerator and denominator polynomials, $N(s)$ and $D(s)$, have real coefficients defined by the system's differential equation and $K=\frac{a_n}{b_m}$ is a positive constant, known as scale factor.

From Eq. (2), we can observe the following:

At $s=z_i, i=1,2,3....,n$, the numerator polynomial $N(s)=0$; these complex frequencies are known as the *zeros of the network function* $F(s)$. At zeros, the value of the network function is zero, i.e., $\lim_{s\to z_i} F(s)=0$.

At $s=p_i, i=1,2,3....,m$, the denominator polynomial $D(s)=0$; these complex frequencies are known as the *poles of the network function* $F(s)$. At poles, the value of the network function is infinity, i.e., $\lim_{s\to p_i} F(s)=\infty$.

- **Significance of poles and zeros** The values of poles and zeros of $F(s)$ and their locations in the $s$-plane completely specify a network function. All the coefficients of polynomials $N(s)$ and $D(s)$ are real, therefore the poles and zeros must be either purely real, or appear in complex conjugate pairs. In general for the poles, either $p_i=\sigma_i$, or else $p_i, p_{i+1}=\sigma\pm j\omega_i$. The existence of a single complex pole without a corresponding conjugate pole would generate complex coefficients in the polynomial $D(s)$. Similarly, the system zeros are either real or appear in complex conjugate pairs.

  The poles and zeros are properties of the transfer function, and therefore of the differential equation describing the input–output system dynamics. Together with the gain constant $K$ they completely characterize the differential equation, and provide a complete description of the system.

**2. What do you understand by driving point impedance of a two-port network? Enumerate important properties of driving point impedance functions of a two-port passive network.**

For a two-port network in zero state with no internal energy sources, the driving point impedance is the ratio of transform voltage at any port to the transform current at the same port.

$$Z_{11}(s)=\frac{V_1(s)}{I_1(s)};\quad Z_{22}(s)=\frac{V_2(s)}{I_2(s)}$$

- **Properties of driving point impedance**
  1. The coefficients in the polynomials $N(s)$ and $D(s)$ are real and positive.
  2. Complex and imaginary poles are conjugate.
  3. The real part of all poles and zeros is either negative or zero. Thus, all poles and zeros lie on the left half of the $s$-plane or on the imaginary axis.
  4. If the real part of any pole or zero is zero then that pole or zero is simple, not repeated. This includes the origin.
  5. Polynomials $N(s)$ and $D(s)$ do not have missing terms between those of highest and lowest degree unless all even terms or all odd terms are missing.
  6. The degree of $N(s)$ and $D(s)$ differs by zero or one only.
  7. The lowest degree of $N(s)$ and $D(s)$ may differ by one at most.

**3. What do you understand by the transfer function of a system? State its properties.**

It is defined as the ratio of an output transform to an input transform, with zero initial condition and with no internal energy sources except the controlled sources.

For a two-port network, having the variables $I_1(s)$, $I_2(s)$, $V_1(s)$, and $V_2(s)$, transfer function can take the following four forms

1. Voltage transfer function

$$G_{12}(s)=\frac{V_1(s)}{V_2(s)};\quad G_{21}(s)=\frac{V_2(s)}{V_1(s)}$$

2. Current transfer function

$$\alpha_{12}(s)=\frac{I_1(s)}{I_2(s)};\quad \alpha_{21}(s)=\frac{I_2(s)}{I_1(s)}$$

3. Transfer impedance function

$$Z_{12}(s)=\frac{V_1(s)}{I_2(s)};\quad Z_{21}(s)=\frac{V_2(s)}{I_1(s)}$$

4. Transfer admittance function

$$Y_{12}(s)=\frac{I_1(s)}{V_2(s)};\quad Y_{21}(s)=\frac{I_2(s)}{V_1(s)}$$

- **Properties of transfer function**
  1. The coefficients in the polynomials $N(s)$ and $D(s)$ are real and positive.
  2. Complex and imaginary poles are conjugate.
  3. The real part of all poles and zeros is either negative or zero. Thus, all poles and zeros lie on the left half of the $s$-plane or on the imaginary axis.
  4. If the real part of any pole or zero is zero then that pole or zero is simple, not repeated. This includes the origin.
  5. The polynomial $D(s)$ may not have any missing terms between those of highest and lowest degree unless all even terms or all odd terms are missing.
  6. The polynomial $N(s)$ may have missing terms between those of highest and lowest degree; some coefficients of $N(s)$ may be negative, too.
  7. The degree of $N(s)$ may be as small as zero, independent of the degree of $D(s)$.
  8. (a) For the voltage ratio transfer functions, $G$, and current ratio transfer functions, $\alpha$, the maximum degree of $N(s)$ is the degree of $D(s)$.
     (b) For transfer impedance $Z$ and transfer admittance $Y$, the maximum degree of $N(s)$ is the degree of $D(s)$ plus one.

**4. What is natural response of a network? Define open-circuit natural frequency (OCNF) and short-circuit natural frequency (SCNF)? How does the natural response of a network depend on the position of poles in the *s*-plane?**

**Natural response of a network** The *natural response* or *source-free response* of a network is the response with no source excitation; it is the response of the network to its initial conditions. This response is also known as the *transient response*.

The natural response of a system is also described as the response to unit impulse excitation. It is characterized by the *natural frequencies of oscillation* of the system.

We consider a driving-point impedance function as $Z_{11}(s)=\frac{V_1(s)}{I_1(s)}=K\frac{(s-z_1)(s-z_2)\cdots(s-z_n)}{(s-p_1)(s-p_2)\cdots(s-p_m)}$. For two different cases, two different natural frequencies are defined.

**Open-circuit natural frequencies (OCNF)** If $I_1 = 0$, i. e., the input is open-circuited then from the above equation, $V_1 = 0$ unless $Z_{11}(s)$ is infinite. Therefore, with input open-circuited, for a voltage to exist, it is necessary that $Z_{11}(s)$ is infinite.

In this case, the poles of $Z_{11}(s)$, i. e., $p_1, p_2, \ldots, p_m$ are known as the *open-circuit natural frequencies (OCNF)* of the network.

**Short-circuit natural frequencies (SCNF)** If $V_1 = 0$, i. e., the input is short-circuited then from the above equation, $I_1$ will be zero unless $Z_{11}(s)$ is zero. Therefore, with input short-circuited, for a current to exist, it is necessary that $Z_{11}(s)$ is zero.

In this case, the zeros of $Z_{11}(s)$, i.e., $z_1, z_2, \ldots, z_n$ are known as the *short-circuit natural frequencies (SCNF)* of the network.

We can summarize as

(i) the poles of an impedance function are OCNF of the network
(ii) the zeros of an impedance function are SCNF of the network
(iii) the poles of an admittance function are SCNF of the network
(iv) the zeros of an admittance function are OCNF of the network

**Pole Position and natural response** As the transfer function completely represents a system differential equation, its poles and zeros effectively define the system response. In particular, the system poles directly define the components in the homogeneous or natural response. The unforced or natural response of a linear BIBO system to a set of initial conditions is

$$y(t) = \sum_{i=1}^{n} K_i e^{\lambda_i t}$$

where, the constants $K_i$ are determined from the given set of initial conditions and the exponents $\lambda_i$ are the roots of the characteristic equation of the system. The characteristic equation is,

$$D(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

and its roots are the system poles, that is $\lambda_i = p_i$.

Therefore, the natural response may be written as $y(t) = \sum_{i=1}^{n} K_i e^{p_i t}$

**5. Explain how the stability of a network depends on the position of poles in the *s*-plane.**

**Or,**

**Prove that for BIBO stability, the roots of the characteristic equation or the poles of the system transfer function cannot be located in the right half of *s*-plane or on the $j\omega$-axis**

The stability of a linear system may be determined directly from its transfer function. An $n^{th}$ order linear system is asymptotically stable only if all of the components in the homogeneous response from a finite set of initial conditions decay to zero as time increases, i.e.,

$$\lim_{t \to \infty} \sum_{i=1}^{n} K_i e^{p_i t} = 0 \qquad (1)$$

where $p_i$ are the system poles. In a stable system, all components of the homogeneous response must decay to zero as time increases. If any pole has a positive real part, there is a component in the output that increases without bound, causing the system to be unstable.

*For a linear system to be stable, all of its poles must have negative real parts, i.e., they must lie within the left half of the s-plane.*

A system having one or more poles lying on the imaginary axis of the *s*-plane has non-decaying oscillatory components in its natural response, and is defined as a *marginally stable system.*

The location of the poles in the *s*-plane therefore defines the *n* components in the homogeneous response and the stability of the system as described below:

1. A real pole $p_i = -\sigma$ in the left half of the *s*-plane defines an exponentially decaying component, $Ke^{-\sigma t}$, in the homogeneous response. The rate of the decay is determined by the pole location; poles far from the origin in the left half plane correspond to components that decay rapidly, while poles near the origin correspond to slowly decaying components. In this case, the system is a stable system.
2. A pole at the origin $p_i = 0$ defines a component that is constant in amplitude and defined by the initial conditions. In this case, the system is a marginally stable system.
3. A real pole in the right half *s*-plane corresponds to an exponentially increasing component $Ke^{\sigma t}$ in the homogeneous response; thus defining the system to be unstable. In this case, the system is an unstable system.
4. A complex conjugate pole pair $(\sigma \pm j\omega)$ in the left half of the *s*-plane combine to generate a response component that is a decaying sinusoid of the form $Ae^{-\sigma t} \sin(\omega t \pm \varphi)$ where $A$ and $\varphi$ are determined by the initial conditions. The rate of decay is specified by $\sigma$; the frequency of oscillation is determined by $\omega$. The system is a stable system.
5. An imaginary pole pair, i.e., a pole pair lying on the imaginary axis, $\pm j\omega$, generates an oscillatory component with a constant amplitude determined by the initial conditions. The system is a marginally stable system

6. A complex pole pair in the right half-plane generates an exponentially increasing component. the system is an unstable system

These results are summarized in Fig. 12.70.

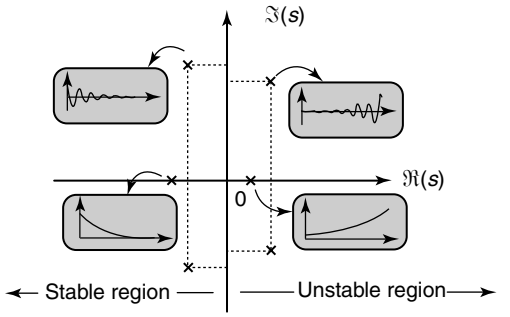



**Fig. 12.70** *Pole position and system stability*

From the above discussion, we conclude that for BIBO stability, the roots of the characteristic equation or the poles of the system transfer function cannot be located in the right half of the *s*-plane or on the *jω*-axis.

**6. What is a Bode plot? What are the advantages of using Bode plot?**

Bode plot or Bode diagram is the curve that provides variation of magnitude and phase of a network function with frequency in logarithmic scale.

We consider a network function given as

$$F(s)=\frac{N(s)}{D(s)}=K\frac{(s-z_1)(s-z_2)(s-z_3)\cdots(s-z_n)}{(s-p_1)(s-p_2)(s-p_3)\cdots(s-p_m)}$$
$$=|F(s)|\angle F(s)=M\angle\phi=Me^{j\phi}$$

Taking the natural log,

$$\text{Log}_e\left[F(s)\right]=Log_e\left[|F(s)|\right]\angle F(s)=\log_e\left[Me^{j\phi}\right]$$

or $$\ln\left[F(s)\right]=\ln M+j\phi$$

where, ln *M* is the logarithmic gain or simply gain in neper, and $\phi$ is the phase angle in radian.

Alternately, the gain is expressed in decibel (*db*), where, gain in $db = 20\log_{10} M$

- **Advantages of bode plot**

1. The product factors in $f(j\omega)$ become additive terms because of the use of logarithms.
2. The shape of the Bode plot is such that for most of the commonly used plots in servo systems, it can approximately represent the exact function by straight-line aysmptotes.

## Exercises

1. Calculate the driving-point impedance of the network shown in Fig. 12.71 and obtain the pole–zero plot.

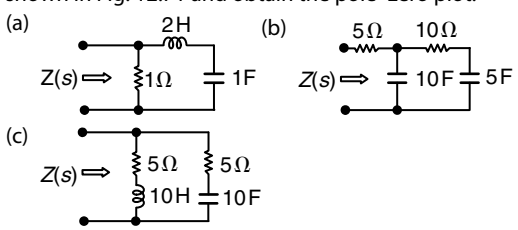
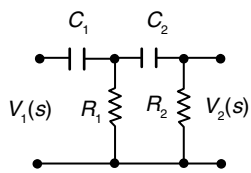



**Fig. 12.71**

[Ans: (a) $\frac{2s^2+1}{2s^2+s+1}$ (b) $\frac{(s+0.04)(s+0.01)}{s(s+0.03)}$ (c) $\frac{5(1+2s)(1+50s)}{100s^2+100s+1}$]

2. Determine the voltage transfer ratio $\frac{V_1(s)}{V_2(s)}$ of the networks shown in Fig. 12.72.

a)




[Ans: $\frac{R_1R_2C_1C_2s^2}{(R_1+R_2)C_2s+1+R_1C_1s(R_2C_2s+1)}$]

b)

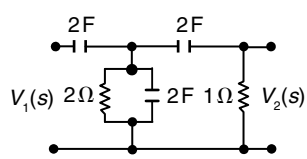



**Fig. 12.72**

[Ans: $\frac{8s^2}{12s^2+12s+1}$]

3. For the resistive two-port network of Fig. 12.73 determine: (a) $G_{21}$, (b) $Z_{21}$, (c) $Y_{21}$, and (d) $\alpha_{21}$.

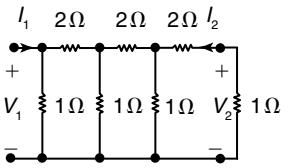



**Fig. 12.73**

$$[\frac{1}{41};\ \frac{1}{56};\ -\frac{1}{41};\ -\frac{1}{56}]$$

4. For the ladder network shown in Fig. 12.74,
   (a) obtain driving-point impedance function at the port 1
   (b) find voltage transfer function $\dfrac{V_2(s)}{V_1(s)}$

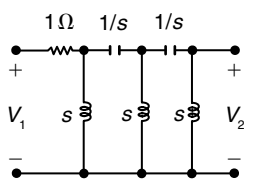



**Fig. 12.74**

$$[\frac{s^5+3s^4+3s^3+4s^2+s+1}{3s^4+4s^2+1};\ \frac{s^5}{s^5+3s^4+3s^3+4s^2+s+1}]$$

5. For a T-twin network as shown in Fig. 12.75, compute $Y_{12}(s)=\dfrac{I_1(s)}{V_2(s)}$ with $V_1(s)$ set to zero.

   Find the relationship between $C_1R_1$ and $C_2R_2$ for $Y_{12}(s)$ to exhibit a complex pair of zeros in the $j\omega$-axis.

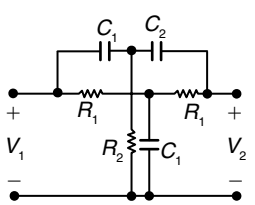



**Fig. 12.75**

$$[R_1C_1 = 2R_2C_2]$$

6. A network and a pole–zero pattern for driving-point impedance $Z(s)$ are shown in Fig. 12.76.

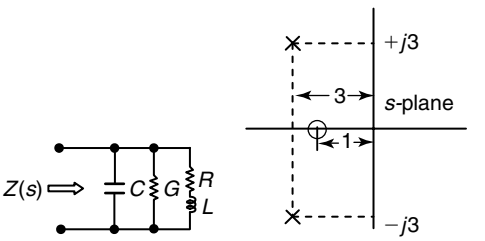



**Fig. 12.76**

   (a) Write an expression for $Z(s)$.
   (b) Specify the parameter $R, L, G$ and $C$ if $Z(0) = 1$.

$$[\text{a) } Z(s)=18\left(\frac{s+1}{s^2+6s+18}\right);\ \text{b) } R = 18\ \Omega;\ L = 18\ \text{H};\ G = \frac{17}{18}\ \Omega^{-1} C = \frac{1}{18}\ \text{F}]$$

7. The transform voltage of a network is given as

$$V(s)=\frac{4(s+1)}{(s+2)(s+3)}$$

   Draw its pole–zero diagram and hence obtain $v(t)$.

$$[\text{Ans: } v(t)=4e^{-2t}+8e^{-3t}]$$

8. The transform current $I(s)$ in a network is given by the following equation:

$$I(s)=\frac{2s}{(s+1)(s+2)}$$

   Plot the poles and zeros in the $s$-plane and hence obtain the time–domain response.

$$[\text{Ans: } i(t)=4e^{-2t}-2e^{-t}]$$

9. The zeros and poles of a network are given below:

   *Zeros*: $-1, (-1\pm j1)$ *Poles*: $-2, (-1\pm j2)$

   If $H(0) = 4$, find the network function $H(s)$ and draw the pole–zero plot.

10. Draw poles and zeros for $V(s)=\dfrac{(s+1)(s+3)}{(s+2)(s+4)}$ and evaluate $v(t)$ either analytically or by making use of pole zero diagram. $\left[\text{Ans: } v(t)=\delta(t)-\frac{1}{2}e^{-2t}-\frac{3}{2}e^{-4t}\right]$

11. Draw the asymptotic Bode plot for the following function:
    (a) $G(s)=\dfrac{1}{s(1+0.5s)(1+0.05s)}$
    (b) $V(s)=\dfrac{200(s+2)}{s(s^2+10s+100)}$

12. Find the frequency response plot of the following network function:

$$F(s)=\frac{s^2+1.03}{s^2+1.23}$$

13. Find the frequency response plot from the pole–zero plot for the following function:

$$F(s)=\frac{s^2+1.21}{s^2+1.44}$$

14. The Bode diagram of a function is shown in Fig. 12.77. Find the function. [Ans: $\frac{100}{(s+1)(s+5)}$]

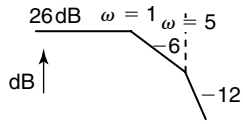



**Fig. 12.77**

## Questions

1. (a) Write short notes on driving point impedance and transfer impedance.

   or,

   Define driving point impedance and transfer impedance of a network.

   (b) Define precisely the following network functions with reference to a two-port network:

   (i) Driving-point impedance

   (ii) Transfer admittance

   (iii) Voltage transfer ratio

   (vi) Current transfer ratio

   (v) Transfer impedance

2. (a) Define precisely the various transfer functions of a two-terminal pair passive network.

   (b) What do you understand by driving point impedance of a two-port network? Enumerate important properties of driving point impedance functions of a two-port passive network.

   (c) What do you understand by the transfer function of a system? State its properties.

3. What are poles and zeros? What information do they provide in respect of the network to which they relate?

4. Explain how time-domain response of a system can be determined from the *s*-plane plot of the poles and zeros of its network function and from the transform of the network sources.

5. (a) Briefly discuss the restriction on pole–zero location in the *s*-plane for driving point impedance functions.

   (b) Briefly discuss the restriction on the pole–zero location in the *s*-plane for transfer functions.

6. (a) Explain how the frequency-domain response of a system can be determined from the *s*-plane plot of the poles and zeros of its network function.

   (b) What information do you get about the sinusoidal frequency response of a linear system from the pole–zero plot?

7. What is a Bode plot? Explain the technique for drawing it. What are the advantages of using a Bode plot?

## Multiple-Choice Questions

1. For the network shown in Fig.12.78, which of the following statements are true?

   1. $\lim_{s\to 0} Z_d(s)=1\ \Omega$
      $\lim_{s\to \infty} Z_d(s)=10\ \Omega$
   2. $\lim_{s\to 0} Z_d(s)=10\ \Omega$
      $\lim_{s\to \infty} Z_d(s)=1\ \Omega$

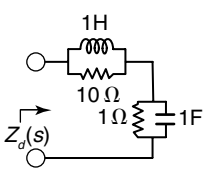



**Fig. 12.78**

   3. $Z_d(s)$ has two real poles and two complex conjugate zeros.
   4. $Z_d(s)$ has two complex conjugate poles and two complex conjugate zeros.

   Select the correct answer using the codes given below:

   (i) 1 and 3 (ii) 1 and 4 (iii) 2 and 3 (iv) 2 and 4

2. In Fig. 12.79, $Z_1(s)$ and $Z_2(s)$ are the driving point impedances of the networks. Match List 1 with List 2

and select the correct answer using the codes given below the lists:

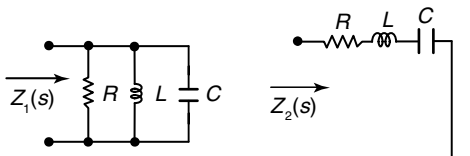



**Fig. 12.79**

| *List 1* (Stipulated condition for the impedance) | *List 2* (Value of the impedance) |
|---|---|
| A. $\lim_{s\to 0} Z_1(s)$ | 1. $R$ |
| B. $\lim_{s\to \infty} Z_1(s)$ | 2. $sL$ |
| C. $\lim_{s\to 0} Z_2(s)$ | 3. $\frac{1}{sL}$ |
| D. $Z_2(s)$ at $s = j\omega = \frac{j}{\sqrt{LC}}$ | 4. $\frac{1}{sC}$ |

Codes:

| | | | | |
|---|---|---|---|---|
| (i) | A | B | C | D |
| | 2 | 4 | 4 | 1 |
| (ii) | A | B | C | D |
| | 1 | 2 | 3 | 4 |
| (iii) | A | B | C | D |
| | 2 | 4 | 3 | 1 |
| (iv) | A | B | C | D |
| | 1 | 3 | 4 | 2 |

3. Consider the following statements regarding the two-port network shown in Fig. 12.80. The transfer functions of the network would include

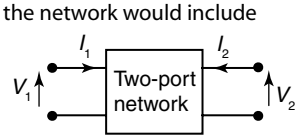



**Fig. 12.80**

(i) $V_2/V_1$, $I_2/I_1$, $V_2/I_1$ and $I_2/V_1$
(ii) $V_2/I_2$ and $V_1/I_1$
(iii) $V_2/V_1$ and $I_2/I_1$
(iv) $V_2/I_1$ and $I_2/V_1$.

4. For the network shown in Fig. 12.81, Z(0) = 3 Ω and Z(∞) = 2 Ω. The values of $R_1$ and $R_2$ will respectively be

(i) 2 Ω, 1 Ω
(ii) 1 Ω, 2 Ω
(iii) 3 Ω, 2 Ω
(iv) 2 Ω, 3 Ω.

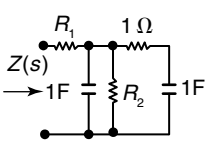



**Fig. 12.81**

5. The voltage ratio transfer function for the network shown in Fig. 12.82 under sinusoidal steady conditions is $G(j\omega)$. Match List I with List II and select the correct answer using the codes given below the lists:

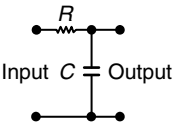


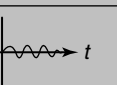
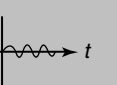
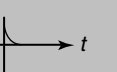
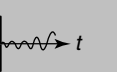
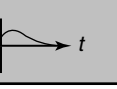

**Fig. 12.82**

| List I (Frequency) | List II ($G(j\omega)$) |
|---|---|
| 1. 0 | A. $1\angle 0°$ |
| 2. $\frac{1}{RC}$ | B. $0\angle -90°$ |
| 3. $\infty$ | C. $0.707\angle -45°$ |

Codes:

| | | | |
|---|---|---|---|
| (i) | A | B | C |
| | 1 | 3 | 2 |
| (ii) | A | B | C |
| | 1 | 2 | 3 |
| (iii) | A | B | C |
| | 2 | 1 | 3 |
| (iv) | A | B | C |
| | 3 | 2 | 1 |

6. Match List I with List II and select the correct answer using the codes given below the lists:

| *List I* (Location of poles on *s*-plane) | *List II* (Type of response) |
|---|---|
| A.  | 1.  |
| B.  | 2.  |
| C.  | 3.  |
| D.  | 4.  |
| | 5.  |

| | | | | |
|---|---|---|---|---|
| (i) | A | B | C | D |
| | 4 | 1 | 2 | 3 |
| (ii) | A | B | C | D |
| | 5 | 1 | 4 | 3 |

(iii) A B C D
2 3 5 4

(iv) A B C D
2 1 3 4

7. Consider the following statements:
   1. Transfer impedance is the reciprocal of transfer admittance.
   2. One can derive transfer impedance of a network if its driving-point impedance and admittance are known.
   3. Driving point impedance is the ratio of the Laplace transform of voltage and current functions at the input.

   Of these statements,
   (i) 1, 2 and 3 are correct
   (ii) 1 and 2 are correct
   (iii) 2 and 3 are correct
   (iv) 3 alone is correct

8. Consider the following statements regarding driving point admittance function having two complex conjugate poles:
   1. Closer the poles to $j\omega$-axis, higher the $Q$ of the circuit.
   2. Value of $Q$ varies inversely as the damping ratio.
   3. A circuit with low $R$ has low $Q$.

   Of these statements,
   (i) 1 and 3 are correct
   (ii) 2 and 3 are correct
   (iii) 1 and 2 are correct
   (iv) 1, 2 and 3 are correct

9. The pole–zero configuration of a network transfer function is shown in Fig. 12.84. The magnitude of the transfer function will

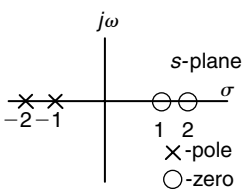


**Fig. 12.84**

   (i) decrease with frequency
   (ii) increase with frequency
   (iii) initially increase and then decrease with frequency
   (iv) be independent of frequency

10. Which one of the pairs of poles and responses is correctly matched?

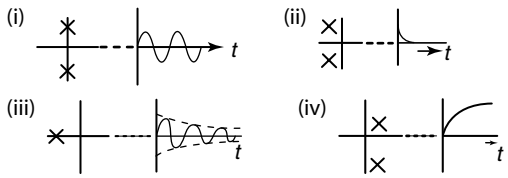


**Fig. 12.85**

11. If an impedance has the pole–zero pattern as shown in Fig. 12.86, it must be composed of

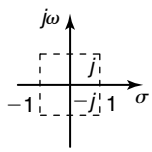


**Fig. 12.86**

   (i) *RC* elements only
   (ii) *RL* elements only
   (iii) *LC* elements only
   (iv) *RLC* elements

12. The voltage transfer function of the two-port network given in Fig. 12.87 has

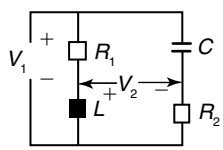


**Fig. 12.87**

   (i) a zero at the origin
   (ii) a zero at ∞
   (iii) no zero
   (iv) a zero at $j1$

13. For a transfer function $H(s) = P(s)/Q(s)$ where $P(s)$ and $Q(s)$ are polynomials in $s$
   (i) the degree of $P(s)$ is always greater than the degree of $Q(s)$
   (ii) the degree of $P(s)$ and $Q(s)$ are the same
   (iii) the degree of $P(s)$ is independent of the degree of $Q(s)$
   (iv) the maximum degree of $P(s)$ and $Q(s)$ differ at the most by one

14. A network has its response with time as shown in Fig. 12.88. Which one of the following diagrams represents the location of the poles in this network?

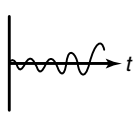
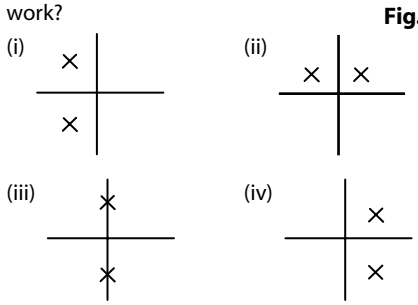


**Fig. 12.88**



15. Which one of the following circuits has a driving–point impedance of $Z(s)=\dfrac{2\left(s^2+s+\frac{1}{2}\right)}{s^2+s+1}$?

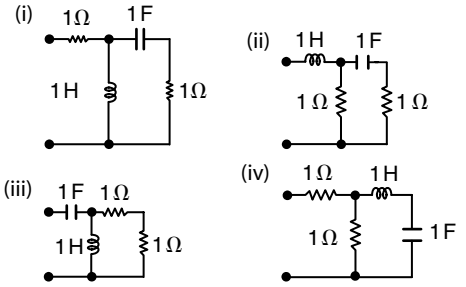



**Fig. 12.89**

16. A pole–zero pattern of a particular filter is shown in Fig. 12.90. It is that of a/an

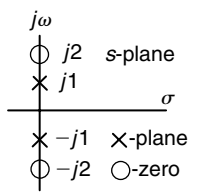



**Fig. 12.90**

(i) low-pass filter
(ii) high-pass filter
(iii) band-pass filter
(iv) all-pass filter

17. An *RC* driving point impedance function has zeros at $s=-2$ and $s=-5$. The admissible poles for the function would be
(i) $s=0; s=-6$
(ii) $s=-1; s=-3$
(iii) $s=0; s=-1$
(iv) $s=-3; s=-4$

18. Figure12.91 indicates the locus for the total current taken by a two-branch parallel circuit fed from a constant voltage ac source *V*, when one element of the circuit is varied. Each branch contains two elements (*R*, *L* or *C*) in series. Consider the following combinations:

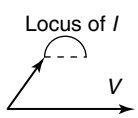



**Fig. 12.91**

1. Fixed *R* and *C*
2. Fixed *R* and variable *L*
3. Variable *R* and fixed *R*
4. Fixed *R* and variable *C*

The arrangements at the two branches are respectively
(i) 1 and 2 (ii) 2 and 3
(iii) 1 and 4 (iv) 2 and 4

19. The admittance locus of the circuit shown in Fig.12.92 is

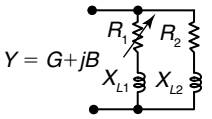
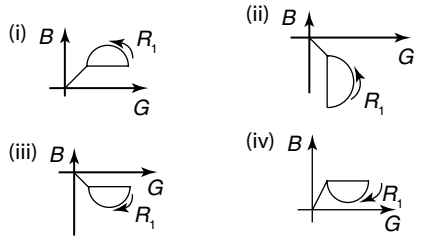



**Fig. 12.92**




20. A system has a single pole. The constant multiplier '*K*' is 1. For the given excitation sin *t*, the response is $\sqrt{2}$ with 45° lagging. The system has a pole and a zero respectively at
(i) zero and 1
(ii) infinity and –1
(iii) –1 and zero
(iv) zero and –1

21. Consider the following functions:

1. $\dfrac{s^3+3s}{2\left(s^2+1\right)}$
2. $\dfrac{s^3+3s+1}{5\left(s^2+1\right)}$
3. $\dfrac{s^2+1}{3s}$
4. $\dfrac{3\left(s^3+\frac{7s}{2}\right)}{s^4+14s^2+30}$

Which of these are valid driving point impedance functions?
(i) 1, 2 and 3 (ii) 1, 3 and 4
(iii) 1, 2 and 4 (iv) 2, 3 and 4

22. A driving point impedance function is given by

$$F(s)=\frac{(s+k_1)(s+k_2)(s+k_3)}{(s+1)(s+2)(s+3)(s+4)}$$

The product $k_1k_2k_3$ cannot be more than
(i) 48 (ii) 24
(iii) 10 (iv) 4

23. A pole of driving point admittance function implies
(i) zero current for a finite value of driving voltage
(ii) zero voltage for a finite value of driving current
(iii) an open-circuit condition
(iv) none of (i), (ii) and (iii) mentioned in the question

24. The driving point function of the circuit shown in Fig.12.93 when $s \to 0$ and $s \to \infty$, (the elements are normalized) will respectively be

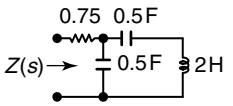



**Fig. 12.93**

(i) $\frac{1}{s}$ and $\frac{2}{s}$ (ii) $\frac{1}{s}$ and 0.75

(iii) 0.75 and $\frac{2}{s}$ (iv) $\frac{2}{s}$ and 0.75

25. When the input frequency is varied from zero to infinity causing variations in magnitude and phase angle, the traced locus by phasor is called the

(i) polar plot (ii) Bode plot

(iii) locus plot (iv) none of these

26. The frequency at which two asymptotes meet is known as

(i) corner or break frequency

(ii) threshold frequency

(iii) cut-off frequency

(iv) critical frequency

27. For a constant gain $k$, log-magnitude curve in Bode plot is

(i) zero (ii) negative

(iii) positive (iv) negative or positive depending upon the value of $k$

28. Slope of a factor $k$, in plotting the Bode magnitude curve is

(i) 0° (ii) ∞

(iii) 90° (iv) none of these

29. A system has fourteen poles and two zeros. The slope of its highest frequency asymptote in its magnitude plot is

(i) −40 dB/decade

(ii) −240 dB/decade

(iii) −280 dB/decade

(iv) −320 dB/decade

30. A network function can be completely specified by

(i) real parts of zeros

(ii) poles and zeros

(iii) real parts of poles

(iv) poles, zeros and a scale factor

31. The frequency response for a network function $H(s)$ is given in Fig. 12.94. $H(s)$ is given by

(i) $\frac{s}{s+1}$

(ii) $\frac{10s}{s+1}$

(iii) $\frac{10}{s+1}$

(iv) $\frac{1}{s(s+1)}$

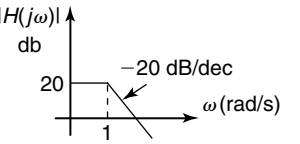



**Fig. 12.94**

32. The driving point impedance of the network shown in Fig. 12.95 is given by

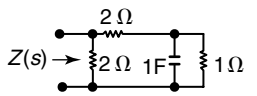



**Fig. 12.95**

(i) $\frac{4\left(s+\frac{3}{2}\right)}{\left(s+\frac{5}{4}\right)}$ (ii) $\frac{\left(s+\frac{3}{2}\right)}{\left(s+\frac{5}{4}\right)}$

(iii) $\frac{\left(s+\frac{5}{4}\right)}{\left(s+\frac{3}{2}\right)}$ (iv) $\frac{1}{4}\frac{\left(s+\frac{3}{2}\right)}{\left(s+\frac{5}{4}\right)}$

33. The *R-L-C* series circuit shown is supplied from a variable frequency voltage source. The admittance-locus of the *R-L-C* network at terminals *AB* for increasing frequency $\omega$ is

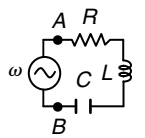
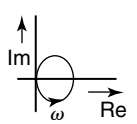
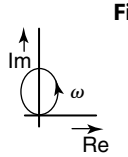
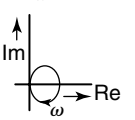
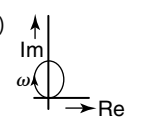



**Fig. 12.96**

(i)



(ii)



(iii)



(iv)



34. In Fig. 12.97, all phasors are with reference to the potential at the point *O*. The locus of the voltage phasor $V_{YX}$ as $R$ is varied from zero to infinity is shown by

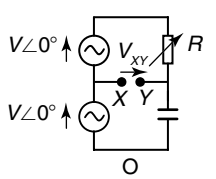
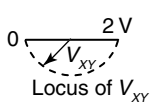
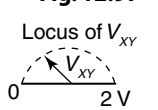
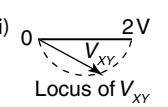
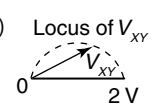



**Fig. 12.97**

(i)



(ii)



(iii)



(iv)


35. Consider the Bode magnitude plot shown in Fig. 12.98. The transfer function $H(s)$ is

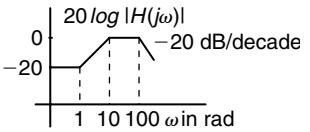



**Fig. 12.98**

(i) $\dfrac{(s+10)}{(s+1)(s+100)}$ (ii) $\dfrac{10(s+1)}{(s+1)(s+100)}$

(iii) $\dfrac{10^2(s+1)}{(s+10)(s+100)}$ (iv) $\dfrac{10^2(s+100)}{(s+10)(s+10)}$

36. A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at $s = -2$ and $s = -4$, and one simple zero at $s = -1$. A unit step $u(t)$ is applied at the input of the system. At steady state, the output has a constant value of 1. The impulse response of this system is

(i) $\left[\exp(-2t)+\exp(-4t)\right]u(t)$

(ii) $\left[-4\exp(-2t)-12\exp(-4t)-\exp(-t)\right]u(t)$

(iii) $\left[-4\exp(-2t)+12\exp(-4t)\right]u(t)$

(iv) $\left[-0.5\exp(-2t)+1.5\exp(-4t)\right]u(t)$

## Answers

1. (i)
2. (i)
3. (i)
4. (i)
5. (i)
6. (iv)
7. (iv)
8. (iii)
9. (iv)
10. (i)
11. (iv)
12. (i)
13. (iii)
14. (iv)
15. (i)
16. (iii)
17. (ii)
18. (iii)
19. (ii)
20. (iv)
21. (ii)
22. (ii)
23. (ii)
24. (ii)
25. (i)
26. (i)
27. (iv)
28. (i)
29. (iii)
30. (iv)
31. (iii)
32. (ii)
33 (i)
34. (ii)
35. (iii)
36. (iii)